UNDERWRITE FOR COLLECTION

A proposed Architecture for MSME Credit LifeCycle System

Building credit around sustainable cash outcomes,

temporal evidence and responsible intervention

Draft compilation: Parts 1, 2a, 2b, 3, 4 and 5

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DRAFT COMPILATION

This reading edition compiles the accepted first drafts of Parts 1, 2a, 2b, 3, 4 and 5. Part 6 will add the pilot, quasi-experimental and ecosystem execution programme. Citations and conclusions remain subject to the final series-level reconciliation.

Part 1

The MSME Credit Lifecycle and Collection Cascades

Underwrite for Collection Series

Abstract

A credit decision is usually recorded at origination, but its economic quality is revealed through a sequence of later cash events. This is especially important for micro, small and medium enterprises whose customer receipts, inventory purchases, payroll, taxes, supplier obligations and household withdrawals often move on different calendars. A viable business can therefore experience a payment-timed liquidity shortage, while a weak product structure can convert a temporary shortage into repeated arrears and eventual loss.

This paper develops **underwrite for collection** as a lifecycle architecture for Kenyan MSME credit. The proposition is that product design, underwriting evidence, monitoring, intervention, restructuring and recovery should be connected through one cash-flow and credit-state history while retaining clear institutional responsibilities. Collection is treated as the production of sustainable contractual or restructured cash flows, measured alongside customer outcomes, operating cost, expected loss, capital and funding consequences.

Part 1 establishes the economic problem, interprets Kenyan portfolio evidence through consistent populations and denominators, maps collection cascades, and identifies the ecosystem required for responsible implementation. The later parts develop temporal data and feature engineering, hierarchical Bayesian and causal inference, regulatory orchestration, bank-controlled research architecture, KESONIA-linked pricing and balance-sheet consequences, and a programme of randomised pilots and quasi-experimental evaluation.

1. When a profitable month still misses Tuesday's instalment

Imagine a small wholesaler serving kiosks, restaurants and informal traders on the edge of a Kenyan town. The business begins the month with stock purchased partly in cash and partly on seven-day supplier terms. Some customers pay immediately through mobile money. Others receive goods on short trade credit because maintaining those relationships matters more than insisting on cash at every delivery. Two larger customers settle only after completing their own approval cycles. Payroll falls on the twenty-eighth day. Rent, electricity, transport and tax obligations follow separate calendars. The owner also draws money for household needs because the boundary between enterprise and family liquidity is economically real even when the accounting records treat them separately.

At month end, sales may exceed the cost of stock and operating expenses. The income statement can show a profit. Yet the Tuesday loan instalment may still fail if the larger customer payment arrives on Friday, the supplier debit falls on Monday, and the owner replenishes fast-moving stock before the weekend. The problem is not captured fully by monthly income, average account balance or a static debt-service ratio. It is a timing problem inside a wider operating cycle.

The lender experiences the same business through a different calendar. Its system knows the disbursement date, contractual schedule, interest accrual, due amount and days past due. The borrower experiences inventory days, receivable days, supplier terms, wage dates, seasonality and the timing uncertainty of each customer payment. Underwriting becomes stronger when those two calendars are designed to meet rather than left to collide after disbursement.

This paper uses **MSME** as a practical portfolio term while preserving the distinctions inside it. The Central Bank of Kenya's 2024 supply-side survey classified enterprises by turnover, employment and investment, drawing on Kenya's Micro and Small Enterprises Act and the relevant public-finance framework. In that survey, micro enterprises had annual turnover not exceeding KSh500,000 and fewer than ten workers; small enterprises had turnover between KSh500,000 and KSh5 million and 10 to 49 workers; medium enterprises had turnover between KSh5 million and KSh100 million and 51 to 250 workers, with additional investment thresholds [1]. The Micro and Small Enterprises Act itself establishes statutory definitions for micro and small enterprises and recognises enterprises across production and services, including formal and informal activity [2].

Those categories support reporting and policy, but they do not make the underlying businesses homogeneous. A market trader, transport operator, manufacturer, pharmacy, farm-input dealer and professional practice can occupy the same broad size category while presenting different cash-conversion cycles, fixed-cost burdens, seasonality, concentration and recovery options. A bank may also segment enterprises differently for product, risk or relationship management. Every empirical result in this series will therefore state the population definition used by its source or participating institution.

The economic unit of analysis is not simply "the MSME" or "the loan". It is the relationship among a business model, a financing purpose, a contractual facility and a sequence of cash obligations. The same enterprise may use an overdraft to bridge receivables, asset finance to acquire productive equipment, invoice finance against a particular debtor, a term loan for expansion, and a digital working-capital facility for short stock cycles. A sound assessment asks whether each product fits the cash flow it is meant to finance and how the facilities interact when conditions weaken.

Let the business's liquid cash position immediately before the contractual payment at time (t) be represented as:

$$L_t = B_{t^-} + C_t^{\text{collected}} - O_t^{\text{essential}} - W_t^{\text{committed}} - D_t^{\text{senior}},$$

where B_{t^-} is available cash before the payment window, $C_t^{\text{collected}}$ is cash actually settled and available, $O_t^{\text{essential}}$ is essential operating expenditure, $W_t^{\text{committed}}$ is other committed working-capital use, and D_t^{senior} is any obligation contractually or operationally paid before the facility being assessed. This is a descriptive liquidity identity, not a universal affordability formula. Its purpose is to show why booked revenue, invoiced sales and average monthly inflows do not necessarily equal cash available for a particular instalment.

The first architectural task is therefore to align observable business clocks with the credit clock.

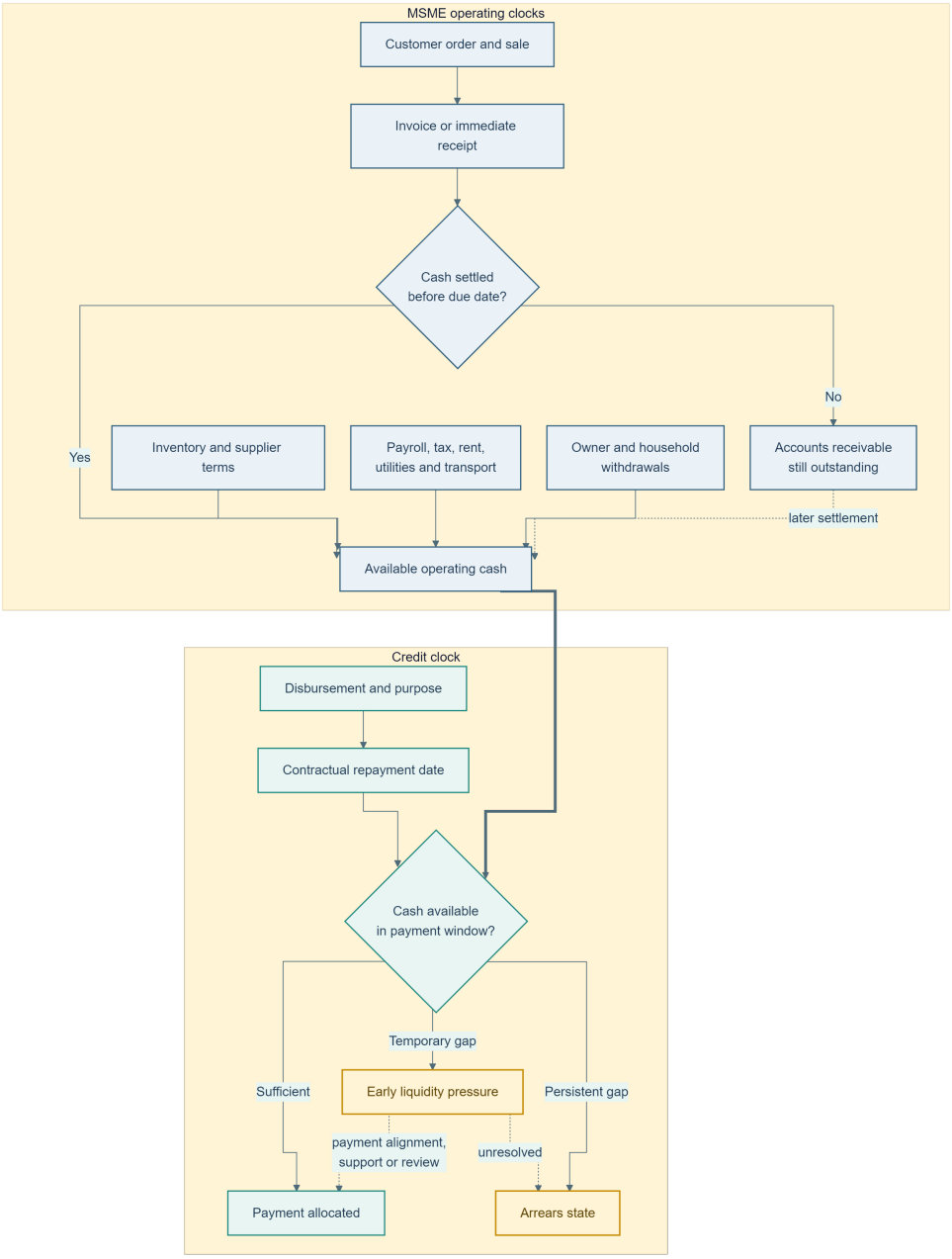


Figure 1: Two calendars determine whether a viable MSME can meet a particular instalment. The business clock records when economic activity turns into available cash. The credit clock records when contractual cash is due. Product design and lifecycle monitoring connect them.

This starting point changes the underwriting question. Instead of asking only whether the enterprise appears able to repay on an average day, the lender asks how repayment will be produced through the facility's life, which events could interrupt it, which evidence would reveal deterioration, and which actions could restore alignment while the enterprise remains viable. That is the practical foundation of underwriting for collection.

2. Read the portfolio through matched populations

The national evidence gives this question material scale. The CBK survey reported approximately 0.89 million active MSME loan accounts worth KSh784.4 billion at December 2024. Micro enterprises represented about 80 percent of accounts across the surveyed banking industry, while medium enterprises held the largest share of MSME credit by value within commercial banks. The average loan therefore conceals a portfolio with many small exposures and a smaller number of much larger balances [1].

Asset quality also changes according to the unit being measured. The same survey recorded 252,502 non-performing MSME accounts worth KSh149.8 billion. These represented 28.4 percent of MSME accounts and 19.1 percent of outstanding MSME value. An account-based ratio is sensitive to the number of smaller facilities in arrears. A value-based ratio is sensitive to the balances of larger problem exposures. Both are useful, but they answer different management questions [1].

The KBA MSME dashboard provides a later and separately scoped view. Its selected June 2026 portfolio reported a value-based NPL ratio of 24.5 percent and KSh234.67 billion of NPL outstanding [3]. The dashboard also separates product, institution, borrower and count views. Its figures should travel with their reporting month, filter state, covered institutions, numerator and denominator. That discipline allows the dashboard to support comparison within its defined population without being spliced into a differently constructed CBK survey series.

The full banking-sector picture moved on another path. CBK reported that gross loans rose from KSh4,453.0 billion in March 2026 to KSh4,645.5 billion in June 2026. Over the same quarter, the sector's gross NPL ratio improved from 15.6 percent to 14.8 percent as gross NPLs declined by 1.3 percent and gross loans increased by 4.3 percent [4]. This broader ratio describes all commercial-bank lending rather than the MSME dashboard population. CBK respondents also expected recovery efforts to intensify across nine economic sectors during the quarter ending September 2026. That is an expectation reported by credit officers, and later realised outcomes require their own observation [4].

Figure 1A turns these statements into three deliberately different views. Panel A holds the CBK December 2024 MSME population constant and asks how non-performance looks by account count and outstanding value. Panel B places three value-based NPL ratios beside one another while preserving their different dates and coverage. Panel C shows why account-weighted analysis is especially important in a portfolio where micro enterprises comprise approximately four-fifths of active accounts. The panels are read together, but they are not merged into one trend or denominator.

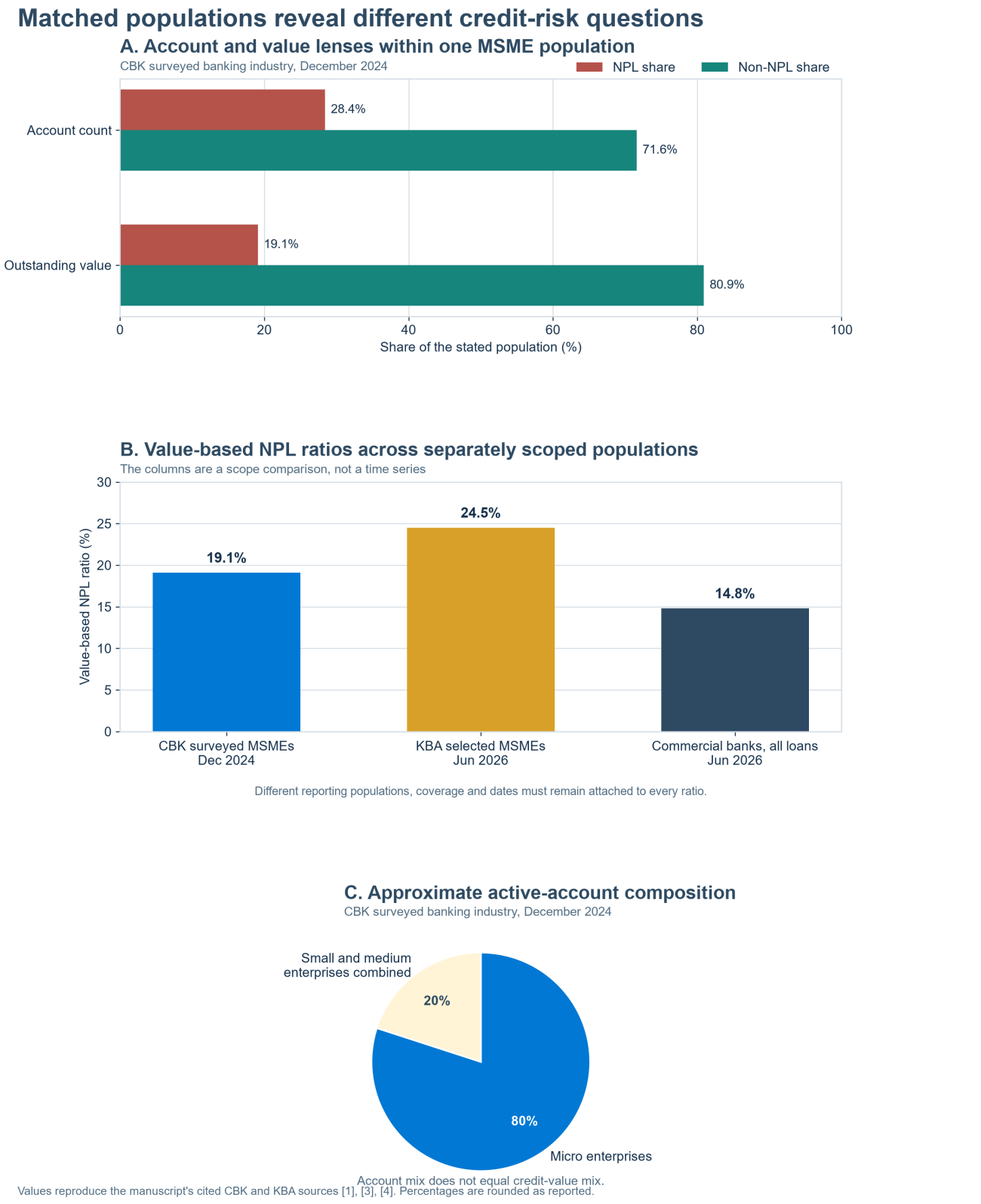


Figure 1A: Matched-population views of Kenyan MSME credit evidence. Panel A holds one CBK 2024 MSME population; Panel B compares separately scoped value ratios, not a time series; and Panel C shows active-account composition. Sources: [1], [3], [4].

These views form a more useful evidence chain when their identities remain visible:

Measure	Economic meaning	Minimum reporting information	Question it can answer
New lending during a period	Origination flow	Start and end date, product, institution and amount	How much qualifying credit was advanced?
Outstanding performing balance	Stock at a date	Reporting date, accrued amounts and classification	How much current exposure remains?
NPL value ratio	Problem balance divided by the stated loan balance	Definition of NPL, numerator, denominator and date	What share of value is non-performing?
NPL account ratio	Problem accounts divided by stated accounts	Account unit, multiple facilities and closure treatment	How widespread is non-performance by count?
Vintage default rate	Defaults from one origination cohort divided by that cohort's exposure	Origination window, maturity, censoring and horizon	How did a particular lending cohort perform?
Cure and re-default	Movement out of and back into delinquency	State rules, performance window and modification treatment	Did apparent improvement remain durable?
Net recovery	Realised recovery less attributable workout cost	Cash dates, cost allocation and discount basis	What economic value remained after default?

This denominator discipline changes management conversation. Origination, delinquency, cure and recovery become connected observations rather than competing headlines. A bank can follow a defined cohort from application and disbursement through scheduled payments, actual receipts, arrears, modification, cure, default and recovery. KBA can support common industry definitions and aggregate comparison. Regulators and public institutions can interpret access and stability using transparent populations. Researchers can identify which questions public aggregates can answer and which require controlled institution-level data.

The resulting research object is a **lifecycle vintage**. It contains facilities sharing a defined origination window, product, borrower segment and observation horizon. Within that vintage, account transitions and cash outcomes can be measured consistently. This creates the empirical bridge between lending activity and collection performance that a stock-versus-flow comparison cannot supply on its own.

3. Collection begins in product architecture

A collection outcome is partly designed before the first shilling is disbursed. The financing purpose determines when productive value can emerge. The tenor determines how long the business has to convert that value into cash. Repayment frequency determines how closely debt service follows the receipt cycle. Grace periods, draw conditions, limit controls, security, guarantees, covenants and pricing determine how risk is divided and how much room remains when conditions change.

Berger and Udell describe an SME lending technology as more than a score. It combines primary information sources, screening and underwriting policies, the structure of the loan contract, and mon-

itoring strategies. Their wider framework also connects lending technology to the legal, information, regulatory and institutional environment in which finance is provided [5]. This is a useful foundation for the present argument. A bank does not underwrite an enterprise in isolation. It selects a technology through which a particular product will gather evidence, allocate rights, observe performance and respond to deterioration.

Consider four common mismatches. A short digital loan can fund inventory whose sale and customer settlement take longer than the contractual tenor. An overdraft can finance equipment that produces returns over several years, leaving the facility permanently utilised. Equal monthly instalments can burden a seasonal enterprise during its low-revenue period even when annual cash generation is sufficient. An automatic sweep can collect efficiently while removing the liquidity needed to purchase the next unit of stock, weakening the activity from which repayment must ultimately come.

Each mismatch can appear initially as a borrower-risk problem. Yet part of the risk originates in the relationship between purpose and structure. Product architecture should therefore answer seven questions before approval:

1. What productive or liquidity need is the facility funding?
2. When should that use generate settled cash rather than booked revenue?
3. Which cash flows are realistically available for debt service after essential operating needs?
4. How do tenor, repayment dates and amortisation follow that conversion process?
5. Which other lenders, suppliers, tax claims or account sweeps compete for the same cash?
6. Which observable events indicate a temporary timing gap, a persistent viability problem or a servicing error?
7. Which actions remain proportionate and useful at each stage?

The answer is not necessarily a more flexible product in every case. Flexibility can postpone recognition of a structurally weak facility. The objective is an explicit fit among purpose, cash generation, payment structure, control rights and remedial options. A productive asset with a long payback period may require term finance. A verified invoice may support a self-liquidating facility linked to its settlement. Fast-turning stock may support a shorter working-capital cycle. Seasonal activity may justify a repayment profile aligned with observable peaks and troughs. Each design still requires affordability, credit approval, pricing and portfolio controls.

The current Basel credit-risk principles place these activities in one governance chain. They connect a sound credit-granting process with ongoing administration, grading, monitoring, independent review and early remedial action. They also call for prompt identification of deterioration and recognise that specialised workout capacity may improve rehabilitation and ultimate repayment. The principles warn that remuneration linked narrowly to volume, short-term profit or portfolio indicators can create conflicts with sound credit management [6].

This matters because an organisation can optimise each department locally while weakening the whole credit relationship. Sales can maximise disbursement. Credit can minimise approval-time default estimates. Operations can maximise automated deductions. Collections can maximise this week’s receipts. Finance can minimise current provisions. Treasury can minimise apparent funding cost. Sus-

tainable collectability requires a shared lifecycle scorecard so that one function cannot improve its headline measure by transferring risk or cost to another stage.

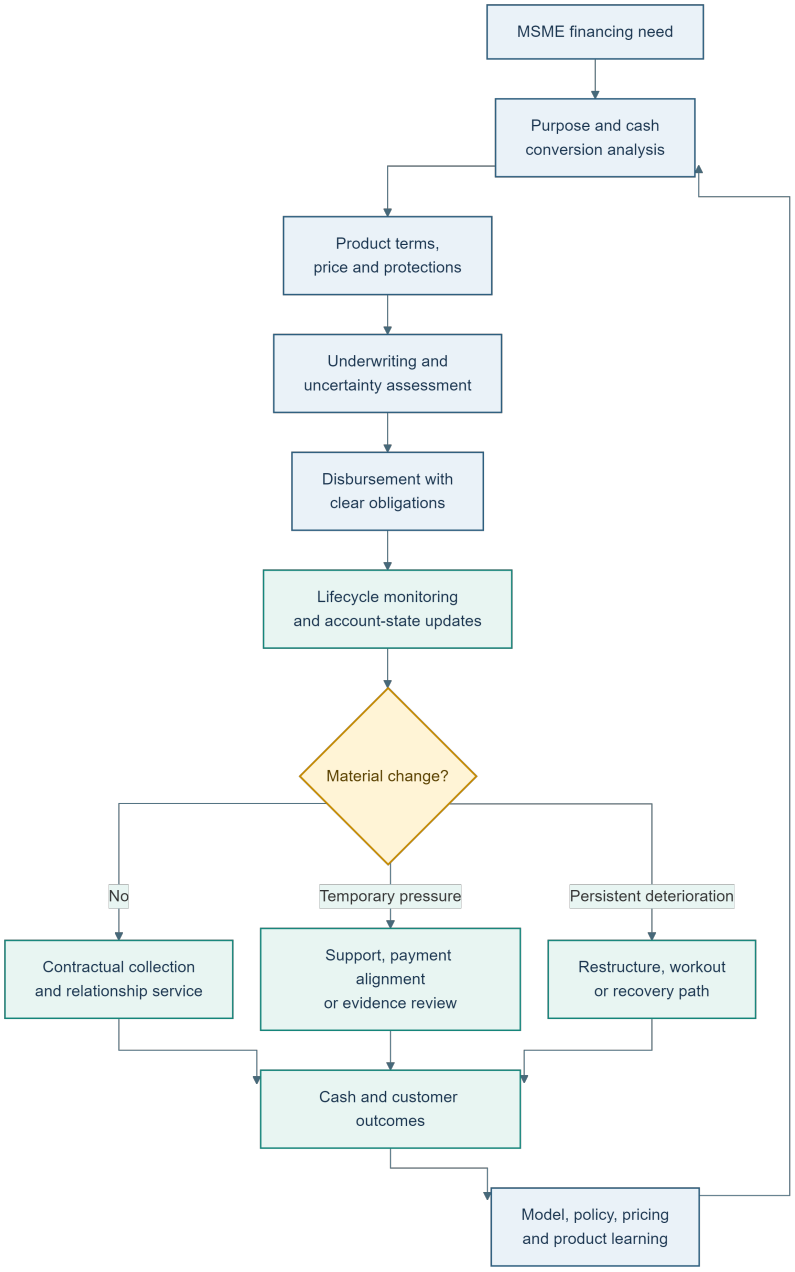


Figure 2: Product architecture begins and completes the collection loop. Realised payment, cure, recovery, cost and customer outcomes return to product design through governed learning.

Underwriting for collection can now be defined more precisely. It is the governed design and assessment of a credit relationship so that expected contractual or restructured cash flows remain compatible with the enterprise's economic activity, while deterioration can be recognised and addressed through lawful, proportionate action. Approval remains important, but it becomes one gate within a longer productive system.

4. From cash-flow pressure to a collection cascade

An MSME default is often preceded by a chain of smaller adaptations. A restaurant loses a major corporate customer and initially absorbs the decline through its cash buffer. It then buys smaller quantities from suppliers, losing volume discounts. The owner delays maintenance, draws an overdraft more heavily and pays one obligation with cash intended for another. Stock-outs reduce sales. A missed instalment attracts charges and collection contact. Suppliers shorten terms. The business can move from a temporary revenue shock to a self-reinforcing liquidity and credit problem.

The sequence matters because the same observed arrears balance can arise from different mechanisms. A delayed receivable may produce a short timing gap with a credible cure path. A permanent loss of customers may indicate reduced operating viability. A payment-allocation error may create apparent delinquency even though cash was received. Multiple automatic deductions may generate avoidable competition for liquidity. Product misuse may place long-term investment inside a facility designed for short working-capital turnover. A lifecycle architecture seeks evidence that distinguishes these mechanisms early enough for the response to fit the problem.

The credit state at time (t) can be represented at a high level as:

$$S_t \in \{\text{current},\ \text{early arrears},\ \text{late arrears},\ \text{modified},\ \text{cured},\ \text{defaulted},\ \text{recovery},\ \text{closed}\}.$$

This set is a management vocabulary rather than the final statistical model. Part 2b will define transition, censoring and competing-risk mechanics. For Part 1, its value is to prevent several economically different conditions from being compressed into a single performing or non-performing label.

A **cure** occurs when an account returns from a defined delinquent state to a defined performing state. A **durable cure** remains performing through a specified follow-up window. A **re-default** occurs when material delinquency or default returns after cure. A **recovery** records value realised after default and should be reported gross and net of attributable workout costs. These definitions make short-term cash receipt, restored payment capacity and final economic resolution visibly different outcomes.

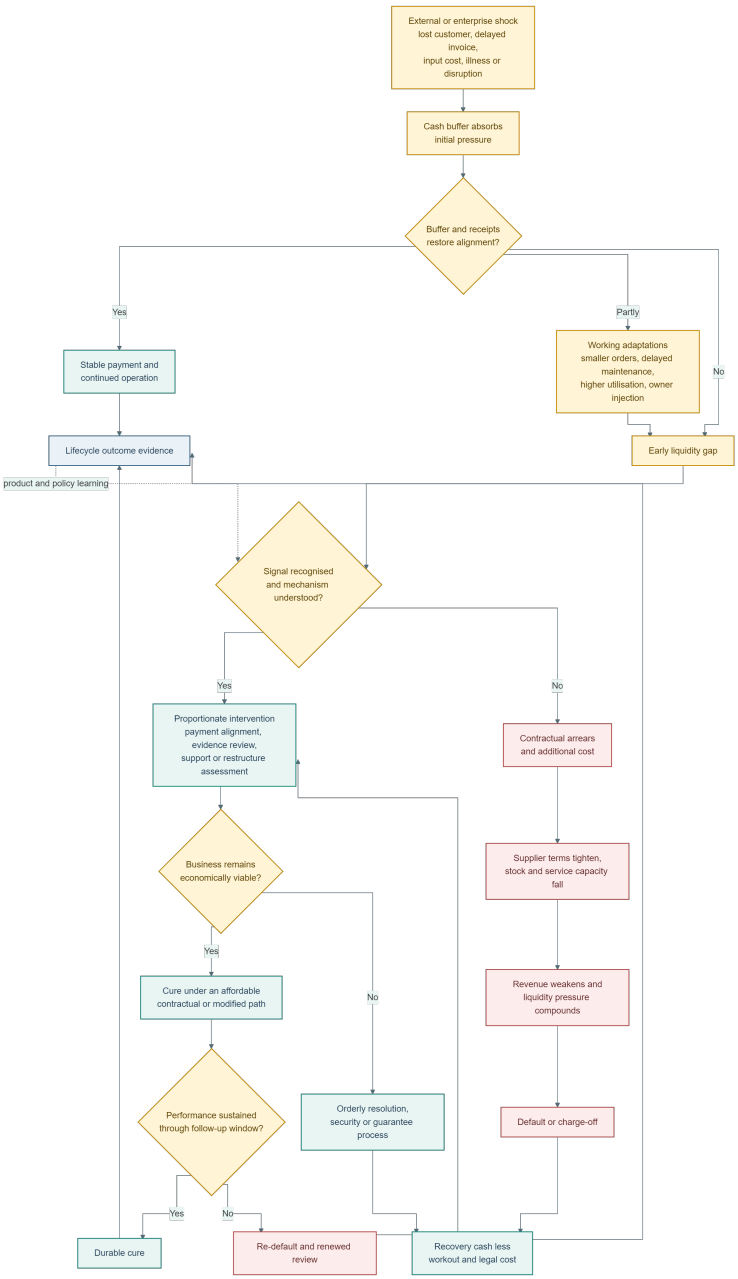


Figure 3: An MSME collection cascade with stabilising intervention paths. The architecture distinguishes an external shock from the adaptations, account states and feedback loops through which the shock affects repayment. Earlier diagnosis preserves a wider range of constructive actions.

The diagram also shows why collection conduct and credit performance are connected. Contact that consumes the owner's operating time, creates reputational harm or demands cash needed to complete a profitable order can weaken the repayment source. A well-timed reminder, accurate reconciliation, temporary alignment of payment dates, or a viability-based restructuring review may preserve more value. Part 3 will translate this principle into regulated action design and customer safeguards.

The bank should measure the entire path. A short-term payment following intensive contact may be followed by renewed arrears. A modification may lower the instalment yet extend exposure and total financing cost. A recovery may look strong before legal, repossession and staff expenses are recognised. A durable lifecycle scorecard therefore includes time to arrears, cure, durability, re-default, cash collected against cash due, modification performance, gross and net recovery, complaint outcomes and the continuing operation of the financed business where this can be measured lawfully.

This creates a productive learning loop. Each completed or censored account history adds evidence about the fit of the original product, the timing of deterioration, the response to intervention and the eventual cash outcome. That evidence can improve future design, but it must be reconstructed point in time and interpreted with the policy that produced it. Parts 2a and 2b develop those requirements.

5. One borrower, many institutions, one economic outcome

The MSME experiences one financing relationship even when the supporting capabilities are distributed across many organisations. The bank may own the facility and credit decision. A payment provider records cash movement. A credit-reference bureau holds shared credit information. An accountant or enterprise platform may hold invoices and operating records. A guarantee provider may absorb a defined share of loss. A specialist servicer may support restructuring or recovery. Each participant sees a different projection of the same economic relationship.

Fragmentation becomes costly when definitions, incentives and clocks fail to meet. The payment system can show a receipt that the loan ledger allocates a day later. A relationship manager can know that a major invoice is delayed while the collections engine sees only a broken instalment. A bureau can record an exposure without the lender's current hardship context. Finance can calculate an allowance from one state history while operations report cure from another. The borrower can receive several messages from teams that do not share the same case record.

The World Bank's analysis of Kenya's entrepreneurship ecosystem connects MSME finance to product diversity, financial information, payment delays, managerial capacity, guarantees and wider institutional infrastructure [7]. Underwrite for collection adds a lifecycle question to that ecosystem view: how can each capability contribute to a coherent and fair path from financing need to final cash outcome?

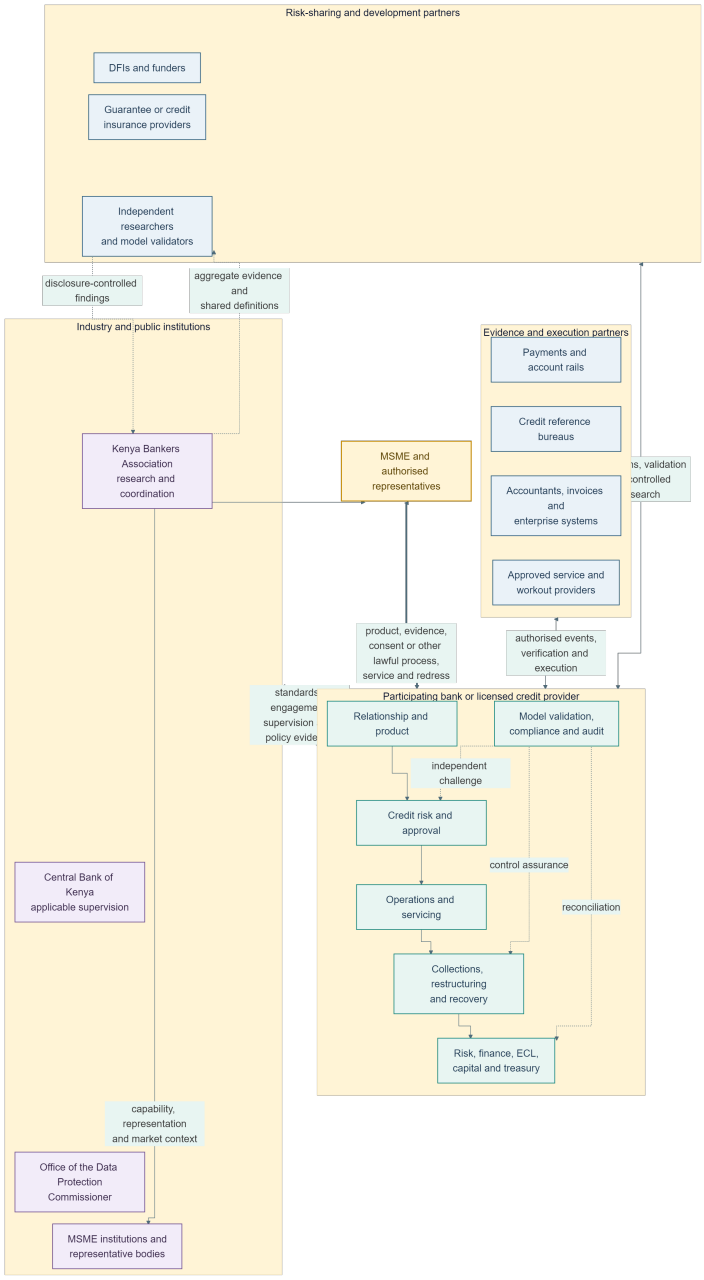


Figure 4: Proposed MSME lifecycle-credit ecosystem. The participating lender remains accountable for its product, credit decision, servicing and controls. Partners contribute authorised evidence, execution, risk sharing, research or oversight through defined interfaces.

The MSME remains an active participant rather than a data source at the edge of the diagram. Product purpose, repayment expectations, data use, price, available support, consequences of default and routes for correction or complaint should be intelligible. Enterprise representatives can also help define outcomes that matter beyond immediate repayment, including continuity of trade, employment, supplier relationships and the burden created by an intervention.

Within the lender, ownership must remain precise. The business function understands the relationship and financing purpose. Credit approves exposure within policy. Operations maintains the account and allocates payments. Collections and restructuring manage deterioration through approved actions. Finance and risk translate cash outcomes into allowance, capital and management information. Treasury assesses funding and liquidity. Validation, compliance and audit provide independent challenge. A shared event and state history connects these roles without collapsing their responsibilities.

KBA can support common research definitions, aggregate evidence and convening. CBK and other relevant authorities retain their statutory mandates. DFIs, guarantee providers and credit insurers can support risk sharing or research when the contractual structure, incentives and claims process are explicit. Technology and data partners can provide tools within the lender's control environment. Independent researchers can evaluate performance through approved, disclosure-controlled access.

This is a proposed collaboration architecture. Participation, data access, regulatory engagement and publication rights will be established through written agreements in any actual pilot. Part 4 specifies the bank-controlled technical boundary, while Part 6 turns the ecosystem into a staged research and implementation programme.

6. From a strong proposition to testable evidence

The lifecycle thesis creates hypotheses rather than predetermined results. Payment-date alignment may improve durable cure for enterprises with predictable receipt cycles. A temporary working-capital adjustment may preserve more value than repeated collection contact when a verified customer payment is delayed. Additional transaction evidence may improve risk differentiation for one product and add little for another. A guarantee may expand responsible access, alter screening incentives, or shift losses without improving the underlying facility. Each proposition needs an estimand, comparison, horizon and outcome.

Historical data provide an essential starting point, but treatment decisions shape the history being analysed. Accounts receiving intensive collection may have been selected because they were already deteriorating. Borrowers approved under one policy differ from rejected applicants whose repayment outcomes are unobserved. Restructured facilities differ from untreated facilities in both recorded and unrecorded ways. A model can forecast these observed patterns accurately while leaving the effect of a proposed action unresolved.

Randomised pilots can identify incremental effects when two or more lawful and professionally credible options exist. Quasi-experimental methods can support inference when assignment follows a genuine threshold, phased rollout, external rule or other mechanism with defensible assumptions. Both approaches begin with prospective definitions of treatment, comparison, eligibility, outcome and

time zero. They also require operational adherence, customer safeguards, contamination analysis and enough follow-up to observe durable rather than momentary performance.

Evidence from microfinance illustrates why outcome breadth matters. Banerjee, Duflo, Glennerster and Kinnan's randomised evaluation of a group-lending expansion in India found increased take-up and investment in existing businesses, while effects differed across consumption, business and longer-term outcomes [8]. Its setting and product differ from Kenyan MSME bank credit, so its value here is methodological. A credit intervention can change several outcomes by different amounts and over different horizons. A Kenyan pilot should therefore measure repayment, cure, re-default, net recovery, business continuity, access, cost, complaints and distributional effects rather than searching for one universal success statistic.

The practical route begins within a participating institution's control environment. Public CBK and KBA data establish definitions, portfolio context and reproducible aggregate baselines. Synthetic data can test software, power calculations and disclosure controls. Institution-level analysis then occurs through a bank-controlled research environment in which authorised data remain governed by the bank. Retrospective point-in-time reconstruction precedes prospective shadow scoring. Customer-affecting trials begin only after product, model, conduct, privacy and operational gates are satisfied.

The remaining parts build this programme in sequence:

- **Part 2a** defines event time, data lineage, explicit lifecycle variables and residual AI representations.
- **Part 2b** develops hierarchical Bayesian risk, timing and dependence models, then separates prognostic prediction from causal identification.
- **Part 3** maps model evidence into fair interventions through the Credit Policy and Compliance Gate.
- **Part 4** specifies the bank-controlled research, integration, validation and audit architecture.
- **Part 5** reconciles lifecycle cash outcomes with KESONIA-linked pricing, expected credit loss, capital, liquidity, ALM, FTP and risk transfer.
- **Part 6** defines randomised pilots, quasi-experimental alternatives, ecosystem governance, scale-up and publication.

The evaluation object will be a vector rather than a single collection rate. For horizon (h), a programme may report:

$$\mathcal{O}_h = \left(P(\text{current}_h), P(\text{durable cure}_h), P(\text{redefault}_h), \text{NPV}(\text{net cash}), C_{\text{workout}}, H_{\text{customer}}, B_{\text{continuity}}\right).$$

Here, C_{workout} represents attributable servicing and recovery cost, H_{customer} represents defined conduct or customer-harm outcomes, and $B_{\text{continuity}}$ represents a pre-specified measure of continued business operation where lawful data are available. The vector does not assign an automatic common monetary value to every component. It makes the trade-offs visible so that the statistical analysis, credit committee and conduct review can apply their respective decision standards. Each pilot will identify one primary estimand and a limited family of secondary outcomes before assignment begins.

Together, these parts will test a central research question:

> Which product structures, evidence sources and proportionate interventions improve sustainable collection for defined Kenyan MSME populations, at acceptable risk, cost and customer impact?

That question can support a bank pilot, an industry research programme and a policy discussion because it does not require all participants to share one model or pool raw customer data. It requires common outcome definitions, reproducible evidence, explicit responsibilities and a willingness to measure what happens after disbursement.

Conclusion

The wholesaler who misses Tuesday's instalment may have a failing business, a temporary settlement gap, an unsuitable repayment calendar, a competing-liquidity problem or an account error. A lender learns which explanation applies by connecting product purpose, event-time evidence, contractual state and intervention history. It then measures whether the chosen response produces durable cash performance and a fair customer outcome.

Underwrite for collection turns that reasoning into an institutional architecture in practice. It aligns the business clock with the credit clock, follows matched vintages through their lifecycle, recognises deterioration while constructive options remain available, and returns realised outcomes to product and policy design. The next part begins by constructing the point-in-time evidence required to make that lifecycle observable.

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Part 2a

Temporal Evidence, Behavioural Signals and Point-in-Time Feature Engineering

Underwrite for Collection Series

Abstract

Part 1 followed a Kenyan MSME from an operating cash-flow mismatch into arrears, intervention, cure or loss. Part 2a asks what the lender could legitimately have known at each point in that journey. A customer receipt has an economic event time, a settlement time, a ledger-posting time and an analytical availability time. A repayment arrangement has an agreement date and an effective date. A macroeconomic observation has a reference period, publication date and possible later revision. If these clocks are collapsed, a model trained on the latest database state can learn from information that was unavailable when the historical decision was made.

This paper develops a bank-controlled temporal evidence architecture for MSME behavioural credit scoring. It separates source events from derived credit states, establishes a point-in-time feature contract, and combines a versioned historical plane with low-latency online state. Interpretable lifecycle variables retain explicit economic definitions and ownership. Artificial intelligence, where it adds validated value, learns complementary sequence representations from authorised event histories. Those representations are generated out of fold and residualised against the explicit design matrix before entering the hierarchical model developed in Part 2b.

The contribution is an evidence discipline rather than a promise that greater data volume produces better lending. Every candidate signal must have a permitted purpose, provenance, unit, window, availability time, correction rule, missingness treatment and validation path. This allows the same history to support underwriting, monitoring, intervention evaluation and collection analysis without confusing prediction with causation or present knowledge with hindsight.

1. A payment is not one timestamp

Return to the wholesaler from Part 1. A restaurant customer initiates a KSh180,000 payment on Monday afternoon. The wholesaler sees a confirmation and expects to use the money to meet Tuesday's loan instalment. The payment rail settles the transfer on Tuesday morning. The business account reflects cleared funds at 09:06. The lending ledger allocates the receipt at 09:13. A credit-monitoring job runs at 09:10, while the allocation is still pending. At 11:30, operations corrects an erroneous transaction reference that had prevented automatic matching.

What was the wholesaler's liquidity at 09:10? Economically, the customer had paid and the funds had cleared. Operationally, the loan system had not yet recognised the receipt. A model reading the corrected end-of-day table would see a timely payment. A model using the state actually available at 09:10 would see cleared cash in one authorised source and an unpaid instalment in another. Neither view is complete without the timestamps and reconciliation state that explain the difference.

This is the basic temporal problem of lifecycle credit. Credit administration and monitoring depend on information that remains current after origination [1]. For MSMEs, that information may include account transactions, sales settlements, invoices, inventory movements, scheduled obligations, facility utilisation, repayment allocation, servicing contacts and verified external context. The Central Bank of Kenya's 2024 MSME credit survey also identifies the importance of financing that fits enterprise cash-flow dynamics and of information capital that can broaden credit assessment [2]. Applied evidence

from other markets suggests that sales, expenses, orders, invoices and platform activity can complement formal credit histories for defined micro and small enterprise populations [3]. These findings motivate investigation. They do not make every transaction accurate, lawful, stable or predictive for a Kenyan bank portfolio.

The architecture therefore begins with clocks, not model layers. Let (i) identify an entity or facility, (e) an event, and (d) a decision. A minimum temporal envelope records:

$$\mathcal{T}_{ied} = \left(T_{\text{event}}, T_{\text{effective}}, T_{\text{commit}}, T_{\text{ingest}}, T_{\text{publish}}, T_{\text{decision}}, T_{\text{correct}}, T_{\text{mature}}, T_{\text{cash}}\right).$$

These clocks describe different facts:

- **Economic event time** records when the underlying activity occurred, such as a sale, payment initiation, account debit or customer receipt.
- **Effective time** records when a contractual, policy, product or reference state begins to apply.
- **Source commit or posting time** records when the source system accepted the transaction into its own ledger or operational database.
- **Ingestion time** records when the controlled analytical platform received the source event.
- **Publication time** records when an external observation became available to users, which may be later than its economic reference period.
- **Decision time** is the governed cutoff for a particular approval, review, intervention or monitoring calculation.
- **Correction time** records when a reversal, restatement, remapping or source correction became knowable.
- **Outcome-maturity time** records when enough follow-up exists to classify an outcome such as 30-day default, durable cure or re-default.
- **Cash-realisation time** records when collection or recovery value was actually received and cleared, which may differ from the accounting recognition date.

Not every source carries all nine clocks, and a bank need not manufacture false precision. It must, however, preserve the clocks needed to answer the decision question. A repayment event may need event, settlement, posting, ingestion and correction times. A sector index may need reference period, publication time and revision vintage. A model label needs an observation window and maturity time. A recovery needs receipt, clearance, allocation, cost and reporting dates if the analysis is to distinguish nominal recovery from realised net cash.

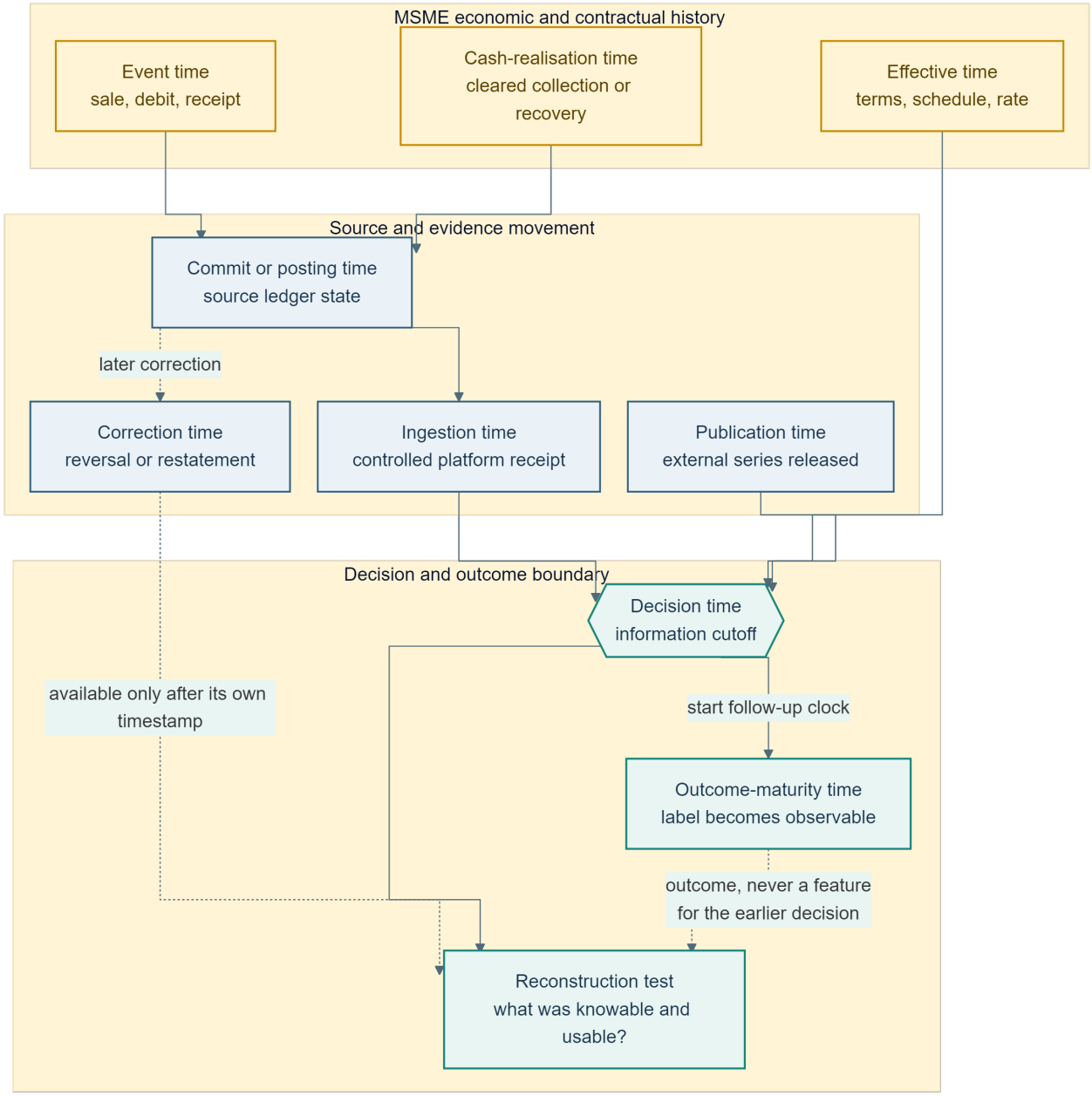


Figure 1: Nine clocks surrounding one MSME credit decision. The usable information set ends at the governed decision cutoff. Corrections and matured outcomes can improve later analysis, but they cannot be inserted retrospectively into the features attributed to the earlier decision.

The qualifying rule is stricter than selecting rows whose economic timestamp precedes the decision. A feature value $x_{ij}(T_d)$ is eligible only if its source event had become available through the authorised process, its effective interval covered the relevant observation, its feature definition was approved for that use, and any maximum-age rule remained satisfied:

$$\mathbb{1}_{ij,d}^{\text{eligible}} = \mathbb{1}\left\{T_{\text{event}} \le T_d, T_{\text{available}} \le T_d, T_{\text{effective_from}} \le T_d < T_{\text{effective_to}}, A_{ij} = 1, P_{ij,d} = 1\right\}.$$

Here, A_{ij} indicates that the value remains within its approved age or time-to-live, and $P_{ij,d}$ indicates that the source and feature are permitted for the decision purpose. For a corrected record, the reconstruction uses the version known by T_d, even when a later version is more accurate. The current corrected record remains valuable for present operations and outcome analysis, while the historical version preserves an honest account of the earlier decision.

This distinction turns data lineage into economic evidence. Without it, model performance can be inflated by backfilled payments, revised macroeconomic series, post-decision collection notes, later bureau pulls or labels that have not matured consistently. With it, the bank can ask a more useful question: given the information lawfully available at that moment, what did the system infer, which action was permitted, and what happened afterwards?

2. The MSME event and state ledger

A useful credit history cannot be built from one wide customer table whose columns are repeatedly overwritten. That table may be convenient for today's operations, but it conceals the path by which the present balance, arrears status or limit was reached. Lifecycle analysis needs a governed event history and a separate state reconstruction. The event records what a source asserted at a time. The state is a calculation based on the events and rules available at a specified cutoff.

This distinction matters because an enterprise, a person and a facility are not interchangeable entities. One proprietor may operate several businesses. One enterprise may have several authorised representatives, accounts and facilities. A single facility may finance several invoices or drawdowns. A payment may cover principal, interest, fees and arrears across more than one obligation. A collateral item or guarantee can secure several exposures under defined priority. A contact may concern the business relationship rather than one account. The entity model should therefore preserve relationships among at least:

- the customer or authorised representative;
- the enterprise and its governed size, sector and legal-form classifications;
- the application, decision and override record;
- the facility, limit, drawdown and contractual schedule;
- the bank account, mobile-money account or approved settlement source;
- the invoice, order, sale or platform activity where access is authorised;
- the payment, allocation, reversal and reconciliation record;
- the collateral, guarantee or risk-transfer instrument;
- the servicing case, contact, promise, modification and hardship review;

- the delinquency, cure, re-default, write-off, recovery and closure outcome.

Stable pseudonymous keys can connect these entities inside an approved analytical environment without distributing direct identifiers throughout every model table. The re-identification map remains separately controlled. Pseudonymisation reduces unnecessary exposure of identity; it does not make a record non-personal when the data can still be attributed using separately held information. Kenya's Data Protection Act requires lawful and transparent processing, explicit and legitimate purposes, data minimisation, accuracy, retention limits and safeguards by design [4]. The General Regulations add practical expectations for automated individual decision-making, including meaningful information about the logic involved, error prevention, appropriate statistical procedures, measures to reduce bias, and routes for human intervention [5]. Part 3 develops the decision and customer-rights implications. Part 2a makes those obligations executable in the data design.

Every accepted source event should therefore arrive in a canonical envelope such as:

$$e_k = (\text{event_id}, \text{entity_keys}, \text{event_type}, \text{value}, \text{unit}, \mathcal{T}_k, \text{source}, \text{schema_version}, \text{purpose}, \text{quality}, \text{correction_of}) .$$

The `event_id` supports deduplication. Entity keys bind the record to the permitted analytical unit. Event type and unit prevent a KSh balance from being confused with a KSh flow or a percentage. The temporal envelope preserves the applicable clocks. Source and schema version establish provenance. Purpose records the approved use rather than relying on later inference. Quality contains validation status, reconciliation state and uncertainty. `correction_of` links a reversal or replacement to the event it changes without erasing what the platform originally received.

The received-evidence zone is append-oriented, but the word *immutable* should be used with care. Source systems can correct transactions, courts can alter enforceability, customers can dispute data, and retention rules can require restriction or deletion. The architecture preserves auditable versions and controlled corrections. It does not turn a mistaken upstream assertion into truth by writing it to durable storage.

Derived states are produced by versioned rules. If $E_i^{\leq d}$ is the eligible event set for facility i at decision cutoff d, and g_v is state rule version v, then:

$$S_{i,d}^{(v)} = g_v\left(E_i^{\leq d}\right).$$

The rule may reconstruct contractual balance, amount due, payment allocation, days past due, available limit and lifecycle state. Re-running the rule after a correction produces a new state version. Re-running a later rule definition on the same historical events produces a research comparison, not a silent replacement of the state used by the original decision.

Table 1: Proposed source, access, latency and authority matrix

Source domain	Candidate evidence	Access and purpose condition	Natural latency	Principal quality and correction risks	Authoritative record
Core lending and product subledger	Facility, schedule, draw, accrual, payment allocation, arrears, modification, write-off	Internal role-based access for servicing, risk, finance and approved research	Transactional to daily close	Misallocation, backdating, product-rule change, reversals	Product subledger for contractual position
Bank account and payment rails	Credits, debits, settlement, return, channel and reconciliation	Customer relationship, contract and other applicable lawful basis; purpose-specific fields	Seconds to settlement-cycle	Pending versus cleared cash, duplicated messages, failed or reversed transactions	Account or payment ledger for booked movement
Customer-supplied enterprise records	Invoices, orders, expenses, inventory and accounting extracts	Informed, documented and revocable access where applicable; relevance to stated assessment	Event, daily or periodic	Selective coverage, manual edits, invoice cancellation, inconsistent accounting	Source enterprise system for its own record, subject to verification
Approved platform or merchant partner	Sales, fulfilment, settlement and operating activity	Written data-sharing role, purpose, retention, security and customer transparency	Event to daily	Platform dependence, identifier changes, outages, commercial-policy shifts	Partner system for originating activity
Credit reference bureau	Shared facilities, enquiries and repayment history	Applicable statutory and contractual process; dispute and correction path	Pull or scheduled update	Reporting lag, identity matching, unresolved disputes	Bureau record as received, with source institution context
Collections and servicing case system	Contact, promise, fulfilled promise, hardship, complaint, restructure and recovery steps	Need-to-know servicing use with conduct controls	Event to daily	Free-text inconsistency, strategy-driven selection, unrecorded contact, outcome leakage	Case system for recorded action; ledger for realised cash
Collateral and guarantee records	Security, valuation, perfection, cover, claim and proceeds	Contractual, legal and risk-management purpose	Event to periodic	Stale valuation, disputed priority, unperfected interest, claim rejection	Relevant legal register, custodian, guarantee or claims record
Public Kenyan and multilateral data	Rates, inflation, exchange rates, sector and wider economic indicators	Public-source terms and documented analytical relevance	Daily, monthly, quarterly or release-driven	Revision, publication lag, series break, changed methodology	Publishing authority and retained release vintage

Source domain	Candidate evidence	Access and purpose condition	Natural latency	Principal quality and correction risks	Authoritative record
Derived feature and model stores	Approved features, masks, representations and version identifiers	Approved model purpose only	Streaming, scheduled or decision-triggered	Training-serving skew, stale state, duplicate ownership, unapproved backfill	Reconstructible analytical artefact, not the underlying legal ledger

Authority in this table is deliberately distributed. The model feature store is authoritative for the value it served under a specific version, but it is not authoritative for the contractual loan balance. A partner platform may be authoritative for the order status it recorded, but the bank must still decide whether the record is relevant, permitted and sufficiently reliable for credit use. A servicing note records that an agent made a promise arrangement, while cleared account cash determines whether the promise was fulfilled.

The ledger also prevents behavioural scoring from becoming character scoring. A late payment is an account event whose meaning depends on schedule, cash timing, allocation and available support. A fall in sales may reflect seasonality, temporary closure, lost demand, a data outage or migration to another payment rail. A broken promise may indicate hardship, disagreement, operational failure or unwillingness. The feature architecture can measure these patterns. It should not convert them into moral attributes.

The output of this section is therefore not a maximal customer profile. It is a minimum sufficient evidence graph for a defined credit purpose. Fields without a defensible economic role, lawful basis, quality process or decision owner remain outside the feature contract. That boundary supports both model discipline and practical data minimisation.

3. From operational systems to a bank-controlled evidence plane

The ledger design becomes useful when it can absorb events without making the analytical platform the master of every business fact. The bank retains its core, product, payment, customer, servicing, finance and risk systems as authoritative for their assigned records. Approved partners remain responsible for the records they originate. The evidence plane carries governed copies, source metadata and transformations into two coordinated views: a low-latency state used at a decision moment, and a versioned historical view used for reconstruction, research and validation.

No single ingestion mechanism fits every source. Committed changes from supported relational product or servicing databases may be captured from database logs. Debezium's SQL Server connector, for example, takes an initial snapshot and then emits committed row-level insert, update and delete events. Its fault-tolerance documentation also notes that an outage can lead to duplicate events after restart because offsets are committed periodically [6]. That makes unique source identifiers, log positions, idempotent writes and reconciliation essential. A payment switch or mobile-money integration may already publish native messages. An invoice platform may expose an API. A guarantee

administrator may provide a controlled file. Public statistics arrive through websites, APIs or scheduled publications. Treating all of these as CDC would obscure their different failure and correction modes.

After validation at the boundary, events can be published to domain topics such as `facility-events`, `payment-events`, `servicing-events`, `enterprise-activity-events` and `external-context-events`. Apache Kafka provides a practical event-streaming backbone. Records with the same key can be assigned to the same topic partition, and ordering is guaranteed within a topic partition rather than across the entire topic [7]. The architecture should therefore choose stable keys at the level whose order matters. A facility key is appropriate for contractual state; an account key is appropriate for booked cash; an enterprise key may be needed for consolidated authorised activity. Cross-key relationships are reconciled by explicit event and allocation identifiers rather than an assumed global sequence.

A stateful processor such as Apache Flink then validates schemas, deduplicates events, applies event-time windows, maintains eligible state and routes exceptions. Event time describes when an event occurred, while processing time describes when the processing operation ran. Flink uses watermarks to represent progress through event time in the presence of out-of-order records [8]. A watermark is therefore a policy about tolerated lateness, not proof that no earlier event will arrive. The bank should set it by source and use case. A five-minute payment watermark and a one-day invoice watermark may both be reasonable, while a monthly macro series needs a publication schedule rather than a second-by-second stream.

Temporal reference joins require equal discipline. Flink supports event-time temporal joins against versioned tables, but their semantics depend on time attributes, keys, connectors and late-record behaviour [9]. A sector classification, repayment schedule or product rule joined to an event should be the version effective and available at that event or decision cutoff. Late records outside an approved watermark are not silently discarded from the evidence history. They move to a correction or reconciliation path, after which current state may change and the original decision snapshot remains preserved.

The historical plane stores received source events, cleaned canonical events and approved analytical outputs at different levels. Delta Lake provides ACID table transactions, schema enforcement, batch and streaming support, and access to historical table versions [10]. Its change data feed can expose inserts, updates, delete images, commit versions and commit timestamps after the feed has been enabled, subject to retention constraints [11]. These capabilities support reproducible processing, but they do not recover evidence that was never retained. The bank must align transaction-log retention, change-feed retention, model-validation horizons, dispute requirements and legal retention policy before relying on time travel for a long historical study.

The proposed layers have precise roles:

- **Received evidence** retains source payload, source position, schema, timestamps, access basis and validation status.
- **Canonical events** standardise identifiers, units, time zones, event types and correction links without losing source lineage.

- **Reconciled states** apply approved contractual and business rules to produce balances, schedules, arrears and lifecycle states.
- **Feature history** stores versioned explicit variables, masks and approved AI representations exactly as generated.
- **Research and reporting views** apply population, maturity, disclosure and purpose controls to defined analyses.

The labels Bronze, Silver and Gold may be used operationally, but colour does not establish evidential authority. A Gold feature remains derived. The product subledger remains authoritative for the contractual balance; the bank account ledger remains authoritative for booked cash; the retained release remains authoritative for what an external publisher reported at that time.

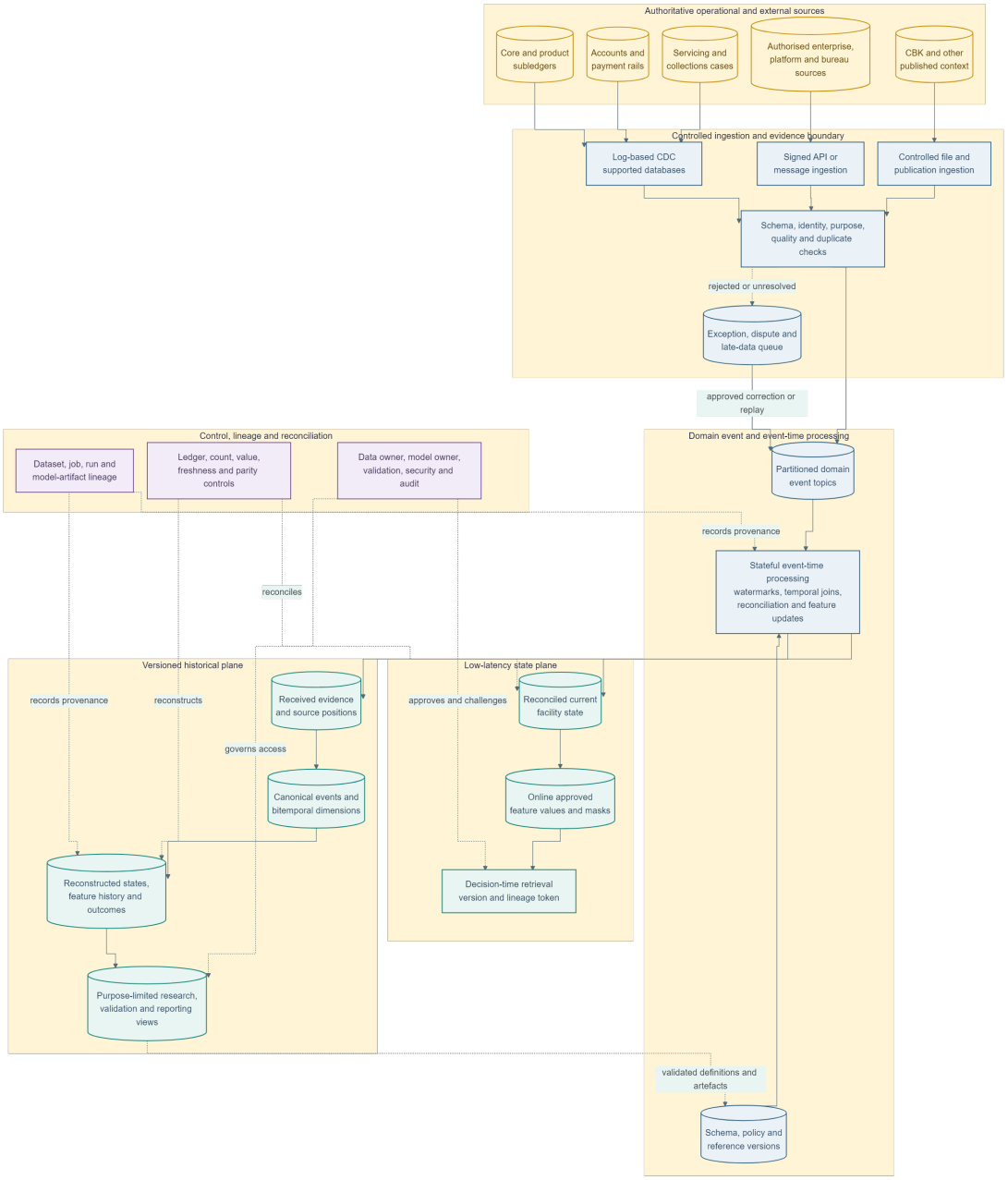


Figure 2: Proposed bank-controlled temporal evidence plane. Authoritative systems keep their assigned responsibilities. The streaming plane supports current state, while the historical plane preserves reconstructible evidence. The control plane links both through lineage, reconciliation and approved versions.

Lineage must connect more than tables. OpenLineage models jobs, their runs, and input and output datasets, with extensible facets for such matters as schema, versions and data quality [12]. A lender can extend that vocabulary to record the source positions, transformation code, configuration, feature definition, model artefact, decision purpose and approval package associated with a scoring event. This permits an independent reviewer to trace a served variable back to the canonical events and source versions from which it was calculated.

The online plane stores only what the decision service needs at low latency: the latest eligible feature values, their as-of times, masks, definition version and lineage token. A feature-store implementation such as Feast separates historical retrieval from online serving. Its point-in-time join scans backward from each entity timestamp within a feature view's time-to-live, while its online store is designed to serve recent feature values rather than preserve the full history [13]. The design lesson is broader than one product: historical reconstruction cannot be delegated to a latest-value cache, and low-latency serving should not depend on scanning the full analytical history.

Online and offline calculations must implement the same approved economic definitions. Equivalence is demonstrated by replaying sampled decision events through both paths and comparing feature values, timestamps, missingness masks and versions within stated tolerances. A shared feature name is insufficient. Differences in window closure, late-event handling, rounding, time zone, reference-table version or zero-denominator rules can create training-serving skew even when both systems appear correct in isolation.

This architecture also creates a practical research boundary. A participating bank can keep customer-level source data inside its controlled environment, expose purpose-limited analytical views to approved code, and release only disclosure-checked results. Synthetic data can test pipelines and experimental logic before any protected data are used. Part 4 develops the secure research and integration pattern. Part 2a remains concerned with the prerequisite: the historical and online evidence must describe the same decision world.

4. Reconstruct the state the decision could actually see

Historical model development often begins with a table of accounts and a request to "join the latest information before each observation date." That instruction is incomplete. The latest effective value may not have been published, received or approved when the decision occurred. A transaction posted before the cutoff may later be reversed. A customer classification may be corrected retrospectively. A servicing note may describe an event that occurred earlier but was recorded after an intervention. The reconstruction must therefore represent both the world being described and the lender's knowledge of that world.

A bitemporal record gives each version two intervals. Its **valid interval** records when the value applies to the underlying economic, contractual or operational state. Its **knowledge interval** records when that version was available within the controlled system:

$$b_k = \left(i_k, x_k, [T_k^{\text{valid_from}}, T_k^{\text{valid_to}}), [T_k^{\text{known_from}}, T_k^{\text{known_to}}), v_k\right).$$

For a decision at T_d concerning an observation at $T_o \leq T_d$, the eligible version is the one whose valid interval contains T_o and whose knowledge interval contains T_d. If several records still qualify, the deterministic source sequence and correction order resolve the version. This is different from loading today's most accurate value and assigning it to yesterday.

Consider a monthly sector indicator for July. July is its reference period. It is published on 20 August, ingested on 21 August, and revised on 15 October. A limit review on 10 August cannot use the value at all. A review on 1 September may use the original release. A backtest of that September review must not use the October revision, although a present-day economic analysis may prefer the revised series. Both versions have legitimate uses, so both need labels that reveal their vintage.

The same logic applies to internal data. Suppose a KSh40,000 payment clears before a 10:00 review but is allocated to the facility at 10:07. If the approved feature definition uses cleared account cash and the account event was available by 09:59, the liquidity feature may include it while the contractual arrears state remains unchanged. If the bank's approved process recognises only allocated loan payments, neither repayment velocity nor arrears cure should include it at 10:00. The correct answer comes from the feature contract, not from whichever database updated first.

A vendor-neutral point-in-time retrieval can be expressed as follows:

```
SELECT
    d.facility_key,
    d.decision_time,
    f.feature_name,
    f.feature_value,
    f.unit,
    f.event_time,
    f.known_from,
    f.definition_version
FROM decision_events AS d
LEFT JOIN feature_history AS f
  ON f.facility_key = d.facility_key
 AND f.event_time <= d.decision_time
 AND f.known_from <= d.decision_time
 AND d.decision_time < COALESCE(f.known_to, TIMESTAMP '9999-12-31 00:00:00')
 AND f.definition_version = d.approved_feature_set_version
QUALIFY ROW_NUMBER() OVER (
    PARTITION BY d.decision_id, f.feature_name
    ORDER BY f.event_time DESC, f.source_sequence DESC
) = 1;
```

The syntax must be adapted to the selected engine. Its control logic is the important part. Each training row begins with a decision event and retrieves feature versions known by that decision time. The query records the source event time, knowledge time and definition version returned. A time-to-live rule

is added where an old observation should expire. A purpose and permission filter is applied before retrieval, not after the training set has been exported.

Table versions and temporal feature joins solve related but different problems. A query such as `VERSION AS OF` or `TIMESTAMP AS OF` selects a historical table snapshot. Inside that snapshot, the training builder must still select the appropriate event for each entity and decision time. Conversely, a feature-store point-in-time join can select the last eligible event timestamp but cannot reproduce a correction history that the offline source did not retain. A defensible process tests both the snapshot and the row-level join.

Leakage controls should address at least the following cases:

Leakage path	How it enters	Required control
Later payment or allocation	Current account table overwrites the pre-decision balance	Event history plus decision-time reconstruction
Revised macroeconomic value	Latest published series replaces the original release	Release-vintage table with publication and ingestion time
Post-decision servicing note	Narrative records contain later knowledge of hardship or default	Note creation time, event assertion time and restricted text window
Unmatured negative label	Recent accounts with insufficient follow-up are treated as performing	Outcome-maturity rule and censoring flag
Policy-version leakage	Today's eligibility or arrears rule is applied to an earlier production decision	Retain production rule version; run restatement as a separate analysis
Entity leakage across folds	Facilities from one enterprise appear in both training and validation	Split by enterprise or connected group before feature fitting
Intervention leakage	Action or response after scoring becomes a predictor of the preceding score	Define time zero and censor features at assignment or decision
Correction leakage	A later dispute resolution is inserted into the original record	Bitemporal knowledge interval and versioned correction link

Labels require the same discipline as features. If $Y_{i,d}^{(h)}$ denotes default within horizon h after decision d, the label is observable only after the earlier of the full horizon or a qualifying terminal event under the approved definition. An account with 20 days of follow-up cannot be assigned a clean 90-day non-default label. It is immature or censored. Cure, durable cure, re-default and recovery each need a time origin, window and cash convention. Part 2b will model selected outcomes, and Part 6 will define intervention estimands, but their credibility begins with this maturity record.

Reconstruction tests should use deliberately difficult examples rather than random rows alone. The suite should include a late payment, a reversed receipt, a schedule modification, a backdated correction, a changed customer-enterprise link, a daylight-saving or time-zone boundary where relevant, an external release followed by revision, a feature at its exact expiry time, and two events sharing a timestamp. For each example, reviewers should be able to state the expected value before executing the query.

The resulting training row is a documented information set, not merely a vector. It identifies the entity, decision, feature values, masks, source vintages, definition versions and outcome-maturity status. That record allows the bank to distinguish a model error from a stale source, a late event, a policy change or a reconstruction defect. It also gives the next section a reliable base on which to define explicit lifecycle features.

5. Explicit lifecycle features with economic meaning

Behavioural scoring becomes credible when each variable describes an observable part of enterprise cash generation, contractual obligation or payment response. The model should not begin with thousands of convenient aggregations and search for a story afterwards. It should begin with an economic question, define the event population and clock, and then test whether the resulting measure is stable and incrementally useful.

Let $C_{i,d}^{+}$ be eligible cleared inflows for enterprise i on day d, $C_{i,d}^{-}$ eligible outflows, and $N_{i,d} = C_{i,d}^{+} - C_{i,d}^{-}$ net observed cash flow. Eligibility excludes internal transfers and recognised reversals and applies documented treatment to financing inflows, taxes, payroll, owner drawings and protected funds. Coverage across payment rails is recorded, since an observed account may represent only part of the business.

Over a trailing window (w), cash-flow velocities are:

$$V_{i,t}^{+,w} = \frac{\sum_{d=t-w+1}^{t} C_{i,d}^{+}}{n_{i,t}^{w}}, \qquad V_{i,t}^{-,w} = \frac{\sum_{d=t-w+1}^{t} C_{i,d}^{-}}{n_{i,t}^{w}},$$

where $n_{i,t}^{w}$ is the number of covered calendar or business days specified by the feature. The unit is KSh per day. A seven-day decline relative to a preceding 90-day baseline can define **inflow velocity**:

$$v_{i,t}^{\text{in}} = \frac{V_{i,t}^{+,7}}{V_{i,t-7}^{+,90} + \varepsilon}.$$

A value below one states that recent observed inflows are below the earlier daily average. It does not state why, whether the enterprise is insolvent, or whether the decline will persist.

Cash-flow asymmetry describes the shape of daily net cash around its median. One transparent candidate is:

$$\text{CFA}_{i,t}^{w} = \frac{Q_{0.50}(N_i^{w}) - Q_{0.10}(N_i^{w})}{Q_{0.90}(N_i^{w}) - Q_{0.50}(N_i^{w}) + \varepsilon}.$$

It is dimensionless. Values above one indicate that the lower-tail span is larger than the upper-tail span in the observed window. Minimum coverage, treatment of zero-activity days and clipping are fixed before estimation. Seasonal restocking can produce legitimate downside concentration, so the feature's relationship with later repayment must be learned by product, sector and time.

Liquidity measures must match the horizon of the obligations. Let $D_i^{\text{due}}(t, t+h)$ be contractual principal, interest and governed charges due over horizon h. Let $L_i^{\text{available}}(t)$ contain cleared unrestricted cash plus eligible confirmed settlements after approved haircuts, less protected or unavailable balances. The **dynamic debt-to-liquidity ratio** is:

$$\text{DLR}_{i,t}^{h} = \frac{D_i^{\text{due}}(t, t+h)}{L_i^{\text{available}}(t) + \varepsilon}.$$

Unconfirmed invoices do not become cash merely because they have been issued. Where validated invoice or order evidence is used, its amount, obligor, dispute state, expected settlement and haircut remain visible. A complementary **buffer-days** measure divides available liquidity by a robust estimate of essential daily outflow. Both measures report capacity at a stated horizon rather than mixing total debt with one day's receipts.

Repayment features use realised allocations:

$$v_{i,t}^{\text{repay},h} = \frac{P_i^{\text{eligible paid}}(t-h, t)}{P_i^{\text{scheduled}}(t-h, t) + \varepsilon}.$$

The registry states whether the numerator includes principal only or all scheduled debt service, how prepayment and modification are treated, and how reversals change the value. A period with no scheduled payment is not a zero repayment. It is not applicable. Revolving products also require utilisation $U_{i,t} = B_{i,t}^{\text{drawn}} / L_{i,t}^{\text{approved}}$, headroom, days near limit, drawdown velocity and repayment after draw. A rising utilisation path has a different meaning from stable working-capital use at the same ending ratio.

Lifecycle features then describe transitions rather than only levels: days since full payment, episodes of early arrears, movement between arrears bands, time since modification, cure duration, re-default history and cleared recovery after default. Servicing features can include contact attempts, successful contacts, promises, fulfilled promises, hardship reviews and completed arrangements. They require particular care because policy selects who receives an action. A promise-kept ratio can be prognostic after a promise exists, but it cannot be used as though every account had an equal chance to make one.

Table 2: Minimum governed feature contract

Field	Required content	Example
Name and economic question	Stable label and the mechanism being tested	`repayment_velocity_30d`: has recent realised payment kept pace with scheduled debt service?
Entity and source	Facility, enterprise or account key; originating systems	Facility allocations plus contractual schedule
Formula and unit	Numerator, denominator, exclusions and scale	Eligible KSh paid divided by KSh due; dimensionless
Observation window	Start, end, calendar, timezone and closure rule	Trailing 30 calendar days ending strictly before decision time

Field	Required content	Example
Availability and expiry	Timestamp used, lag and time-to-live	Latest reconciled allocation known by cutoff; expires after approved age
Missingness	Zero, not applicable, unavailable, insufficient history and disputed	No scheduled amount produces an N/A mask, not zero
Transformation	Winsorisation, scaling, knot support and out-of-range handling	Raw audit value retained; modelling copy clipped at trained bounds
Ownership and version	Named calculation owner and immutable definition identifier	Credit data product, version `2.1.0`
Permitted purpose	Decisions and analyses for which use is approved	Monitoring and shadow intervention model only
Validation	Reconciliation, stability, fairness and outcome tests	Monthly parity, coverage and sector drift tests

The Explicit Lifecycle Feature Path owns these named metrics. Raw transaction and repayment events may also enter an AI sequence branch, but CFA, DLR, inflow velocity, repayment velocity, utilisation and buffer days do not reappear there as separately engineered inputs. Pre-specified interactions can be added where their economic mechanism is clear, such as utilisation pressure with prior delinquency, or a sector downturn with dependence on one buyer. Strong heredity requires the associated main effects to remain in the design.

Part 2b will determine how continuous variables enter the HLR. The main specification uses governed P-splines so that nonlinear relationships can be learned without turning every candidate threshold into a policy rule. The present task is prior to inference: retain the raw audit value, define the supported range, specify units and windows, and ensure that a ratio does not conceal an unstable denominator. Once this contract is fixed, model flexibility can increase without sacrificing meaning.

6. What artificial intelligence adds after behavioural scoring is established

Artificial intelligence enters this architecture for a specific purpose: to test whether the order, duration and interaction of authorised events contain useful information beyond the explicit lifecycle variables. A GRU is a candidate recurrent encoder for shorter ordered histories [14]. A Transformer is a candidate attention-based encoder for longer or more heterogeneous sequences [15]. Their original research settings are not evidence that either will improve MSME credit decisions. Sequence length, sample size, latency, stability and out-of-time performance determine whether a simpler aggregation, GRU, Transformer or no neural branch is appropriate.

The AI input is a sequence of minimally transformed events, with event type, signed amount or scaled magnitude, time gap, channel, source-quality flag and masks. It does not receive separate engineered copies of CFA, DLR, repayment velocity or other explicit features. Each encoder produces a fixed-dimensional representation H_i, which is frozen before the HLR is fitted.

Representations are generated out of fold. For fold k, the encoder $f^{(-k)}$ is trained without the enterprises and outcomes assigned to that fold, then applied to its records:

$$H_i^{\text{oof}} = f^{(-k(i))}(X_i), \qquad i \in k.$$

Folds respect enterprise or connected-group identity and time. Randomly splitting rows from the same enterprise would allow repeated history to cross the boundary. Cross-fitting is adapted here as a sample-separation discipline [16]. The manuscript does not invoke the causal guarantees of double machine learning, which additionally require an appropriate orthogonal score and identification assumptions.

The out-of-fold representation is then projected on the explicit design matrix Q using training-fold weights and regularisation. The residual block is:

$$\widetilde{H}_i = H_i^{\text{oof}} - Q_i \widehat{B}^{(-k(i))}.$$

This reduces linear redundancy between learned and explicit evidence. Optional whitening can improve conditioning within $\widetilde{H}$, with the transformation fitted on training folds and frozen for validation and production. Residualisation does not establish statistical independence, eliminate nonlinear overlap, correct confounding or turn the representation into a causal variable. Part 2b applies regularised horseshoe shrinkage so unsupported residual dimensions can contribute little.

Approval depends on incremental value. The comparison begins with contractual, bureau and explicit lifecycle baselines, then adds the residual AI block. Evaluation includes out-of-time discrimination, calibration, expected-loss-weighted performance, stability, subgroup behaviour, missingness sensitivity and operational cost. If the AI branch does not improve the approved decision objective reliably, the explicit model remains the production architecture.

7. Macroeconomic context moves on another clock

An enterprise can maintain the same internal behaviour while its operating environment changes. Interest rates affect financing cost and demand. Inflation changes input prices and household purchasing power. Exchange-rate movements affect imported stock. Sector conditions, regulated fuel prices, rainfall and public-payment cycles may alter cash conversion for particular businesses. These variables enter only when the economic mechanism, geography, sector mapping and validation plan are stated.

Kenyan context should begin with Kenyan authorities. CBK's *Monthly Economic Indicators* provide a release-driven set of monetary, banking and economic tables [17]. KESONIA offers a particularly clear clock example: CBK publishes the benchmark each business day for the preceding business day [18]. The observation date and publication date are therefore different. World Bank indicator APIs can provide complementary international and long-horizon series with metadata [19], while IMF datasets are accessible through SDMX interfaces [20]. Their availability does not make dissimilar frequencies, definitions or revisions directly comparable.

Each retained macro observation has the form:

$$M_{j,\tau}^{(v)} = \left(m_{j,\tau}^{(v)},\ T_{\text{reference}},\ T_{\text{publish}},\ T_{\text{ingest}},\ v,\ q\right),$$

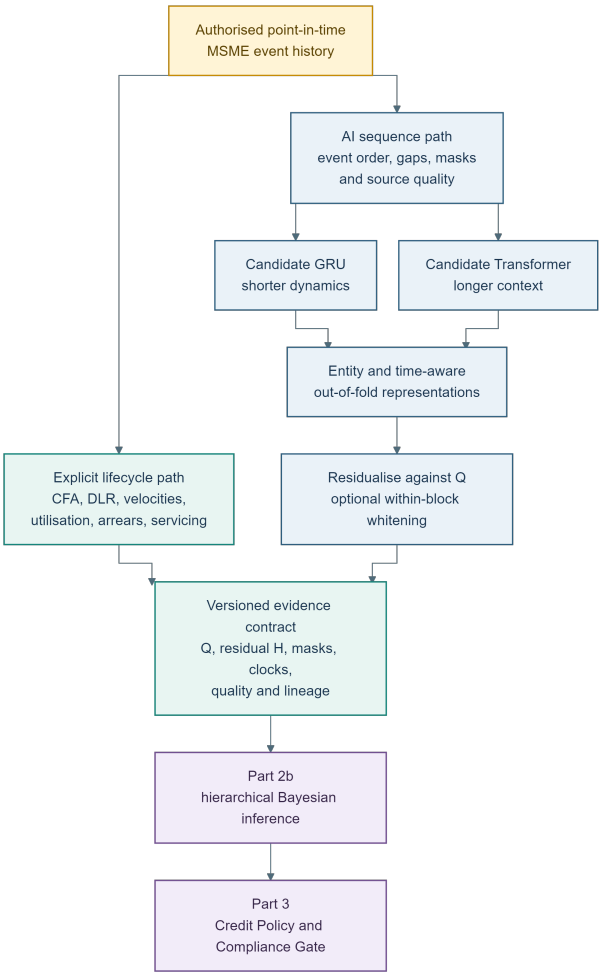


Figure 3: Explicit-feature and residual-AI evidence paths. The AI branch is complementary and contestable. Both branches use point-in-time events, but named engineered metrics have one owner and the residual representation reaches inference only through a versioned contract.

where (j) identifies the series, (au) its reference period, (v) the release vintage and (q) a quality or status flag. A decision uses the latest permitted vintage published and ingested by its cutoff. Later revisions support restated analysis, while the original vintage preserves the production information set.

Daily, monthly and quarterly context should not be duplicated across every transaction as though it were high-frequency borrower behaviour. It can be represented through lagged levels, changes, rolling states, release masks and sector interactions aligned to the decision clock. The model must also test whether a context feature adds information beyond cohort and calendar effects. Part 5 will use KESONIA for pricing and funding. Here it remains an example of how an economically important observation acquires a usable timestamp.

8. Missingness, quality and lineage are model inputs

A zero-value transaction, a day without observed activity and a missing record are different states. Missingness may mean that the enterprise did not trade, used another payment rail, withheld permission, joined recently, experienced a system outage, submitted an unreadable file, or had a delayed source posting. The feature contract records these reasons separately. No-activity evidence can enter a model only when coverage is sufficient to support that interpretation.

Fallback proceeds by approved levels. The service first uses a fresh reconciled value. It may then use a still-valid older value with an age mask, a reduced feature set, or a conservative manual review route. A stale, disputed or structurally incomplete source can suspend the affected feature or model branch. Unavailability alone does not justify assigning the borrower a worse behavioural value.

Quality controls operate at several layers:

- schema, type, unit, timezone and identifier tests at ingestion;
- duplicate, sequence, watermark and late-event tests in the stream;
- opening-to-closing balance and payment-allocation reconciliation;
- coverage, freshness, missingness-reason and correction-rate monitoring;
- online-to-offline parity using sampled decision events;
- distribution, population and outcome drift by product, sector, institution and time;
- feature-owner approval for definition changes;
- model-validation review of transformations, exclusions and subgroup effects.

Exclusions are reported by both account count and exposure value. A test that drops many low-value micro-enterprises can appear small by balance while materially changing inclusion. Conversely, a few missing large facilities can dominate value-based results.

Lineage binds each output to the source datasets, transformation job, run, code version, configuration, feature definition and model artefact used. A correction does not erase the earlier run. It creates a linked version and a defined response: no action, current-state repair, decision review, outcome restatement or model revalidation. This allows operational error, source error and model error to be investigated separately.

9. The serving contract passed to Part 2b

At decision time (d), Part 2a delivers one versioned evidence object:

$$\mathcal{F}_{i,d} = \left(Q_{i,d}, \widetilde{H}_{i,d}, M_{i,d}, \mathcal{T}_{i,d}, V_{i,d}, L_{i,d}\right),$$

where Q is the explicit design row, $\widetilde{H}$ the residual AI representation when approved, M the availability and quality masks, $\mathcal{T}$ the relevant clocks, V the feature and representation versions, and L the lineage token. The object also records the entity, product, decision purpose and operating mode.

Training mode reconstructs mature historical rows. Shadow mode scores current rows without changing customer treatment. Production mode serves only approved artefacts and permitted actions. Backfill mode repairs history under a separate run identity and never masquerades as the original production state.

Before release, sampled events must produce equivalent feature values across historical and online paths within approved tolerances. The test compares values, masks, timestamps and versions, not only final scores. A missing critical source, incompatible schema, expired feature set or unresolved identity conflict invokes the documented fallback rather than an improvised value.

Part 2b consumes this object and estimates posterior risk and uncertainty. It may apply governed P-spline bases, hierarchical pooling and shrinkage, but it does not redefine source events or repair their clocks. Part 3 then applies law, contract, conduct, risk appetite and operational constraints through the Credit Policy and Compliance Gate. Keeping these interfaces separate allows an evidence defect, inference result and policy decision to be challenged on their own terms.

Conclusion

The wholesaler's Monday receipt becomes useful credit evidence only when the lender can distinguish economic occurrence, settlement, posting, availability, correction and decision time. That discipline turns fragmented transactions into a reproducible lifecycle history without confusing data volume with knowledge. Explicit features preserve financial meaning; residual AI representations may add tested sequence information; lineage preserves the path between them. Part 2b now receives an evidence object whose contents, clocks, ownership and limitations are already defined, allowing hierarchical inference to concentrate on uncertainty and pooling rather than repair hidden temporal defects or leakage.

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Part 2b

Hierarchical Bayesian Underwriting, Causal Identification and Lifecycle Dependence

Underwrite for Collection Series

Abstract

Part 2a ended at an evidence boundary. It produced explicit lifecycle variables, residual artificial-intelligence representations, contractual context, data-quality indicators and the exact information cutoff for an MSME credit decision. Part 2b develops the statistical architecture that receives those objects. Its champion is a hierarchical Bayesian logistic regression that estimates a defined-horizon outcome under observed policy, pools information across related but heterogeneous portfolios, retains nonlinear economic effects through governed P-splines, and controls the residual AI block through cross-fitting and regularised-horseshoe shrinkage.

The model is embedded in a wider research design. A prognostic probability describes what is likely under a stated information set and policy. A causal estimand asks what would change under a permissible alternative product or intervention. A policy rule selects an action within legal, conduct, affordability, operational and portfolio constraints. Randomised pilots provide the primary causal route where they are justified; regression discontinuity, difference-in-differences, instrumental variables, synthetic control and target-trial emulation supply narrower alternatives when their identification conditions are credible.

Fixed-horizon default is only one collection outcome. Hierarchical piecewise-exponential, competing-risk and multi-state challengers add default timing, modification, cure, durable cure, re-default, prepayment and recovery. Marginal models for exposure at default, loss severity and recovery timing then convert probability into collectible cash. Shared factors are represented before any residual copula so that economic shocks are not counted twice. Offline Pólya-Gamma or reference NUTS inference produces a signed posterior artefact for low-latency scoring. The result is a disciplined handoff from temporal evidence to intervention governance, expected loss, capital, pricing and funding.

1. When a probability becomes a collection question

Return to the wholesaler whose customer payment cleared on Tuesday morning but reached the lending ledger after the scheduled monitoring run. By the next week, the operational mismatch has been repaired. The facility is current, the large customer continues buying, and the business remains profitable. A statistical system looking back from month end can now see the corrected allocation. The system that advised the lender at 09:10 on Tuesday could see only the information permitted at that earlier cutoff.

Suppose that system produced a 7.4 percent probability of entering 30-day arrears within the next 60 days. The number sounds precise, yet four separate questions immediately follow. What event counts as entry into 30-day arrears? Is the forecast conditional on the existing repayment calendar and servicing policy? Does the posterior remain wide because the enterprise belongs to a thinly observed sector or because two current signals conflict? Would changing the payment date reduce the risk, or would it merely move the same shortage into another week? Which action remains suitable after affordability, customer communication, operational capacity and portfolio limits are considered?

These questions establish the statistical centre of underwriting for collection. A forecast is useful when its outcome, horizon, population, information cutoff and policy context are explicit. Its uncertainty

must travel with it. An observed association becomes evidence for intervention only through a design that supports the relevant counterfactual comparison. The final action remains a governed decision, not a synonym for either prediction or causation.

Hierarchical Bayesian modelling is well suited to this architecture because it represents the uncertainty around portfolio, group and borrower-level quantities in one probability system and allows related groups to share information [1]. Hierarchical credit-risk research also demonstrates why outcome maturity and censoring require more care than labelling every account that has not yet defaulted as performing [2]. The value is not a general claim that Bayesian machinery always predicts better. It is the ability to state what has been pooled, what remains uncertain, how prior information enters, and how a posterior predictive distribution changes as evidence matures.

Part 2b therefore separates three objects:

1. **Prognostic distribution.** The champion estimates a distribution for a defined future outcome under the observed or specified operating policy. It may estimate fixed-horizon default, arrears entry, or another approved event.
2. **Causal contrast.** A trial or credible quasi-experiment estimates how an eligible product, servicing or collection intervention changes a stated outcome for a stated population.
3. **Decision policy.** The Credit Policy and Compliance Gate selects from feasible actions using model outputs, causal evidence, law, contract, affordability, conduct, capacity and portfolio constraints.

The architecture also separates a champion from challengers. The hierarchical logistic champion supplies a calibrated fixed-horizon probability. A timing challenger uses censored exposure to estimate when an event may occur. A multi-state challenger estimates transitions through arrears, modification, cure and recovery. Marginal exposure and severity models estimate the cash amount at risk and the value that may be recovered. A dependence layer asks how losses co-move after observed shared drivers have been modelled. Each component earns a role by answering a different question.

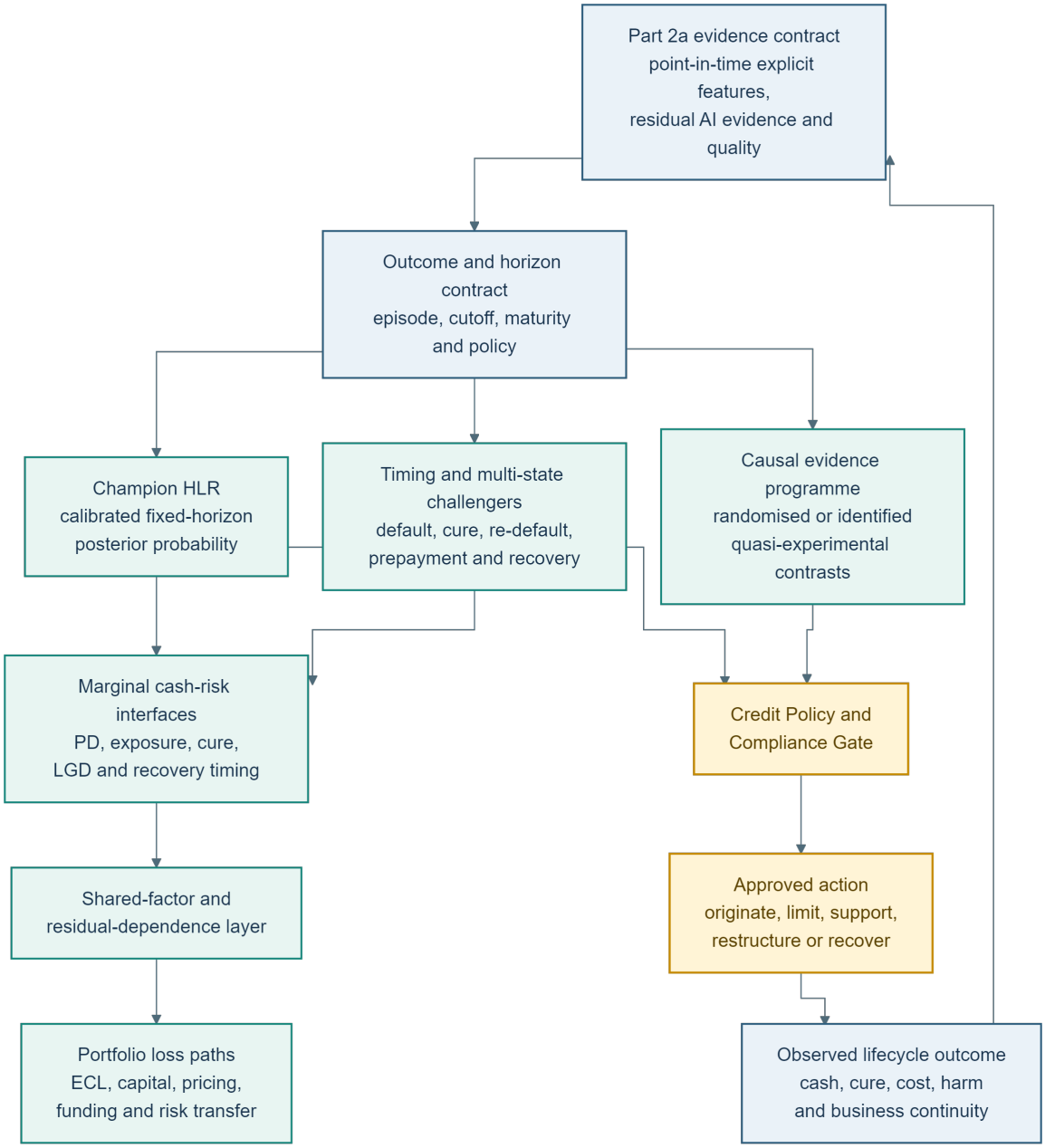


Figure 1: Statistical architecture for underwriting toward collectible outcomes. The champion, timing and state challengers, causal evidence programme, dependence model and policy gate perform distinct jobs. Realised cash and customer outcomes return through the point-in-time evidence process rather than entering earlier decisions retrospectively.

The figure is also a safeguard against modelling escalation. Artificial intelligence enters through the residual evidence contract defined in Part 2a when it improves future-period calibration or decision value beyond the explicit model. A copula is built on calibrated marginal probabilities. A causal forest follows an identified intervention design. A posterior distribution informs a separate decision rule. The chain remains useful because each interface states what the next component can legitimately consume.

The human purpose remains visible. A lender wants to distinguish an enterprise whose current signal reflects a repairable timing mismatch from one whose cash-generating capacity has deteriorated, and to do so early enough for a proportionate response. The borrower benefits when uncertainty is treated as uncertainty rather than silently converted into an adverse label. The collections team benefits when the model forecasts specific states and cash consequences instead of producing a rank with no operational meaning. Finance and risk teams benefit when the same episode definitions can later reconcile default, exposure, cure and recovery with expected loss and capital.

2. Observation units, horizons and selective outcomes

The model does not score an abstract enterprise detached from a facility. Its principal record is a borrower-product-episode-decision tuple:

$$\mathcal{O}_i = (b_i, p_i, e_i, t_{0i}, H_i, \pi_i, \mathcal{I}_i),$$

where b_i identifies the enterprise or connected borrower unit, p_i the product, e_i the legally and operationally defined credit episode, t_{0i} the decision cutoff, H_i the forecast horizon, π_i the policy under which future outcomes will be observed, and $\mathcal{I}_i$ the information set available at the cutoff. Product and episode definitions matter. An invoice-finance draw against a verified debtor, an overdraft review, an asset-finance instalment and a working-capital term loan do not share the same exposure path merely because the borrower is the same.

For product p and horizon H_p, the champion outcome is:

$$Y_i^{(H_p)} = \mathbf{1}\{\tau_i^D - t_{0i} \le H_p\},$$

where τ_i^D is the first time the episode satisfies the approved default definition. The definition must specify the event, materiality threshold, cure treatment, product scope and relationship to modification or forbearance. It must also state whether the intended prediction is first default, any default, entry into a defined arrears state, or another outcome. A probability without that contract cannot be compared across models or passed reliably into accounting and portfolio analysis.

The label is mature only when follow-up reaches the end of the horizon or the event occurs earlier. If the training cutoff is T, an episode with no observed default contributes a known negative champion label only when:

$$t_{0i} + H_p \le T.$$

An open episode whose horizon extends beyond T is right-censored. It is available to a compatible timing model with its observed at-risk interval; it is not converted into a successful repayment. This distinction is especially important when products have different maturities or when a recently originated vintage has had less time to deteriorate. Credit research has long noted that censoring can make future defaulters appear as non-defaulters when classification datasets are constructed too early [2].

One enterprise may contribute repeated decisions, facilities and matured episodes. Those records are statistically dependent through common management, cash flows, ownership, sector and prior lender actions. Randomly distributing them across training and validation folds would let the model learn from another version of the same enterprise. The default validation unit is therefore the connected borrower or economic group. Time-forward validation then holds out later decisions as well as later outcomes. Institution-held-out validation is added when the intended claim concerns transport to another lender.

The binary champion does not collapse the lifecycle into one permanent label. A separate outcome vector records:

$$\mathbf{Y}_i^{\text{life}} = \left(T_i^A, T_i^M, T_i^C, I_i^{DC}, T_i^{RD}, T_i^D, R_i(\cdot), C_i^{\text{work}}, H_i^{\text{cust}}\right),$$

where T^A is time to arrears, T^M time to modification, T^C time to cure, I^{DC} indicates durable cure over a pre-specified window, T^{RD} time to re-default, T^D time to default, $R(\cdot)$ the recovery cash-flow path, C^{work} workout and contact cost, and H^{cust} defined customer-harm or complaint outcomes. These quantities feed different models and decisions. They share the episode spine, which makes reconciliation possible.

Historical approval policy creates another missing-outcome problem. Repayment performance is observed principally for applicants whom the lender approved and who accepted and drew the product. The earlier decision therefore influences which labels exist. This is the selective-label problem: observed outcomes are not a random sample of all applicants, because the decision process determines who becomes observable [3]. Declined applications should not be relabelled as defaults or non-defaults. Likewise, bureau performance on another lender's product may add permitted evidence but does not recreate the exact contract, exposure and intervention path that the declined facility would have produced.

Let $S_i = 1$ denote that the target outcome becomes observable under the historical process. Model development learns directly from:

$$P(Y_i = 1 \mid X_i, S_i = 1, \pi_{\text{hist}}),$$

while a new policy may seek a quantity for a wider population. Transport from the first to the second requires support and assumptions that the approved-only sample cannot manufacture. The programme therefore combines four disciplined routes: retain the champion within its validated population; use shadow scoring to observe score distributions without changing approvals; use external or later outcomes where their definitions are compatible and lawful; and create prospective evidence through carefully bounded randomised or quasi-experimental designs where genuine uncertainty and customer protection permit them.

Selection also continues after disbursement. A limit reduction changes future exposure. A restructuring changes payment opportunities. Intensive contact may change both repayment and complaints. A borrower who prepays leaves the default risk set. A sold or refinanced facility may become unobservable to one institution. Every action is therefore recorded with its assignment mechanism, timing and eligibility. Part 2b models outcomes conditional on that history; Part 6 will design the prospective programme that can learn which actions caused better outcomes.

The observation contract now gives the champion a clear task. It receives only information available at t_0, forecasts a matured product-specific event over H_p under policy π, pools evidence across registered hierarchies, and reports uncertainty. It does not need to answer the timing, intervention and portfolio-dependence questions alone. Those questions receive their own likelihoods and validation tests in the sections that follow.

3. The champion hierarchical Bayesian logistic model

The champion begins with a probability statement rather than a score transformation. For matured episode i, let:

$$Y_i \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \eta_i.$$

The linear predictor is block structured:

$$\begin{aligned}\eta_i = \alpha_{p[i]} &+ \mathbf{q}_i^\top \beta_z + \widetilde{\mathbf{h}}_i^\top \beta_h + \mathbf{r}_i^\top \gamma \\ &+ u_{s[i]} + v_{g[i]} + w_{j[i]} + c_{k[i]} + \delta_{t[i]} + \mathbf{m}_i^\top \xi.\end{aligned}$$

Each block has one declared job:

- $\alpha_{p[i]}$ is the intercept for product p, stated at the registered reference levels and centred feature values.
- $\mathbf{q}_i = Q(\mathbf{z}_i)$ contains centred linear terms and P-spline bases for explicit lifecycle features.
- $\widetilde{\mathbf{h}}_i$ is the cross-fitted residual AI representation delivered under the Part 2a contract.
- $\mathbf{r}_i$ is a small, pre-specified interaction design.
- u_s, v_g, w_j, c_k are partially pooled sector, geography, institution and origination-cohort effects.
- δ_t is an approved calendar-time or regime effect.
- $\mathbf{m}_i$ contains contractual and data-quality metadata that should affect interpretation but are neither behavioural features nor neural evidence.

The explicit block gives the model its economic spine. Candidate quantities include net settled cash generation over registered windows, repayment velocity, cash-flow asymmetry, debt-to-liquidity measures, account volatility, utilisation, days since liquidity depletion, arrears transitions, customer-receipt concentration, operating-obligation coverage and time since the most recent verified business activity. Every transformation retains its unit, window, permitted population and missingness rule from Part 2a. A ratio whose numerator covers seven days and denominator covers ninety days does not silently

change windows during model development. A facility-balance variable does not become an enterprise cash-flow variable because both are denominated in shillings.

The hierarchy reflects repeated structures in the lending problem rather than a search for arbitrary clusters. Sector pools enterprises exposed to related demand and input conditions. Geography may capture lawful and sufficiently aggregated local conditions after proxy and fairness review. Institution effects support research across participating lenders without pretending that their products, populations and definitions are identical. Cohort effects represent origination vintages that share underwriting policy and macroeconomic conditions. Product intercepts keep a short working-capital facility from inheriting the baseline rate of a longer asset-finance contract.

For a generic sector effect u_s, a non-centred representation is:

$$u_s = \sigma_u z_s, \qquad z_s \sim \mathcal{N}(0,1), \qquad \sigma_u \sim \text{HalfStudent-}t(\nu_u, 0, a_u).$$

Equivalent structures apply to geography, institution and cohort effects. The group effects are centred, for example through a sum-to-zero constraint after transformation, so that the intercept retains its meaning. Hyperprior scales are calibrated through prior-predictive simulation on the log-odds and probability scales. A Half-Student-t prior with finite degrees of freedom can retain substantial mass for genuine heterogeneity while controlling the extreme tail more tightly than a Half-Cauchy. A Half-Normal may be a useful sensitivity specification where large group deviations are implausible. The choice is documented and tested because scale-prior behaviour can materially affect weakly identified hierarchical models.

Partial pooling is easiest to understand through the posterior precision. In a Gaussian approximation for an identified group effect, the posterior location has the form:

$$\hat{u}_s \approx \frac{I_s}{I_s + \sigma_u^{-2}} \hat{u}_s^{\text{local}} + \frac{\sigma_u^{-2}}{I_s + \sigma_u^{-2}}\, 0,$$

where I_s represents the group's likelihood information. A group with little matured evidence remains close to the collective mean and carries a wider posterior. A well-observed group can move away when its data support differentiation. In the actual logistic hierarchy the update is joint rather than this scalar approximation, but the economic interpretation remains: differentiation is earned through information rather than assigned because a segment name sounds distinctive.

Random slopes are reserved for relationships with a prior economic reason and adequate cross-group support. If short-term debt-to-liquidity pressure is expected to operate differently across products, the model may use:

$$\beta_{z,p} = \bar{\beta}_z + L_\Omega \,\text{diag}(\sigma_\beta)\mathbf{z}_p, \qquad \mathbf{z}_p \sim \mathcal{N}(\mathbf{0}, \mathbf{I}),$$

where L_Ω is the Cholesky factor of a correlation matrix. Marginal scales and correlations receive separate priors. This separates magnitude from correlation, while the non-centred form improves compu-

tation when group information is weak. Random slopes over the residual AI embedding advance only through a dedicated challenger with adequate group evidence and dimensional control.

Interactions also follow economic ownership. A pre-specified term may represent high revolving utilisation combined with declining settled receipts, or concentration in one customer combined with a longer settlement delay. Strong heredity requires the associated main effects to remain in the model when an interaction is active. Interaction coefficients receive their own shrinkage scale. This prevents the neural block, spline basis and a broad interaction search from competing to express the same local curvature.

The model produces a posterior distribution, not only a fitted coefficient vector:

$$p_i^{(m)} = \text{logit}^{-1}(\eta_i^{(m)}), \qquad m = 1, \dots, M.$$

For a known group, the draw includes its posterior effect. For a genuinely new group, prediction integrates over the registered population distribution or a governed higher-level mapping rather than assigning the most favourable observed group. The output contract can therefore report posterior mean and median PD, credible intervals, tail probabilities above policy thresholds, and uncertainty decomposed by data, group and parameter sources where the approximation supports it.

Calibration is estimated on future or cross-fitted predictions, separately by product when evidence permits:

$$\text{logit}\left(p_{i,\text{cal}}^{(m)}\right) = a_p + b_p \,\text{logit}\left(p_i^{(m)}\right), \qquad b_p > 0.$$

The calibration intercept a_p corrects systematic level error and slope b_p corrects over- or under-dispersion. They are stored with their estimation window and uncertainty. Calibration is fitted on future or cross-fitted outcomes and reported alongside ranking, subgroup stability and time-split integrity.

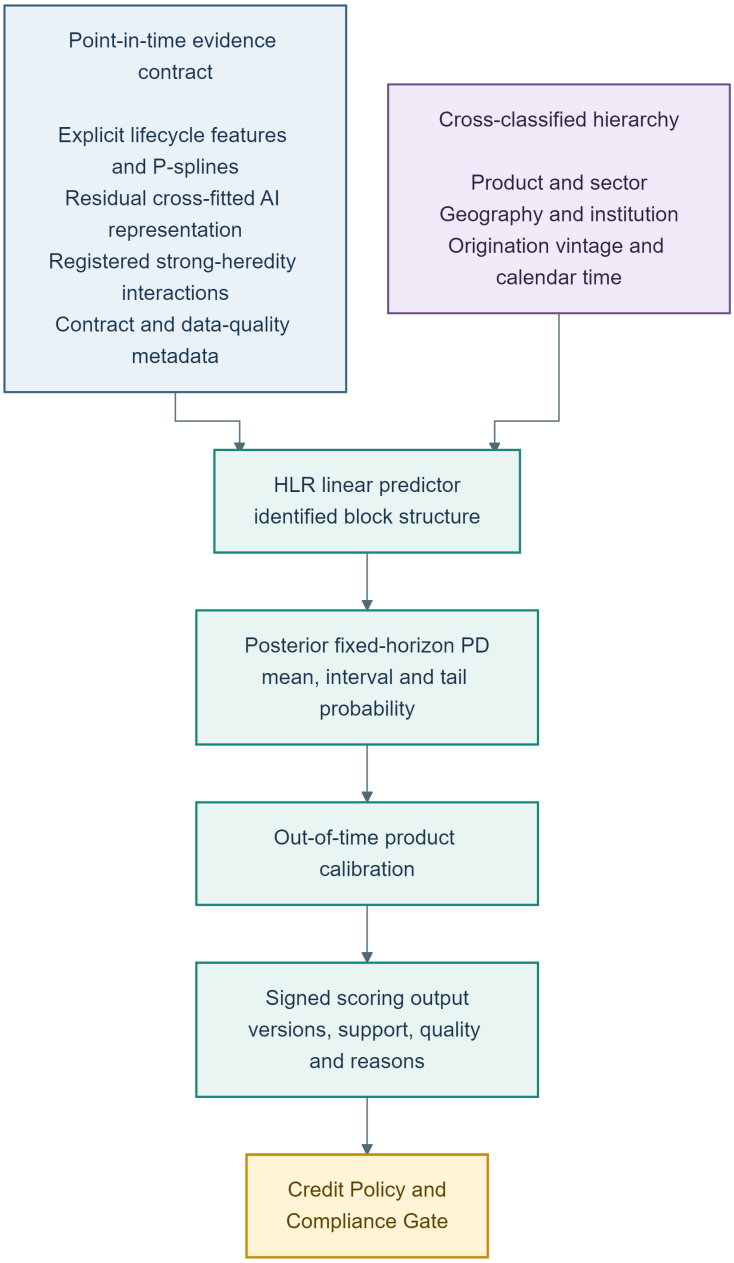


Figure 2: Champion HLR with assigned evidence, hierarchy and calibration interfaces. Explicit lifecycle variables, residual AI evidence, interactions, metadata and group effects enter through separate blocks. The posterior remains an inferential output until calibration, artefact controls and the policy gate are applied.

This champion can be benchmarked against a conventional scorecard, penalised logistic regression and tree-based model using the same point-in-time sample. The comparison is empirical. The champion earns acceptance through calibrated future-period probabilities, stable subgroup behaviour, decision value, reconstructibility and operational fit. Hierarchical structure contributes a principled way to share information; it does not excuse weak labels, incomplete outcomes or a model whose cash implications cannot be reconciled.

4. Governed nonlinearities, hierarchy and time

Many collection relationships are nonlinear. A modest fall in settled receipts may be absorbed by cash reserves, while a similar fall near the repayment boundary may sharply increase arrears risk. Revolving utilisation may carry limited information at low levels and become more consequential when headroom disappears. Repayment velocity can distinguish a business returning toward schedule from one moving away, yet the effect may flatten once the account is comfortably current. A single linear coefficient imposes one slope across all these regimes.

The champion uses a P-spline where nonlinearity is supported. P-splines combine a B-spline basis with an explicit discrete penalty on adjacent coefficients [4]. For continuous feature x, a cubic basis gives:

$$f(x) = \sum_{k=1}^{K} B_{k,3}(x)\zeta_k = \mathbf{B}(x)^{\top}\zeta.$$

A second-order difference penalty controls curvature:

$$\Delta^2\zeta_k = \zeta_k - 2\zeta_{k-1} + \zeta_{k-2}, \qquad \Delta^2\zeta \mid \tau_f \sim \mathcal{N}(\mathbf{o}, \tau_f^2\mathbf{I}).$$

Equivalently, if D_2 is the second-difference matrix, the prior contribution is proportional to:

$$\exp\left[-\frac{1}{2\tau_f^2}\zeta^{\top}D_2^{\top}D_2\zeta\right].$$

The B-spline basis controls representation and the penalty controls roughness. A reasonably rich, fixed basis therefore need not select one fragile set of a few knots. The smoothing scale determines how much curvature the data may support. The penalty has a null space, so the implementation explicitly separates or constrains the unpenalised constant and linear components. The fitted smooth is centred over the development population:

$$\sum_{i\in\mathcal{D}_{\text{train}}} f(x_i) = 0,$$

which prevents the smooth from redefining the product intercept. If a QR, eigen or Demmler-Reinsch transformation improves numerical conditioning, the penalty matrix and back-transformation move

with the basis. Orthogonalising a basis while leaving its prior penalty behind would silently create a different model.

Every accepted smooth has a versioned specification: feature definition, unit, transformations, spline degree, interior and boundary knots, validated support, penalty order, smoothing-scale prior, centring rule, missingness path, clipping policy and reason-code representation. Boundary handling is especially important. A smooth curve inside observed support is not evidence that extrapolation beyond the largest witnessed debt-to-liquidity value is benign. The service flags out-of-support values and applies a governed conservative or referral rule rather than extending an attractive tail mechanically.

Product variation is added through pooled deviations:

$$f_p(x) = f_0(x) + g_p(x), \qquad D_2\zeta_{g,p} \sim \mathcal{N}(\mathbf{0}, \tau_g^2\mathbf{I}).$$

Sparse products remain near the common shape, while sufficiently observed products may differ. Monotonicity is imposed only when the economic definition, contractual mechanism and empirical evidence justify it. For example, increasing obligations relative to verified liquidity may reasonably be expected to increase risk over a defined range, but data errors, numerator changes and product restructuring can complicate an absolute global constraint.

Calendar and vintage variation require a similarly explicit choice. An exchangeable time effect is suitable when periods share a distribution without an ordered dependence. A first-order random walk represents local drift:

$$\delta_t \mid \delta_{t-1}, \sigma_\delta \sim \mathcal{N}(\delta_{t-1}, \sigma_\delta^2).$$

An AR(1) effect represents movement around a long-run level:

$$\delta_t - \mu_\delta = \rho(\delta_{t-1} - \mu_\delta) + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma_\delta^2), \quad |\rho| < 1.$$

Under a stationary initial distribution, the marginal variance is $\sigma_\delta^2/(1-\rho^2)$. A RW1 has no mean-reverting level; an AR(1) does. Neither is selected because a narrative labels the economy as drifting or mean reverting. Prior-predictive trajectories, available time depth, held-out calibration and posterior identifiability determine whether an ordered effect adds value beyond observable macroeconomic variables and cohort intercepts. Appendix A preserves the related B-spline plus RW1 and B-spline plus AR(1) coefficient-prior alternatives for direct comparison.

Binning remains useful for diagnostic plots and operational summaries. It can reveal sparse tails, non-monotonic empirical rates and support gaps when uncertainty and sample counts are shown. It is not the default likelihood transformation. Bernoulli observations may be collapsed exactly to a Binomial count only when the design row, hierarchy assignments, exposure definition and event probability are identical:

$$Y_g = \sum_{i \in g} Y_i \sim \text{Binomial}(n_g, p_g).$$

Grouping continuous features into approximate cells changes the model because members no longer share an identical p_g. Any such quantisation is benchmarked against individual records and labelled an approximation. The champion retains the continuous P-spline representation at scoring time.

The nonlinear and time structures now remain rich but bounded. Each has an economic owner, an identification constraint, a prior-predictive interpretation, a validated support range and a future-period test. That discipline allows curvature and drift to improve collection forecasts without turning flexibility into an untraceable search across every possible shape.

5. Residual AI evidence and regularised horseshoe shrinkage

Part 2a assigned named financial quantities to the explicit path and reserved the AI branch for residual sequence information. This distinction matters because a neural encoder can reconstruct familiar ratios from the same underlying account events even when the ratios are not supplied as named columns. If both representations enter the HLR without control, a declining cash pattern may appear in earnings velocity, a spline over debt-to-liquidity, several transaction-window features and multiple embedding dimensions. Coefficients then compete to explain one economic movement, which weakens stability and reason attribution.

The AI representation is therefore built and tested as an incremental evidence block. Let $\mathbf{h}_i \in \mathbb{R}^d$ be a low-dimensional bottleneck produced from the authorised sequence history. For validation fold $k(i)$, the encoder, bottleneck selection and residualiser are trained without the fold's borrowers and without future periods. The residual representation is:

$$\widetilde{\mathbf{h}}_i = \mathbf{h}_i - \widehat{\mathbb{E}}_{-k(i)} \left[\mathbf{h}_i \mid \mathbf{q}_i, \mathbf{m}_i \right].$$

The projection design includes the explicit P-spline basis and metadata that the AI block could reproduce. A linear ridge projection is often easier to audit; a nonlinear residualiser may be benchmarked when the linear projection leaves material predictable structure. In either case, the mapping is fitted only on training data and replayed unchanged on the held-out fold. Borrower grouping and time-forward splitting remain active. The method borrows the discipline of cross-fitting, which separates nuisance-model fitting from held-out evaluation [5], while its role here is representational control within a prognostic architecture.

Residualisation is tested rather than assumed. The development report measures canonical correlations and out-of-fold predictability between $\widetilde{\mathbf{h}}$ and the explicit design, examines condition numbers and posterior correlations, and checks whether explicit effect shapes change materially when the AI block is added. A small remaining correlation can be acceptable because finite samples and changing populations prevent perfect geometric orthogonality. The objective is stable incremental information, not a ceremonial zero in one development matrix.

Optional whitening addresses a different problem. If S_h is the training-fold covariance of residual embeddings, a regularised transform may define:

$$\mathbf{h}_i^{\star} = (S_h + \epsilon I)^{-1/2}\,\widetilde{\mathbf{h}}_i.$$

Whitening reduces scale and correlation within the residual AI block. It does not project the block away from explicit variables and does not remove confounding. The transform, regularisation ϵ, retained dimensions and sign convention are stored in the artefact. Because embedding axes may rotate when an encoder is retrained, monitoring includes representation-alignment and ablation tests rather than interpreting one latent coordinate as a permanent business concept.

A regularised horseshoe prior controls the residual coefficient vector. The original horseshoe combines a global shrinkage scale with local scales, concentrating weak signals near zero while retaining heavy tails for stronger effects [6]. The regularised form adds a finite slab and a principled way to encode expected sparsity, which is useful for logistic models under separation or weak identification [7]. One parameterisation is:

$$\begin{aligned}
\beta_{h,j} \mid \tau, \tilde{\lambda}_j &\sim \mathcal{N}(0, \tau^2 \tilde{\lambda}_j^2), \\
\tilde{\lambda}_j^2 &= \frac{c^2 \lambda_j^2}{c^2 + \tau^2 \lambda_j^2}, \\
\lambda_j &\sim \mathrm{C}^{+}(0, 1),
\end{aligned}$$

where τ controls global sparsity, λ_j allows local escape from shrinkage and c limits the variance of the largest coefficients through a finite slab. The prior expectation for the effective number of non-negligible dimensions informs τ; it is set with reference to sample size, event prevalence and bottleneck dimension. The slab scale is examined on the log-odds and probability scales so that a weakly informative value corresponds to plausible risk movements.

Continuous shrinkage rarely makes a coefficient exactly zero. Importance is therefore assessed through posterior contribution, ablation and stability, not through a claim that the prior has performed automatic discrete selection. The explicit coefficients, residual AI coefficients, interactions and group deviations receive separate prior families or scales. This prevents a wide AI block from determining how strongly an economically necessary contractual variable is regularised.

Four nested models form the minimum comparison:

1. Explicit linear and contractual baseline.
2. Explicit baseline plus governed P-splines and hierarchy.
3. Hierarchical model plus unadjusted AI bottleneck, used to diagnose duplication.
4. Hierarchical model plus cross-fitted residual AI and regularised horseshoe.

The residual model advances only if it improves out-of-time log score or Brier score, calibration, tail outcomes or defined decision value without unacceptable subgroup instability. If its incremental contribution decays, the explicit model remains a fully operational champion or fallback. This creates an

architecture in which artificial intelligence can add temporal evidence while the lender retains a coherent model when that contribution is unavailable or no longer stable.

6. Pólya-Gamma offline inference and the production artefact

The HLR posterior is computationally demanding because the logistic likelihood is not Gaussian in its coefficient vector. Pólya-Gamma augmentation supplies a useful offline inference strategy for the complete hierarchy [8]. It introduces one latent variable per Bernoulli or Binomial likelihood contribution and converts the coefficient block into a Gaussian conditional while preserving the logistic model and its partial pooling.

For a likelihood contribution with linear predictor $\psi_i = \mathbf{x}_i^\top \theta$, outcome $y_i \in \{0, 1\}$, and $\kappa_i = y_i - \frac{1}{2}$, the Pólya-Gamma identity gives:

$$\frac{\exp(y_i \psi_i)}{1 + \exp(\psi_i)} = \frac{1}{2} \exp(\kappa_i \psi_i) \int_0^\infty \exp\left(-\frac{\omega_i \psi_i^2}{2}\right) p(\omega_i)\, d\omega_i,$$

with:

$$\omega_i \mid \psi_i \sim \mathrm{PG}(1, \psi_i).$$

Conditional on ω, the likelihood kernel is Gaussian in θ. Stack the design rows in X, define $\Omega = \mathrm{diag}(\omega_1, \ldots, \omega_n)$, and use a Gaussian prior $\theta \sim \mathcal{N}(\mathbf{m}_0, V_0)$. The coefficient conditional is:

$$V_n = \left(X^\top \Omega X + V_0^{-1}\right)^{-1}, \qquad \mathbf{m}_n = V_n \left(X^\top \kappa + V_0^{-1} \mathbf{m}_0\right),$$

$$\theta \mid \omega, \cdots \sim \mathcal{N}(\mathbf{m}_n, V_n).$$

In the full HLR, θ can contain product intercepts, explicit linear terms, P-spline coefficients, residual AI coefficients, registered interactions and Gaussian hierarchical or time effects. Sparse P-spline penalties and Gaussian Markov structures can therefore enter the precision matrix directly. A singular difference penalty is combined with proper priors or constraints for its null-space components so that the joint update remains identified.

The model is conditionally conjugate rather than fully closed form. Pólya-Gamma variables must be updated. Hierarchy scales, spline smoothing scales, regularised-horseshoe global and local scales, covariance correlations, AR persistence and calibration parameters retain their own conditional, slice, Metropolis, Hamiltonian or other validated updates. A blocked sampler may update strongly coupled coefficients jointly while using sparse Cholesky factorisation, conjugate-gradient methods or other tested linear algebra. Prior choices for group scales follow the sensitivity discipline developed for hierarchical variance parameters [9].

NUTS remains the reference comparator. On a representative development sample, the team compares posterior means, intervals, tail probabilities, group shrinkage, calibration and predictive quantities from Pólya-Gamma blocked Gibbs with a well-diagnosed NUTS fit. It also records wall-clock time, memory, effective sample size per second, numerical precision and sensitivity under sparse defaults or separation. Agreement is assessed on decisions and posterior functionals, not only on individual coefficients whose parameterisations may differ.

The production scoring path does not run either sampler. It consumes an approved posterior artefact. Three cadences remain distinct:

- **Event-time evidence refresh** updates the point-in-time state when authorised source events arrive.
- **Online score refresh** evaluates the accepted feature and hierarchy mappings against a signed posterior summary or representative draws.
- **Offline model refresh** re-estimates parameters after a scheduled review or a governed trigger, completes validation, and publishes a new artefact only after approval.

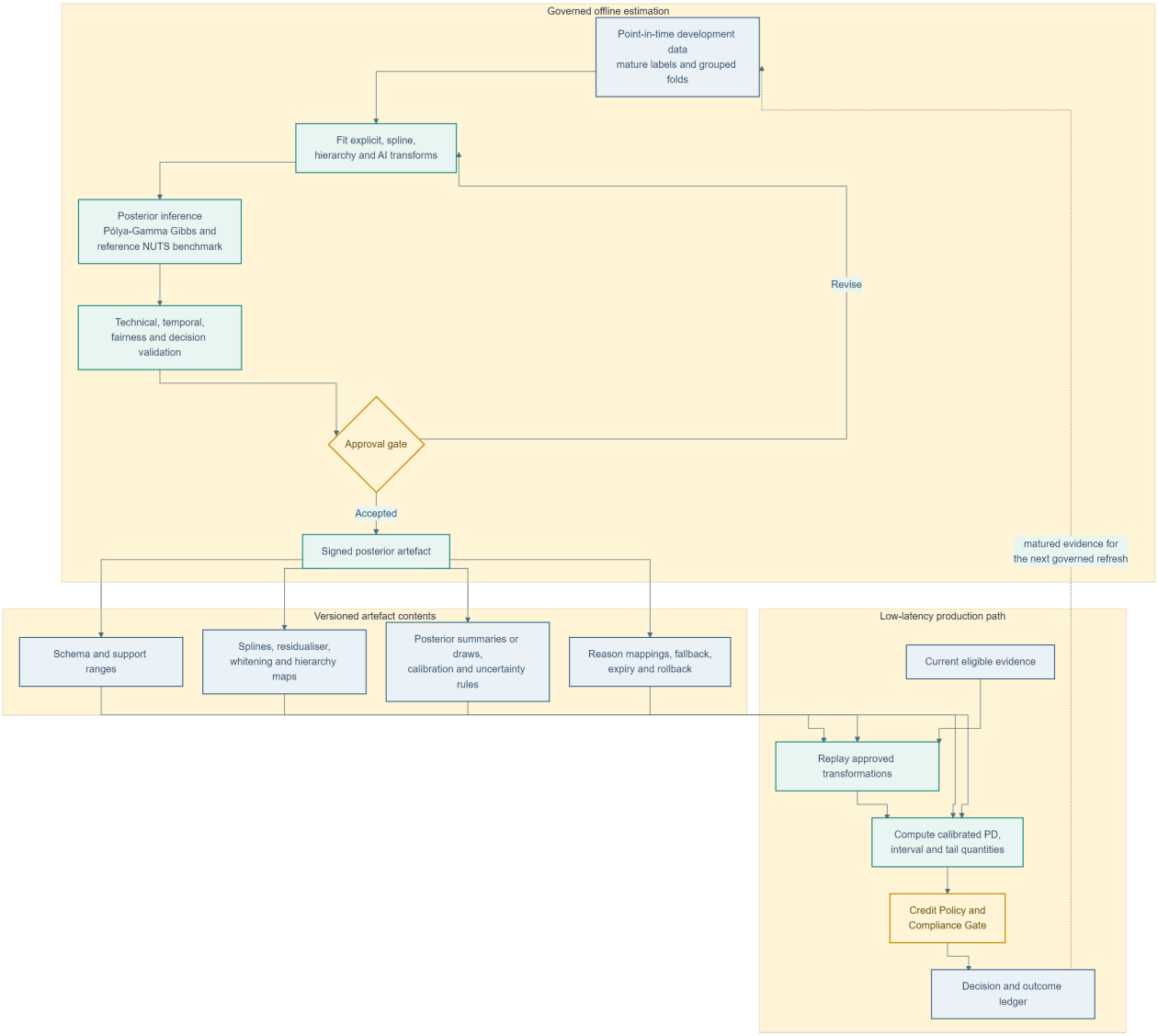


Figure 3: Offline posterior estimation and low-latency scoring remain separate. The signed artefact carries every transformation required to reproduce the accepted model. Current evidence may refresh a score without retraining, while matured outcomes return through a governed development cycle.

The artefact contains model and data-schema versions, feature definitions, spline basis and penalty metadata, encoder and residualiser versions, whitening transform if used, hierarchy mappings, priors, posterior draws or approved analytic summaries, calibration parameters, support limits, uncertainty rules, reason mappings, validation status, effective and review dates, and fallback and rollback identifiers. A new group is scored through the approved population hierarchy. A missing AI representation routes to the tested explicit-only fallback. An out-of-support explicit variable is recorded and handled according to policy rather than silently clipped without trace.

This arrangement makes computational sophistication compatible with operational control. The sampler can use CPUs, GPUs or specialised sparse linear algebra according to the validated implementation. The underwriting request receives a deterministic, replayable interface whose result can be reconstructed from the same artefact and point-in-time evidence. Computation therefore strengthens the model without placing an adaptive, unaudited posterior fit inside a customer interaction.

7. Prognosis, intervention effects and policy value

The champion model forecasts what tends to happen under the lending and servicing policy represented in its development data. That forecast can identify an MSME whose cash conversion is weakening, whose repayment velocity has slowed, or whose exposure is becoming difficult to service. It cannot, by itself, establish whether a payment holiday, a due-date change, a smaller revolving limit, a collections message, or a restructuring offer will improve the outcome. The intervention changes the path being predicted. This is why the collection architecture carries prognosis and causal learning on separate, connected rails.

For an eligible episode i, let $A_i \in \mathcal{A}$ denote an intervention assigned at the decision time and let $Y_i(a)$ be the potential outcome that would be observed under action a. A binary average treatment effect at horizon H is

$$\tau(H) = \mathbb{E}\left[Y_i(1) - Y_i(0)\right].$$

The outcome can instead be a durable cure indicator, days in arrears, net recovery value, or a customer-harm measure. The protocol must state the population, treatment, comparator, assignment unit, decision time, outcome, horizon, estimand and analysis set. For example, an operational question may be: among otherwise eligible Kenyan micro and small enterprises entering 1 to 7 days past due, what is the intention-to-treat effect of a one-cycle flexible repayment offer, compared with standard servicing, on durable cure by day 60 and re-default by day 120? The predictive PD is useful for stratification and precision. The random assignment identifies the effect.

Randomised pilots are the preferred route where uncertainty is genuine, the intervention is lawful and ethically supportable, and no borrower is denied a required protection. Cluster assignment can reduce contamination when relationship managers or branches cannot reliably deliver different scripts to adjacent customers. Individual assignment is more efficient when the treatment is digitally delivered and interference is limited. Randomisation is recorded before exposure, allocation is auditable, and primary outcomes and subgroup analyses are pre-specified. Intention-to-treat preserves the value of

assignment; treatment-on-the-treated analysis additionally requires a defensible instrument or compliance model.

Existing experiments show why contract design deserves this discipline. A Kenyan field experiment with micro-distributors compared a standard fixed-repayment loan with performance-contingent microequity and mutuality designs, illustrating that finance can share business-state risk rather than place the entire adjustment burden on a fixed instalment [10]. Evidence from Bangladesh similarly studies repayment flexibility and enterprise risk taking under a specific microcredit setting [11]. These studies do not prescribe one universal Kenyan bank product. They demonstrate a more important proposition for the present architecture: repayment structure is an intervention that can be assigned, measured and learned from, rather than treated as an invisible constant behind a default label.

When randomisation is not available, a quasi-experiment begins with an explicit source of identifying variation. A regression-discontinuity design estimates a local effect around a threshold only if potential outcomes vary continuously at the cutoff, the running variable is not precisely manipulated, and the policy on each side is stable. Bandwidth, functional form, density, covariate continuity and alternative-cutoff tests accompany the estimate [12]. It is particularly relevant when a registered score threshold changes a limit or support offer, although the result applies locally to cases near that threshold.

A difference-in-differences design can study a staggered servicing or product rollout when untreated potential-outcome trends are credible. Group-time effects, anticipation windows and appropriate comparison cohorts are required because a single two-way fixed-effects coefficient can blend different treatment timings. Pre-trend and event-study evidence, composition stability and sensitivity to concurrent changes form part of the design [13]. Synthetic control can construct a weighted comparison for a branch, region or platform rollout when one aggregate unit is treated and a sufficiently long pre-period and donor pool exist [14]. Target-trial emulation provides a broader protocol for observational comparisons: specify eligibility, treatment strategies, assignment, start of follow-up, outcome, causal contrast and analysis before aligning the observational records [15].

An instrumental-variable route may be considered when an operational mechanism shifts treatment receipt without directly shifting the outcome except through that treatment. Queue rotation, randomised encouragement or a capacity rule can sometimes supply such variation. Relevance, exclusion, independence and monotonicity are substantive claims, not labels inferred from a strong first stage. The resulting local effect belongs to the population whose treatment changes because of the instrument.

Heterogeneous treatment effects follow identification. A causal forest or hierarchical interaction model can estimate how a validated effect varies across pre-treatment characteristics [16]. Policy learning can then search for a constrained allocation rule that maximises an explicitly defined value function [17]. Both stages use held-out or cross-fitted predictions, overlap diagnostics and uncertainty. A subgroup with a high predicted default probability is not automatically the subgroup with the largest benefit from support. Conversely, a lower-risk enterprise may yield a large, low-cost improvement from a small timing adjustment.

The policy value must include both sides of the ledger. Let $a \in \mathcal{A}_i$ be a feasible action and let $L_i(a)$, $C_i(a)$, $H_i(a)$ and $O_i(a)$ denote expected credit loss, intervention cost, customer-harm cost and operational burden under that action. A simple research objective is

$$a_i^\star = \arg\min_{a \in \mathcal{A}_i} \mathbb{E}[L_i(a) + C_i(a) + \lambda_H H_i(a) + \lambda_O O_i(a) \mid \mathcal{I}_i] .$$

The weights and feasible set are governed policy choices. The exercise reports value by borrower segment, product, institution and economic state, together with distributional and conduct outcomes. It also monitors interference, non-compliance, attrition, delayed maturity, repeat assignment, contact fatigue and multiplicity. A pilot stops or adapts under pre-specified safety and operational rules, while its statistical analysis preserves the original estimand.

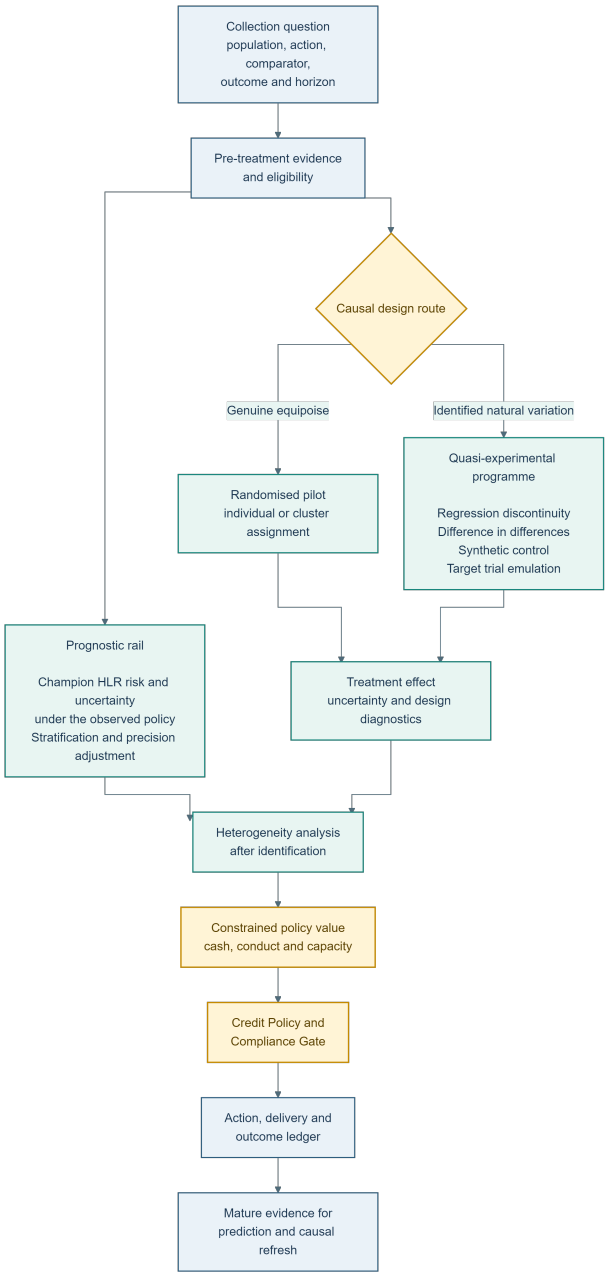


Figure 4: Prediction, causal identification and policy execution answer different questions. The champion score describes risk under an observed policy. Trials and credible quasi-experiments estimate intervention effects. A governed value rule combines the evidence only after its assumptions and operational constraints are visible.

8. The lifecycle timing and multi-state challengers

The fixed-horizon HLR gives a stable champion decision interface, but collections also asks when risk may materialise and how an account moves among recoverable states. A hierarchical piecewise-exponential model is the first timing challenger. Divide follow-up into intervals $j = 1, \ldots, J$. For episode i, let d_{ij} indicate an event in interval j, r_{ij} be at-risk exposure time, and

$$\log \lambda_{ij} = \alpha_j + \mathbf{x}_i^{\top}\beta + u_{s(i)} + v_{g(i)} + w_{j(i)}.$$

The interval likelihood can be written as

$$d_{ij} \mid r_{ij}, \lambda_{ij} \sim \text{Poisson}(r_{ij}\lambda_{ij}),$$

which supplies a computationally convenient representation of a piecewise-constant hazard [18], [19]. Right-censored episodes contribute their observed at-risk time without being labelled non-default. The survival probability through time t is

$$S_i(t) = \exp[-\Lambda_i(t)], \qquad P(T_i \le t) = 1 - S_i(t),$$

where $\Lambda_i(t)$ is cumulative hazard. At the champion horizon H_p, the timing model's $1 - S_i(H_p)$ is compared with the HLR probability on the same outcome definition, eligible population and cutoff. Material disagreement is investigated through censoring, time-varying effects, specification and calibration rather than averaged away.

The reconciliation uses posterior draws. For each eligible episode and draw, the challenger produces a horizon event probability and the champion produces its calibrated counterpart. The review examines their average difference, dispersion, calibration by predicted-risk band, and divergence by product, vintage and remaining tenor. A timing model that agrees at 60 days but separates at 180 days may be revealing genuine duration structure. A model that diverges only in the newest, least mature cohort may be exposing unstable baseline hazards or incomplete follow-up. This comparison gives the challenger a defined route to improvement: it must explain timing, censoring or state occupancy better while preserving or improving common-horizon calibration.

Collections contains more than a first-default clock. A multi-state model represents transitions among current, early arrears, modification, cure, durable cure, re-default, default, recovery and closure states [20]. Its transition intensity

$$\lambda_{rs,i}(t) = \lim_{\Delta t \downarrow 0} \frac{P\{X_i(t + \Delta t) = s \mid X_i(t) = r, \mathcal{H}_i(t)\}}{\Delta t}$$

answers a state-specific question: for an account presently in state r, how quickly does it enter state s, conditional on its history and current evidence? A semi-Markov extension includes time already spent in the present state where memorylessness is implausible. Transition probabilities then support

expected days in arrears, durable-cure probability, re-default risk, collection workload and discounted cash recovery.

Competing events require the estimand to remain explicit. A cause-specific hazard asks about the instantaneous occurrence of one transition among accounts still event-free. A subdistribution approach targets cumulative incidence while retaining competing events in a modified risk set [21]. Neither is a universal replacement for the other. Prepayment, closure, write-off and cure can change both the interpretation and the population remaining under observation.

Transition construction also respects the observation process. A daily state engine can record exact digital payments, while external recovery and legal events may arrive in intervals. The likelihood distinguishes an actual event time from an interval-censored or administratively recorded date. The Markov assumption is checked by adding duration in state and selected history summaries. When duration materially changes the next transition after current state and covariates are known, the semi-Markov version becomes the more credible operational representation.

Persistent borrower frailty is optional. A shared Gamma frailty may be useful when repeated, matured episodes identify stable unobserved heterogeneity, but borrower identity alone is not enough. Product, enterprise, connected exposure and time effects are evaluated first, and frailty sensitivity is compared with alternative dependence and robust-variance treatments. The survival challenger is therefore adopted because timing and censoring matter, while any convenient conjugacy remains secondary to an identified economic question.

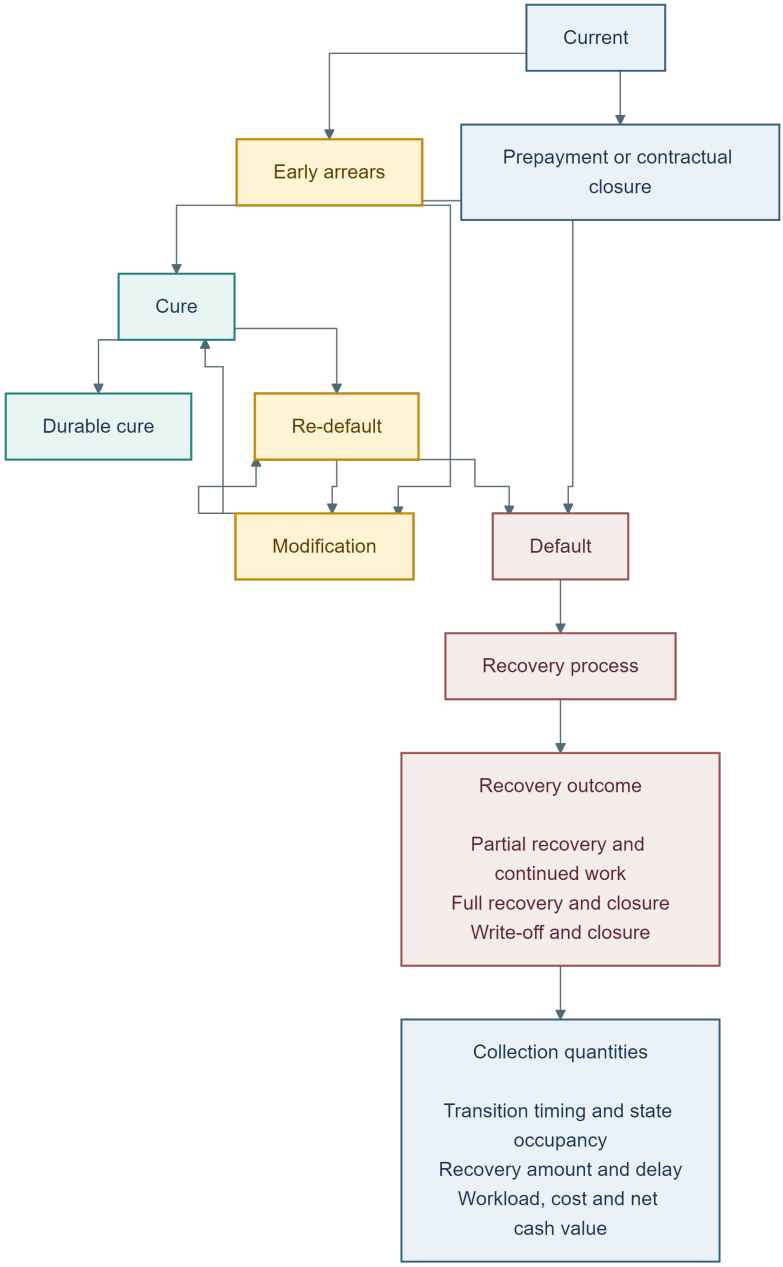


Figure 5: The lifecycle challenger follows timing, cure, re-default and recovery. States are operational definitions with entry, exit and maturity rules. The diagram becomes a transition register, not a decorative funnel, and each permitted movement maps to data, likelihood and cash-flow tests.

9. From default probability to collectible cash

A probability becomes financially useful when it joins a cash-flow model without absorbing quantities that have different meanings. For product p at horizon H, the architecture therefore passes separate posterior distributions for probability of default, exposure at default, cure, prepayment, loss severity, recovery amount, recovery delay and workout cost. A convenient expected-loss identity is

$$\mathrm{EL}_{i,H} = \mathbb{E}\left[\mathrm{PD}_{i,H} \times \mathrm{EAD}_{i,T_D} \times \mathrm{LGD}_{i,H} \mid \mathcal{I}_i\right],$$

where dependence inside the expectation is retained. Multiplying three separately averaged point estimates is permitted only as a disclosed approximation. It can miss the fact that adverse business conditions raise both drawdown and severity, or that an early restructuring reduces future EAD while changing the timing of receipts.

Exposure follows the contract. For an amortising facility it evolves through scheduled principal, verified prepayments, capitalised charges and modification terms. For revolving credit it also includes future utilisation before default. If $B_i(t)$ is drawn balance, $L_i(t)$ limit and CCF_i the conditional conversion of undrawn capacity, a scenario-specific exposure can be written

$$\mathrm{EAD}_i(t) = B_i(t) + \mathrm{CCF}_i(t)\,[L_i(t) - B_i(t)].$$

The conversion factor is estimated by product, utilisation state and horizon with time-forward evidence. It is not borrowed unexamined from a different portfolio. A limit reduction, draw freeze or performance-contingent schedule changes this path, so policy evaluation simulates exposure under each feasible action rather than inserting an intervention effect into an unchanged EAD.

For invoice finance, exposure and collection are also tied to the financed invoice, verified debtor, dilution, dispute and settlement route. One enterprise can remain solvent while a concentrated customer delays payment, so debtor-level concentration and recourse terms affect both EAD and recovery. For asset finance, collateral value, repossession time and business interruption enter the recovery path. The common episode spine permits product-specific mechanics without forcing them into one generic balance multiplier.

Recovery is a dated sequence of gross receipts and costs. For default at T_D, let R_{ik} be cash received at delay τ_{ik}, C_{ik}^{work} the attributable workout cost, and $d_i(\tau)$ the approved discount factor. Net recovery value is

$$\mathrm{NRV}_i = \sum_{k=1}^{K_i} d_i(\tau_{ik})\left(R_{ik} - C_{ik}^{\mathrm{work}}\right), \qquad \mathrm{LGD}_i = 1 - \frac{\mathrm{NRV}_i}{\mathrm{EAD}_i(T_D)}.$$

Collateral realisation, guarantor payment, insurer refund, cash sweep and borrower repayment remain separately coded because their legal rights, delays and costs differ. Cures are not automatically recoveries: an account can return to current status and later re-default. A durable-cure window and re-default observation period therefore precede final performance classification.

Portfolio aggregation retains scenario coherence. Posterior draws are propagated through contractual schedules so that one draw contains aligned default, utilisation, cure, recovery and timing assumptions. This produces a distribution of monthly and cumulative net cash rather than a collection of unrelated percentiles. Finance can reconcile the mean path, risk can examine tail loss and collections can see workload and timing. The same draw identifiers allow Part 5 to compare gross asset cash, operating costs, funding obligations and protection layers under one state of the world.

The handoff to pricing, accounting and structured finance is a versioned distributional output, not one undifferentiated score. It includes horizon PD, transition probabilities, EAD paths, discounted recovery paths, expected loss, tail loss, calibration segment, posterior uncertainty, scenario and model version. Part 5 can then translate the same cash logic into KESONIA-linked pricing, expected credit loss, capital, reserve and ring-fenced funding decisions without redefining the underlying credit event.

10. Common shocks and residual dependence

MSME defaults cluster because firms share customers, suppliers, platforms, neighbourhoods, transport corridors, weather, policy changes, interest rates and demand conditions. The first response is to model these observed channels directly. Macroeconomic regimes enter the contextual evidence set; sector, geography, institution and origination cohort receive hierarchical effects; connected enterprises are grouped; and validation holds out entire common-risk units. Residual dependence is considered only after the marginal models are calibrated under those shared factors.

Let $F_1, \dots, F_m$ be calibrated marginal distributions for portfolio loss or default quantities. Sklar's representation writes the joint distribution as

$$F(y_1, \dots, y_m) = C_\theta\left(F_1(y_1), \dots, F_m(y_m)\right),$$

where C_θ is a copula [22]. The decomposition allows marginal underwriting evidence and residual co-movement to be tested separately. For discrete default indicators, the likelihood uses probability rectangles or an appropriate latent-variable construction because the copula is not uniquely identified between the jumps of discrete marginals.

The Clayton family is a purposeful challenger when lower-tail liquidity states are expected to coincide. In two dimensions,

$$C_\theta(u, v) = \left(u^{-\theta} + v^{-\theta} - 1\right)^{-1/\theta}, \qquad \theta > 0,$$

with lower-tail dependence coefficient

$$\lambda_L = 2^{-1/\theta}.$$

Clayton's construction has a long association with dependent event times [23], but narrative plausibility does not select it. Gaussian, Student t, Gumbel, rotated Archimedean and factor-copula alternatives

are compared according to the portfolio question. A t-copula may represent symmetric tail association; a rotated family may match an adverse orientation defined on survival rather than default scales. Variable definitions and rotations are recorded so that "lower tail" always refers to the same economic direction.

The main control is a shock-allocation register. Fuel-price stress, policy tightening or platform disruption cannot be loaded simultaneously as an explicit scenario shift, a time effect, a latent hierarchy effect and a copula shock without reconciliation. Each systematic driver has a primary owner. The residual layer receives the dependence left after those owners have acted, and stress scenarios state whether they condition the marginals, the copula, or both.

Dependence may also change through time. A static parameter is the baseline because short datasets can mistake one crisis window for a permanent law. A dynamic challenger links θ_t to a small number of registered macro states or a regularised latent process, then tests whether the added movement predicts later co-defaults. The fit preserves the same marginal forecasts used in the independent baseline, which allows the incremental value of time-varying dependence to be measured directly. Scenario analysis can then widen dependence under a named common shock without refitting every borrower probability to the same adverse observation.

Validation compares empirical and posterior-predictive co-default counts, sector and geography concentrations, rank dependence, joint tail exceedances, portfolio loss quantiles and tranche or covenant quantities. Parametric-bootstrap goodness-of-fit tests help evaluate candidate copulas [24], while temporal and cluster resampling test whether apparent tail dependence is stable. A more elaborate dependence model advances only when it improves held-out joint outcomes and materially informs concentration, capital, reserve or risk-transfer decisions. This keeps correlation from becoming an ornamental coefficient detached from collectible cash.

11. Decision quantities and the policy boundary

The underwriting service publishes quantities that a decision can consume: calibrated event probabilities, posterior intervals, tail exceedance probabilities, lifecycle transitions, expected collectible cash, expected loss and data-quality status. It does not collapse policy, regulation and commercial judgement into the fitted coefficient vector. For feasible action set $\mathcal{A}_i^{\text{feasible}}$, the economic layer may calculate

$$a_i^\star = \arg\max_{a \in \mathcal{A}_i^{\text{feasible}}} \mathbb{E}\left[U_i(a, L_i, C_i, H_i) \mid \mathcal{I}_i\right],$$

and can add a tail constraint such as

$$\mathrm{CVaR}_q(L_i(a) \mid \mathcal{I}_i) \le K_i(a).$$

Here U_i states the approved value function, L_i loss, C_i delivery cost, H_i customer outcome and K_i allocated risk capacity. Decision-focused evaluation can show whether a forecast improves choices at relevant thresholds [25], but the utility units and constraints belong to credit governance. They are

calibrated for the lender's and customer's consequences rather than inherited from another decision setting.

The Credit Policy and Compliance Gate receives model version, evidence timestamp, posterior quantities, uncertainty, support and quality flags, causal-evidence status, aggregate exposure and concentration. It then applies the registered legal, contractual, risk-appetite, affordability, fair-treatment, product and operational rules. The output may approve, decline, change a limit, alter price within authorised bounds, offer support, restructure, intensify contact, route to specialist recovery, or defer for missing evidence. Approval and collections are therefore joined by the same lifecycle record without giving one model unrestricted authority over every action.

Uncertainty changes the route, not merely the wording of a report. A wide interval caused by a new sector may support a smaller initial exposure, a shorter review period or additional verified evidence. A wide interval caused by a stale ledger may trigger data repair and temporary reliance on contractual features. A high probability with strong causal evidence for a low-cost support action may route to an offer before arrears deepen. These mappings are registered separately from model parameters and tested for customer and portfolio consequences.

Reason communication follows the same ownership structure. Explicit features and registered interactions supply human-readable economic drivers. A residual AI contribution is described through its governed feature family, direction, stability and supporting explicit context rather than assigning a fictional name to a latent coordinate. The decision record distinguishes the model's influential evidence from the policy rule that determined the action and from any authorised human override.

This boundary also supports supervision. The Central Bank of Kenya's risk-based approach emphasises the identification, measurement, monitoring and control of material risks [26]. The architecture makes those verbs observable: the champion estimates and monitors, the causal programme tests interventions, the cash layer measures consequences, and the policy gate controls action. Exceptions and overrides carry reason, authority, duration and outcome. Aggregate limits remain enforceable even when an individual expected-value calculation appears attractive.

12. Validation as a collection discipline

Validation asks whether the architecture remains truthful at the moments that matter: before disbursement, during deterioration, after intervention, at recovery, and under common stress. It begins before fitting. Prior predictive simulation tests whether intercepts, spline shapes, hierarchy scales, horseshoe sparsity and time effects imply plausible event rates and group variation. Simulation-based calibration on recoverable synthetic settings tests whether the inference implementation can reproduce known generative quantities. Unit tests compare Pólya-Gamma and reference NUTS likelihoods, transforms and predictions.

Posterior computation is assessed with multiple chains, rank-normalised split $\widehat{R}$, folded diagnostics, bulk and tail effective sample sizes, Monte Carlo standard errors and trace behaviour [27]. NUTS runs additionally report divergences, tree-depth saturation and energy diagnostics. Pólya-Gamma runs report latent-variable stability, sparse-solve accuracy and block autocorrelation. Thresholds are defined

in the validation protocol, and exceptions require resolution or a documented restriction before an artefact can be approved.

Predictive validation is time-forward and borrower-grouped. Every connected enterprise remains in one fold. Successive windows reproduce information availability, label maturity, macro vintages and the policy that generated outcomes. Sector-held-out, geography-held-out, institution-held-out and emerging-product tests measure transfer rather than hiding sparse groups inside random splits. Performance is reported overall and by decision-relevant segment using calibration intercept and slope, observed-to-expected ratios, Brier and log scores, ranking, interval coverage and tail-event capture. Proper scoring rules reward honest distributions rather than confident point forecasts alone [28].

Model comparison follows the architecture's ownership rules. The explicit-only champion candidate is compared with explicit plus unadjusted AI, explicit plus cross-fitted residual AI, and residual AI with regularised horseshoe shrinkage. The review examines out-of-time value, calibration, coefficient and reason stability, subgroup reliability, compute cost and incremental decisions. Residual correlations, variance-inflation summaries and projection diagnostics reveal duplicated evidence. Removing the AI block, perturbing it, or replacing it with noise should change results only in ways supported by genuine held-out signal.

The timing challenger is validated on survival calibration, time-dependent discrimination, integrated scoring, cumulative incidence, state occupancy, transition counts and recovery timing. The cash layer reconciles predicted principal, interest, prepayment, cure, default, recovery, cost and write-off to ledger totals. The copula layer is judged on joint events and portfolio loss, not marginal AUC. PSIS-LOO can support within-development comparison when its Pareto diagnostics are acceptable [29], but it does not replace the forward and clustered tests that mirror deployment.

Causal validation follows the design. Randomised pilots report allocation integrity, baseline balance, treatment receipt, attrition, missingness, spillovers, primary estimates and multiplicity control. Regression discontinuity reports density and continuity tests, bandwidth sensitivity and placebo cutoffs. Difference-in-differences reports pre-trends, timing and comparison cohorts. Target-trial analyses inspect eligibility alignment, time zero, overlap, censoring and negative controls. Heterogeneous effects are replicated out of sample before entering a policy rule.

Monitoring continues after approval. Calibration, evidence freshness, missingness, support excursions, population and concept drift, transition rates, overrides, contact outcomes, customer complaints, restructures, cures, re-defaults and net recoveries are reviewed together. A change in data or business process opens a diagnosis: source defect, economic shift, product change, servicing change or model deterioration. The response may repair data, recalibrate, restrict a segment, revert to the explicit-only artefact, initiate a controlled experiment or refresh the model. Independent model risk, credit, collections, legal, compliance, data and business reviewers receive enough lineage to reproduce the finding.

Transfer validation is staged rather than declared once. A new participating institution begins with schema reconciliation, outcome-definition mapping and shadow scoring. Its posterior group effect remains strongly pooled until matured local evidence accumulates. Approval for live use specifies products, limits, monitoring frequency and expiry. Expansion requires stable calibration and collection outcomes in the new setting. This sequence turns external validity into a measurable programme

and prevents an encouraging pooled result from becoming an unsupported claim of universal transportability.

The final acceptance question is economic and human. Does the model support sustainable collection while improving risk selection, timing, product structure and intervention quality across the full lifecycle? A technically converged model that creates unstable limits, avoidable customer harm or uncollectible balances has not passed. A simpler specification that remains calibrated, actionable and auditable may be the stronger production model.

Conclusion

The MSME introduced at the beginning is not a static application row. It is a business whose invoices, costs, withdrawals, repayments and operating shocks unfold under a product and servicing policy. Part 2a established the point-in-time evidence contract. This part turns that evidence into a disciplined inferential architecture.

The champion is a calibrated hierarchical Bayesian logistic regression with governed P-splines, partially pooled group effects, residual artificial-intelligence representations and regularised-horseshoe shrinkage. Pólya-Gamma augmentation makes its Gaussian blocks computationally tractable offline while the production request uses a signed, replayable posterior artefact. Timing and multi-state challengers add censoring, cure, re-default and recovery dynamics. A residual dependence layer addresses joint stress only after observable common causes and calibrated marginals are represented.

Prediction remains connected to, but distinct from, causal evidence. Randomised pilots and credible quasi-experiments test whether product and servicing actions improve durable collection and customer outcomes. The cash layer carries PD, EAD, cure, prepayment, LGD, recovery timing and cost into a common distributional handoff. The policy gate then makes authorised decisions within conduct, capacity and concentration constraints.

Part 3 develops the intervention and governance operating model. Part 4 carries the artefacts, event lineage and controls into enterprise systems. Part 5 converts the forecast distribution into pricing, expected credit loss, capital, liquidity and risk-transfer decisions. Together, the sequence underwrites for collection by treating repayment as a designed lifecycle outcome rather than an after-the-fact statistic.

Appendix A: Preserved spline and time-prior alternatives

The champion uses P-splines: a B-spline basis with a second-difference penalty and separately identified null-space terms. Two alternatives remain available for sensitivity and challenger work.

For ordered coefficient vector $\beta = (\beta_1, \dots, \beta_K)^\top$, a B-spline plus first-order random-walk prior is

$$\beta_k - \beta_{k-1} \sim \mathcal{N}(0, \tau^2_{\mathrm{RW}}), \qquad k = 2, \dots, K.$$

This favours locally smooth coefficient evolution while permitting broad level and slope components. Identification uses an explicit intercept and centring or sum constraints. A B-spline plus stationary AR(1) prior is

$$\beta_k - \mu = \rho(\beta_{k-1} - \mu) + \epsilon_k, \qquad \epsilon_k \sim \mathcal{N}(0, \sigma^2_\epsilon), \qquad |\rho| < 1.$$

It favours reversion toward μ across the ordered coefficient index. Because a coefficient index is not automatically economic time, its interpretation is stated carefully. For calendar effects, the same RW1 or AR(1) structures can apply directly to δ_t, where order has an explicit temporal meaning.

All variants use the same knots, degree, boundary policy, feature scaling and out-of-support rules during comparison. The validation set tests calibration, smooth shape, boundary stability, effective degrees of freedom, extrapolation, posterior uncertainty and decisions. The alternatives are preserved because different portfolios may support different local smoothness or persistence structures. Promotion depends on evidence and interpretability rather than on prior family alone.

The comparison report plots the function and first derivative on the observed support, with a rug or weighted density showing where evidence exists. It also reports prior and posterior roughness, sensitivity to knot density, and the contribution of null-space terms. Product-specific deviations are overlaid with their uncertainty and effective sample sizes. This makes the distinction between basis choice, penalty choice and economic shape visible to reviewers, while retaining B-spline plus RW1 and B-spline plus AR(1) specifications as fully reproducible challengers.

Appendix B: Blocked Pólya-Gamma computation

For each offline iteration:

1. Compute every linear predictor from the current explicit, spline, residual-AI, interaction, hierarchy and time blocks.
2. Draw $\omega_i \sim \mathrm{PG}(1, \eta_i)$ for each observation or exact aggregated design row.
3. Update strongly coupled Gaussian coefficient blocks jointly using their likelihood and prior precision matrices.
4. Update hierarchy scales, covariance parameters, spline smoothing scales, horseshoe scales and time-process parameters using their validated conditional or numerical kernels.
5. Apply identification constraints or the equivalent parameterisation and calculate calibration and posterior-predictive quantities.

6. Record diagnostics and retain draws according to the approved warm-up, thinning if any, storage and convergence protocol.

Fixed seeds, environment lockfiles, source hashes, fold assignments, transforms and sampler settings form the reproducibility record. The exported artefact contains only approved scoring quantities and transformations. It does not contain an active training loop.

The implementation stores the sparse precision blocks and their ordering, checks positive definiteness after constraints, and verifies selected Gaussian updates against direct dense calculations on small fixtures. Parallel chains use independent random streams. Posterior storage retains enough draws for tail decisions and challenger reconciliation, while a compressed scoring representation is accepted only after its probabilities, intervals and threshold exceedances match the retained draws within approved tolerances. Restart, rollback and numerical-failure tests form part of release validation.

Appendix C: Causal estimand and design register

Treatment family	Assignment unit	Primary estimand	Example horizon	Identification route	Required diagnostics	Conduct constraint
Repayment timing	Enterprise episode	Intention-to-treat effect on durable cure	60 and 120 days	Randomised pilot	Balance, receipt, attrition, spillover	Preserve required protections
Threshold support	Score-threshold episode	Local average effect near cutoff	Product-specific	Regression discontinuity	Density, continuity, bandwidth, placebo cutoffs	Registered threshold and override log
Servicing rollout	Branch, platform or cohort	Group-time treatment effect	Multiple periods	Difference-in-differences	Pre-trends, anticipation, composition	Stable comparison practice
Regional product design	Region or platform	Aggregate treated-unit effect	Pilot period	Synthetic control	Pre-fit, donor sensitivity, placebo units	No hidden concurrent intervention
Observational restructuring	Eligible distressed episode	Per-protocol or treatment-strategy contrast	Cure and re-default windows	Target-trial emulation	Time-zero alignment, overlap, censoring, negative controls	Explicit eligibility and treatment versions
Targeted intervention	Eligible episode or cluster	Constrained policy value	Operational horizon	Validated heterogeneous effects and policy learning	Honest splitting, subgroup stability, capacity simulation	Fair treatment, budget and contact limits

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Part 3

Regulatory Orchestration, Fair Intervention and Collection Governance

Underwrite for Collection Series

Abstract

Continuous underwriting creates value when an earlier warning leads to a better decision, a more sustainable payment path and a more collectible cash outcome. Part 3 defines the governance architecture between statistical evidence and customer treatment. It separates the prognostic distribution, causal evidence, policy eligibility and authorised action; introduces a graduated intervention library; and formalises a Credit Policy and Compliance Gate that carries legal, contractual, affordability, conduct, operational and portfolio constraints into every decision. Kenyan data-protection, banking, digital-credit, credit-information and consumer-protection duties are mapped according to the institution and activity in scope. Fairness is measured across access, error, intervention, contact, complaint, cure, re-default and net-recovery outcomes, with uncertainty made visible. Randomised and quasi-experimental designs are then positioned as governed methods for learning which permissible treatments create durable value. The result is an accountable route from a posterior risk estimate to proportionate human and institutional action.

1. From warning to permission

A small distributor supplies twelve neighbourhood shops. Her largest customer normally pays on the twenty-fifth day of each month, while the lender collects a working-capital instalment on the twenty-third. A two-day misalignment creates a familiar sequence: the account balance falls, the scheduled deduction fails, a reminder is sent and the arrears clock begins. Part 2b may correctly detect that this pattern raises the probability of 30-day arrears. Yet the warning alone cannot decide whether the lender should move the collection date, offer a short grace period, reduce the facility, begin a restructure review or continue the existing schedule.

The decision depends on more than risk rank. The lender must know whether the customer's receipts remain economically viable, whether the product contract permits a change, whether the action is affordable, who bears its cost, whether similar customers receive comparable treatment, whether the communication is clear, and what happens if the customer disagrees. A model may estimate a future state. Permission to act comes from the licensed institution's mandate, the contract, applicable law, approved policy and a named human or committee authority.

That separation brings the paper's governing idea into practice. A loan is underwritten for collection when product structure, monitoring and intervention are designed around sustainable cash performance after disbursement. Collection therefore begins as a design discipline, not as a department that appears after default. Kenya's Central Bank prudential guideline on consumer protection places fairness, reliability, transparency, equity and responsiveness across the institution's relationship with the customer, and assigns the board responsibility for the consumer-protection framework [1]. For digital credit providers within scope, the 2022 regulations add specific requirements for fair and transparent information, complaints handling, transaction records and collection conduct [2]. Data use and automated decision-making also sit within the rights and duties created by Kenya's Data Protection Act [3].

Part 3 operationalises these obligations through four linked objects:

1. **Model inference** estimates a defined future outcome from the information available at the decision cutoff. It includes the posterior mean, interval, tail probability, support-range status and data-quality state.
2. **Causal evidence** estimates how a specified permissible treatment changes an outcome for an eligible population relative to a defined alternative.
3. **Policy eligibility** determines which actions are available under law, contract, conduct standards, affordability rules, operational capacity and portfolio constraints.
4. **Customer treatment** is the actual communication, offer, modification, restriction, support or recovery action executed by its authorised owner.

The same record connects the four objects without collapsing them. A high probability of arrears may prompt an affordability conversation. Evidence from a prior pilot may show that payment-date alignment improves durable cure for enterprises with verified recurring receipts. The contract may permit that alignment without repricing. The policy gate may still route the case to a person because the receipts are unusually concentrated or the model is outside its support. The final action can then be explained in terms of payment timing, contractual conditions and the customer's available choices.

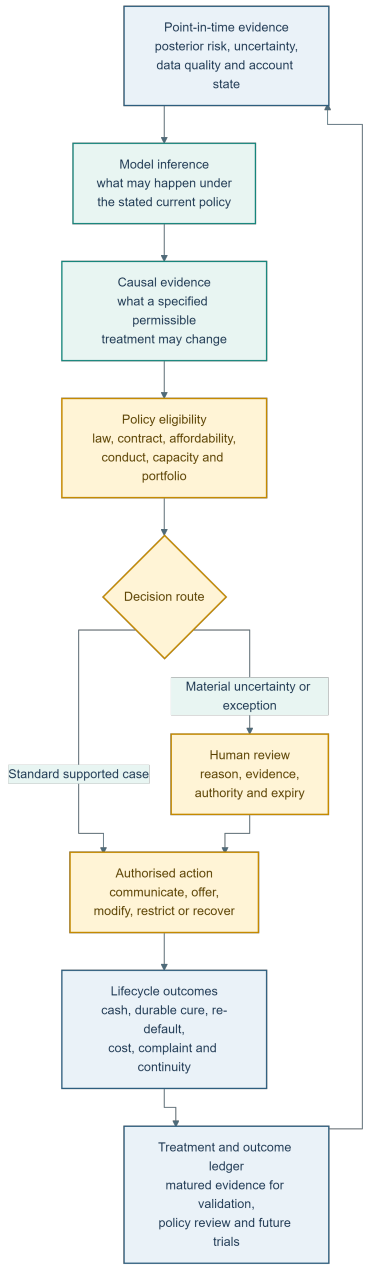


Figure 1: From lifecycle evidence to authorised customer treatment. Prediction, causal learning, policy eligibility and execution remain separate decision objects. Outcomes return through a point-in-time ledger so later evidence can improve the system without rewriting what was known at the earlier decision.

2. The intervention contract

An intervention needs the same discipline as a financial product. It has a target population, a mechanism, an owner, a cost, an expiry and a measurable outcome. Calling an action "supportive" does not establish that it helped. A reminder may reduce accidental lateness for one enterprise and create an excessive contact burden for another. A grace period may protect productive investment, or postpone recognition of a cash-flow problem. A limit freeze may contain exposure while removing the liquidity needed to complete profitable orders. The decision record must make these possibilities testable.

For action (a) considered at decision time (t), define the intervention contract:

$$\mathcal{C}_{a,t} = \left(\mathcal{E}_{a,t}, \mathcal{X}_{a,t}, m_a, o_a, c_a, d_a, \tau_a, \mathbf{Y}_a, \rho_a, \chi_a\right),$$

where $\mathcal{E}$ is the eligible population, $\mathcal{X}$ the mandatory exclusions, m the hypothesised mechanism, o the authorised owner, c the cost bearer, d the customer communication and delivery channel, τ the treatment and outcome horizon, $\mathbf{Y}$ the outcome vector, ρ the monitoring and stop rule, and χ the reconsideration, complaint and escalation route. A policy version links these fields to the exact rules effective at the time.

The outcome vector is deliberately wider than payment received:

$$\mathbf{Y}_a = \left(\mathrm{Cash}_{0:H}, I_H^{\mathrm{DC}}, I_H^{\mathrm{RD}}, \mathrm{Cost}_{0:H}, \mathrm{Complaint}_{0:H}, \mathrm{ContactBurden}_{0:H}, \mathrm{Continuity}_H\right).$$

It records discounted cash, durable cure, re-default, direct intervention and recovery cost, complaint outcomes, contact burden and a product-appropriate indicator of business continuity. The last measure might be continued trading, verified turnover, fulfilment of customer orders or retention of a productive asset. It is defined before the experiment and collected only where a lawful, proportionate data route exists.

Every executed action enters a treatment ledger with the pre-action score and uncertainty, policy and model versions, eligibility, assignment mechanism, communication, customer response, executed terms, cash-flow change, human participation and downstream outcomes. This record addresses three sources of bias. First, the action changes what happens next. Second, the lender observes repayment only for customers who received credit and remained observable. Third, teams may select easier cases for supportive treatment and harder cases for recovery, making a successful treatment appear more effective than it was.

The ledger also preserves customer meaning. A business that pays after one reminder is not automatically "unwilling." A night-time transaction pattern is not intent. A decline in earnings is evidence about cash generation, not a diagnosis of the owner. The system uses observed states and stated uncertainty, while staff establish context through appropriate engagement.

3. A graduated intervention library

The intervention library follows the lifecycle rather than dividing work into disconnected origination and collections systems. A current account can receive information or payment alignment. Early stress may justify a voluntary budgeting conversation or a temporary support offer. Arrears may require a documented cure or modification pathway. Persistent distress may require restructuring, collateral action or recovery. Each step preserves the customer's payment history and contractual state so that assistance and recognition develop together.

Lifecycle state	Illustrative intervention	Primary mechanism	Required evidence and control	Outcome emphasis
Current, stable	Clear schedule and self-service information	Reduce avoidable error and information cost	Accurate balance, due date and channel; customer choice	On-time cash, low contact burden
Current, timing mismatch	Payment-date alignment or split instalment	Match debt service to verified receipt rhythm	Contractual authority, affordability, no hidden fee	Durable performance and business continuity
Early stress	Reminder choice, budgeting prompt or hardship triage	Create time for diagnosis and action	Contact preferences, vulnerability screen, staff route	Avoided arrears, complaints and opt-out
Short arrears	Funded grace period or cure arrangement	Bridge temporary liquidity without obscuring account state	Eligibility, cost bearer, accounting review and expiry	Cure, re-default and net cash
Viable but impaired	Modification or restructure assessment	Restore a payment path supported by cash capacity	Full affordability review, approval, new cash flows and disclosures	Sustainable payment, ECL and continuity
Non-viable or persistent default	Proportionate recovery and security enforcement	Maximise lawful net recovery while limiting harm	Authority, accurate debt, conduct rules, complaint route	Discounted net recovery, time and cost

This taxonomy separates communication from contract change. A message that explains the amount due is not a restructure. A payment-date change is recorded even when it has no fee or principal concession. A temporary grace arrangement remains visible to impairment and portfolio processes. A restructure creates a new contractual cash-flow schedule and passes through the accounting assessment owned by the reporting institution. Recovery uses an accurate balance, an authorised channel, humane conduct and the appropriate notice process.

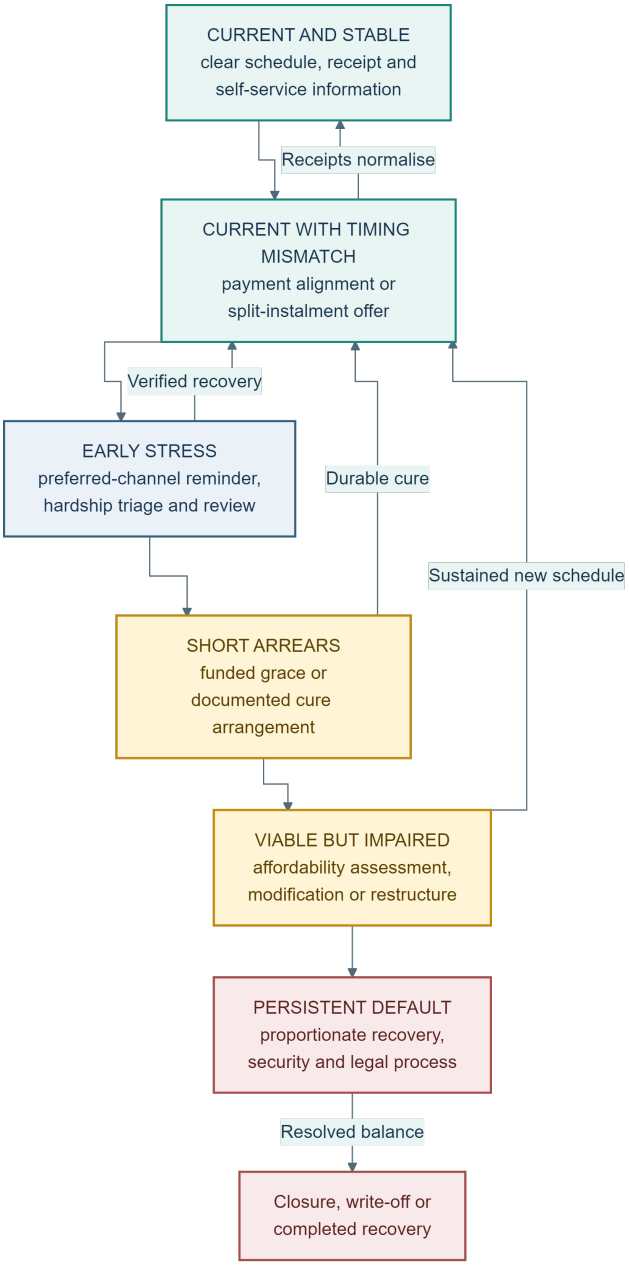


Figure 2: The governed MSME intervention ladder. Accounts can move forward or return to a healthier state. Every transition preserves the contract, account status, intervention, customer communication and outcome history required for accounting, validation and fair-treatment review.

The ladder is neither a universal sequence nor an automated escalation timer. An enterprise can move directly to a restructure review after a verified shock, remain on its original schedule with no intervention, or proceed to recovery when viability and contractual facts warrant it. Mandatory protections and legal processes take priority over an experimental treatment. The action library makes these routes explicit before a model or agent begins recommending them.

4. The Credit Policy and Compliance Gate

The Credit Policy and Compliance Gate is the runtime boundary between evidence and permission. It receives the Part 2b posterior artefact, explicit lifecycle state, product terms, affordability evidence, data-quality status, customer treatment history and portfolio context. It returns an authorised action, a review route or a reasoned deferral. The gate is versioned independently from the statistical model, allowing a lender to change a product limit, contact rule or approval authority without retraining the HLR.

For customer-account state $s_{i,t}$, let $\mathcal{A}$ be the intervention library. The feasible action set is:

$$\mathcal{A}^{\text{feas}}(s_{i,t}) = \mathcal{A}^{\text{law}} \cap \mathcal{A}^{\text{contract}} \cap \mathcal{A}^{\text{conduct}} \cap \mathcal{A}^{\text{afford}} \cap \mathcal{A}^{\text{ops}} \cap \mathcal{A}^{\text{portfolio}}.$$

The terms are implemented as traceable decision tables. The legal table identifies the entity and applicable rule. The contractual table verifies product terms and authority. The conduct table applies contact, disclosure, fair-treatment and vulnerability controls. The affordability table tests the proposed cash path. The operations table verifies channel, staff and servicing capacity. The portfolio table applies exposure, concentration and funding constraints. An empty intersection produces a deferral or human route rather than an improvised action.

Within the feasible set, a policy may rank actions by expected lifecycle value:

$$a^*(s) = \arg\max_{a \in \mathcal{A}^{\text{feas}}(s)} \mathbb{E}[\text{DCF}_a - C_a - \lambda_h H_a - \lambda_b B_a \mid s, \mathcal{D}],$$

where DCF_a is discounted net cash, C_a direct intervention and servicing cost, H_a a registered customer-harm vector, and B_a business-continuity loss. The weights λ_h and λ_b express approved governance priorities; they cannot monetise away a hard legal or policy prohibition. The expectation integrates predictive and, where identified, treatment-effect uncertainty.

The value calculation is reported as a vector before any approved scalar ranking is applied. Credit risk sees expected arrears, default and recovery. Finance sees cash timing and direct cost. Conduct sees communication, complaint and hardship outcomes. The relationship team sees continuity of the enterprise and its trading cycle. Presenting these components separately prevents a small estimated collection gain from concealing a material increase in re-default, customer distress or operating burden. Policy owners approve the horizon and weights through a recorded change process, and sensitivity analysis shows when a recommended action depends on a narrow valuation choice.

Uncertainty creates an operational choice. The gate can route a case to review when the probability of harmful performance exceeds an approved tolerance or when the model is outside support:

$$\text{Review}_{i,t} = \mathbf{1}\Big\{P\big(g_h(H_a) > h_{\max} \mid s_{i,t}, \mathcal{D}\big) > \alpha \;\vee\; q_{i,t}^{\text{support}} = 0 \;\vee\; q_{i,t}^{\text{data}} < q_{\min}\Big\}.$$

Here, g_h maps the registered harm vector to the governed escalation measure. This rule gives uncertainty a destination. A reviewer receives the information that produced the route, the available actions, the contractual and policy constraints, and a time-limited authority. The reviewer records the decision and reason rather than replacing the audit trail with free text.

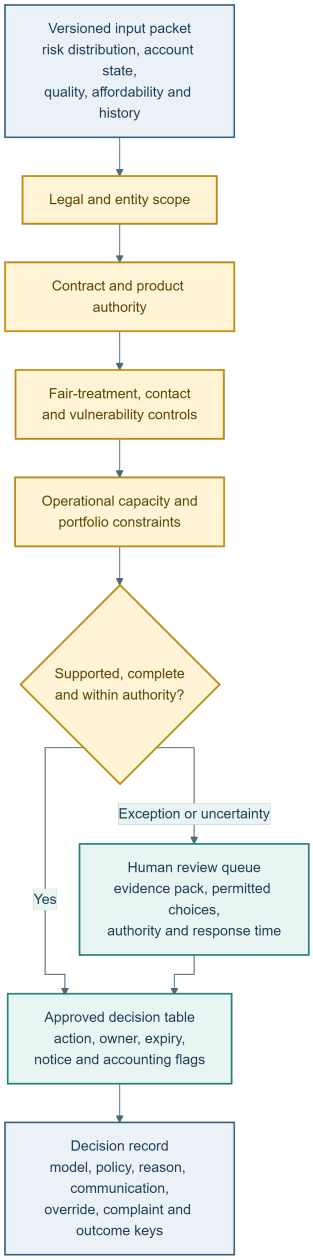


Figure 3: Credit Policy and Compliance Gate with a substantive human-review route. The gate tests entity scope, authority, conduct, affordability, capacity and portfolio constraints before an action is executed. The final record preserves the evidence, owner, reason, communication and future outcome key.

5. A Kenya-first control architecture

Regulatory orchestration begins by identifying the institution, activity, customer and decision in scope. A bank, a licensed digital credit provider, a credit reference bureau, a data controller and a contracted technology provider may participate in the same lifecycle while carrying different duties. The canonical control register therefore records five fields before translating a source into a system rule: the instrument and provision, regulated entity, covered activity, effective date, and operational control owner. Contractual covenants, internal credit policy and statistical monitoring thresholds sit in adjacent registers and retain their own labels.

For a bank licensed under the Banking Act, CBK/PG/22 supplies a direct supervisory foundation. It expects customers to be treated equitably, honestly and fairly throughout their relationship with the institution, calls for humane treatment in face-to-face and remote contact, and addresses prudent lending, understandable product information and complaints [1]. The board owns the framework; credit, servicing and collections teams express it in their controls. This supports an intervention architecture in which suitability, total cost, communication and recourse are checked when an action is designed and again when it is delivered.

The Digital Credit Providers Regulations apply to licensed DCPs while expressly excluding institutions licensed under the Banking Act and other listed categories from their application. Within their scope, they provide particularly concrete lifecycle requirements. Regulation 20 prohibits threats, shaming, unauthorised contact with a customer's contacts and other oppressive or abusive collection conduct. Regulations 21 and 22 address transaction records and complaints. Regulations 25 to 27 address fair and transparent information, reasonably required customer information, terms, charges and complaint channels [2]. A multi-party programme maps these controls to the DCP when one participates and uses them as a documented conduct benchmark only where another institution voluntarily adopts them.

Personal-data governance begins with the Data Protection Act. Its principles include lawful, fair and transparent processing, purpose limitation, minimisation, accuracy, retention discipline and appropriate safeguards. The architecture registers a lawful basis for each purpose rather than treating one consent screen as permanent permission for every later use. Collection servicing, credit assessment, model development, fairness audit and research publication are separate purposes with potentially different data fields, recipients, retention periods and legal analyses [3].

Automated decision-making receives a specific control path. Section 35 concerns a decision based solely on automated processing, including profiling, that produces legal effects or similarly significantly affects the data subject. Where an applicable exception supports such processing, the Act provides notification and a route to reconsideration or a new decision that is not based solely on automation. The General Regulations add operational safeguards, including meaningful information about the logic, significance and envisaged consequences, appropriate statistical procedures, error controls, measures addressing discriminatory effects and bias, and an ability to obtain human intervention and express a point of view [4].

The practical unit of analysis is the decision, not the presence of a model somewhere in the process. A daily portfolio-risk dashboard may inform a manager without deciding a customer's rights. A fully automated decline, material limit reduction or restructure decision can have a very different effect. The

decision inventory records whether processing is solely automated, the effect on the customer, the legal basis, the logic summary, notification, reconsideration path and human authority. A human reviewer must be able to examine the evidence, consider relevant customer information and change the result. Merely clicking an automatically selected outcome would not provide the substantive reconsideration contemplated by the control design.

High-risk processing is assessed before deployment. Under the General Regulations, DPIA triggers include automated decisions with legal or similarly significant effects, large-scale processing, sensitive data, combining datasets from different sources, systematic monitoring and innovative technologies in the stated circumstances [4]. The ODPC's guidance for digital credit providers translates these principles into sector-specific practices around purpose, transparency, data-subject rights, profiling, data sharing, retention, security and DPIAs [5]. For this architecture, the DPIA is a living design artefact linked to the feature register, decision inventory, processors, cross-border transfers, incidents and material changes.

Credit-information controls form another distinct route. The Banking (Credit Reference Bureau) Regulations provide customer rights to obtain and dispute credit information and to understand how a bureau score was arrived at. They also require an institution to issue an adverse-action notice when bureau information contributed to an adverse decision, identifying the bureau and the customer's access and dispute rights [6]. The notice rule is attached to the bureau-data condition in the decision log. A separate, institution-designed explanation standard covers material actions based on its own lifecycle model, ensuring that useful reasons remain available even when CRB information played no part.

The Consumer Protection Act supplies wider rules on consumer agreements, unfair practices and credit agreements where the transaction and borrower fall within its scope [7]. Because MSMEs range from sole traders to incorporated enterprises, the applicability field cannot be inferred from the label "MSME." Product counsel determines the covered relationship. The customer experience can still adopt a consistent higher standard: accurate balances, understandable terms, visible fees, freely chosen channels, documented modifications and meaningful recourse.

Complaints close the control loop. CBK's banking-sector mechanism describes documented channels, prompt acknowledgement, fair investigation, a reasoned outcome and escalation to the Central Bank where appropriate [8]. The architecture therefore treats a complaint as both an individual case requiring resolution and a portfolio signal. Repeated complaints about a reason code, contact channel, branch, model version or intervention reveal a possible design defect. Root-cause analysis feeds the policy owner, model-risk team, data-protection function and product committee while the individual remedy proceeds on its own timetable.

6. Explanation, fairness and vulnerability

An explanation should tell the customer what happened and give an authorised reviewer enough evidence to act. The architecture uses two linked layers. The **decision reason** names the material explicit facts and policy rule, such as a recent scheduled-to-actual payment gap, current facility utilisation and the affordability rule governing a proposed limit. The **model evidence record** gives validators the

feature values, nonlinear contributions, posterior uncertainty, residual AI-block contribution, support status and sensitivity checks. The customer receives plain language and available choices; the reviewer receives the governed evidence package.

Residual AI evidence requires special care. Part 2a strips from the neural representation the linear and nonlinear variation already represented by explicit features, while Part 2b shrinks the residual block through regularised horseshoe priors. An explanation can therefore state that an approved behavioural pattern contributed incremental evidence after the named explicit variables, then show whether the same decision survives without that block. The architecture does not invent a human motive for a latent coordinate. When the residual block becomes decisive and no stable customer-facing reason exists, the gate can route the case to review or restrict the block to monitoring until explanation and outcome evidence mature.

Fairness is evaluated as a system property. The NIST AI Risk Management Framework provides a useful voluntary comparator by placing validity, reliability, transparency, explainability, privacy and harmful-bias management within a continuous govern, map, measure and manage cycle [9]. For MSME credit, the measurement unit must also respect the enterprise form, product, sector, geography, channel and data availability. Audit attributes are processed under an approved lawful basis, separated from production decisioning where appropriate and never inferred from a location proxy merely to complete a metric.

For group g and score bin B_k, group-conditional calibration can be monitored through:

$$\Delta_{g,k}^{\mathrm{cal}} = \mathbb{E}[Y \mid \widehat{p} \in B_k, G = g] - \mathbb{E}[\widehat{p} \mid \widehat{p} \in B_k, G = g].$$

Conditional error gaps can complement calibration. Equal opportunity, for example, compares true-positive rates across groups for a carefully defined favourable outcome, while equalized odds also compares false-positive rates [10]. In collections, the outcome label must match the decision. Approval error, unnecessary limit restriction, failure to offer an eligible support action and escalation to intensive recovery are different events. A metric defined around default prediction cannot alone assess whether contact burden or restructure access was fair.

Calibration and error-rate parity can conflict when outcome prevalence differs across groups, even for informative risk scores [11]. The governance response is a declared measurement portfolio rather than selection of the most convenient statistic. The monthly fairness pack reports sample size and intervals for approval, price, limit, support eligibility, intervention receipt, contact frequency, complaints, durable cure, re-default, write-off and discounted net recovery. Intersectional slices are reviewed where data support them. Small groups receive uncertainty intervals and pooled monitoring rather than unstable league tables.

Vulnerability is handled through service design. A temporary illness, bereavement, business interruption, disability, low literacy or language need may alter the suitable channel and human support route. The system records only the minimum information required for an authorised purpose, places sensitive details behind restricted access and expires temporary accommodations according to policy. A vulnerability flag cannot become an unreviewed risk penalty. Contact policies specify permitted times,

frequency, channel preference, identity verification and escalation, while collectors see the accurate balance, prior promises, disputes and accommodations before engagement.

Fairness review also follows benefits. If a payment-alignment tool is effective, the institution asks who was eligible, who was offered it, who accepted it and who achieved durable cure. Unequal access to a beneficial treatment can matter as much as unequal exposure to an adverse decision. Complaint text and call-quality review provide qualitative evidence alongside rates. Customer interviews can reveal whether the intervention was understood, whether it preserved supplier or employee payments, and whether the revised schedule remained workable after the first successful instalment.

7. Learning which interventions work

Part 2b established the difference between prognosis and causal identification. Part 3 turns that distinction into an intervention protocol. Randomisation occurs only after the gate has removed actions that are unlawful, contractually unavailable, unaffordable, operationally unsupported or required as mandatory protection. The experiment compares two or more permissible alternatives about which the institution has genuine uncertainty. Customers retain the disclosures, service standards, reconsideration and complaint routes applicable to every arm.

Evidence from credit experiments explains why this discipline matters. A Kenyan randomised evaluation of performance-contingent finance for micro-distributors compared fixed debt with equity-like, hybrid and indexed structures, finding that contract design changed business and repayment outcomes [12]. In India, a two-month microfinance grace period increased business investment and long-run profits while also increasing default [13]. The results point in the same methodological direction: product flexibility can redistribute benefit and risk, so a lender should measure enterprise outcomes, repayment, cost and selection together.

The primary estimand for intervention a against comparator a_0 is defined in the eligible population:

$$\tau^{\mathrm{DC}}_{a,a_0}(H) = \mathbb{E}\left[I^{\mathrm{DC}}_H(a) - I^{\mathrm{DC}}_H(a_0) \mid S \in \mathcal{E}_{a,a_0}\right],$$

with companion estimands for re-default, discounted net recovery, business continuity, complaints and contact burden. Durable cure requires a defined arrears exit followed by sustained performance over the registered window. A payment made immediately after intensive contact is an intermediate response, not the complete outcome. The analysis includes take-up, non-compliance, censoring and interference where businesses share customers, suppliers, guarantors or local shocks.

The first controlled pilot uses a stable randomised policy with recorded assignment probabilities. Mandatory hardship or protection routes remain outside randomisation. Allocation can be stratified by product, baseline risk band, enterprise size and relevant cohort variables, with clusters used when treatment may spill across branches or relationship managers. Pre-specified stopping rules cover customer harm, complaints, arrears, model failure, operational incidents and imbalance. A data-monitoring group reviews blinded or appropriately partitioned results under a charter.

Adaptive policies belong to a later challenger stage. A contextual bandit can learn which available action performs best for a state, but only after actions, propensities, costs and delayed outcomes are logged reliably. For a deterministic evaluation policy π, a doubly robust estimate of value is:

$$\widehat{V}_{\mathrm{DR}}(\pi) = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{\mu}(X_i,\pi(X_i)) + \frac{\mathbf{1}\{A_i = \pi(X_i)\}}{\hat{e}(A_i \mid X_i)}\{Y_i - \widehat{\mu}(X_i,A_i)\}\right],$$

where $\hat{e}$ is the logged or estimated treatment propensity and $\widehat{\mu}$ the outcome model [14]. The estimator gains robustness when at least one nuisance component is adequately specified, while credible use still requires overlap, correct timing, consistent treatment definitions and stable logging. Propensity floors, bounded importance weights, time-forward evaluation and subgroup diagnostics protect against a policy whose apparent value depends on rare assignments.

When randomisation is unavailable, the design follows the assignment mechanism. A policy threshold may support regression discontinuity near the cutoff. Staggered branch adoption may support difference-in-differences if pre-trends and concurrent changes are credible. A capacity lottery can serve as an instrument when it affects the outcome through treatment and satisfies the other identifying conditions. Target-trial emulation can structure an observational comparison using explicit eligibility, time zero, strategies, follow-up and censoring. Each route identifies a narrower claim than an RCT, and the evidence label travels with the resulting treatment estimate into the gate.

8. Portfolio triggers and accountable management information

Individual actions accumulate into portfolio consequences. The management pack connects model, policy, treatment, cash and customer outcomes by product, vintage, sector, geography, institution and intervention cohort. It reports calibration, missingness, support-range exceptions, approval and limit movements, arrears entry, modification, cure, re-default, gross and net recovery, contact cost, complaints, human-review volume and override outcomes. Finance can reconcile these states to balances and cash; risk can reconcile them to exposure and loss; conduct teams can examine treatment and recourse.

A portfolio threshold is most useful when uncertainty accompanies it. For metric M_t and approved threshold c, the monitoring engine can report:

$$q_t = P(M_t > c \mid \mathcal{D}_t),$$

alongside the point estimate, interval, denominator and data maturity. Escalation may progress from analyst investigation to policy-owner review, temporary restriction, committee action and documented release. The action depends on cause and consequence. A data-quality failure can suspend a feature path while contractual servicing continues. Calibration deterioration can return uncertain cases to human review. A concentration breach can restrict new exposure. A statistically stable increase in durable re-default can prompt a product or intervention redesign.

Population Stability Index, feature drift and embedding drift remain screening measures. They trigger diagnosis of changes in population, data generation, product rules, economic conditions and outcomes. Release decisions use performance, calibration, economic value and customer impact rather than a standalone heuristic. Each trigger records the owner, evidence window, action, expiry, cure test and committee authority.

9. Conclusion

Continuous underwriting becomes an institutional capability when it connects early evidence to a lawful and accountable choice. The architecture in Part 3 gives that connection a formal shape. Prognostic distributions describe future states. Causal designs estimate the effects of permissible alternatives. The Credit Policy and Compliance Gate defines feasible actions. Named owners communicate and execute those actions. A treatment ledger then measures cash, durable cure, re-default, cost, complaints and business continuity.

This is how underwriting for collection moves from a slogan to a governed operating system. The lender can intervene earlier without converting every signal into restriction, learn from real outcomes without confusing treatment selection with model performance, and preserve a reviewable record from data cutoff to customer resolution. Part 4 will carry this decision contract into the enterprise technology, identity, event and audit architecture. Part 5 will translate its outcomes into pricing, expected loss, capital, liquidity and funding. Part 6 will turn the controls and estimands into a staged research and implementation programme with banks and ecosystem partners.

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Part 4

Bank-Controlled Research Sandboxes and Enterprise Credit Architecture

Underwrite for Collection Series

Abstract

Lifecycle underwriting becomes institutional when a bank can reproduce the evidence, model, policy and human authority behind a decision without releasing an unrestricted customer dataset or turning a research prototype into an ungoverned production service. Part 4 designs that institutional boundary. It establishes a bank-controlled research sandbox, a canonical event and identity fabric, purpose-bound model-to-data access, progressive evidence stages from synthetic development to shadow scoring, and an approved artefact lifecycle with deterministic serving, fallback and rollback. Privacy-enhancing methods are matched to explicit threats rather than treated as substitutes for governance. Experiment assignment, treatment receipt and matured outcomes enter the same lineage as payments, modifications, cures and recoveries. Explicit integration contracts then carry authorised results into origination, core banking, servicing, collections, finance, expected-credit-loss, capital and treasury processes while preserving their separate ownership. The resulting architecture makes rigorous research compatible with bank accountability, customer rights, independent validation, operational resilience and controlled publication.

1. Bring the method to the bank

A relationship manager visits a small food processor whose instalment has just failed. The enterprise is still producing, its retail orders remain visible and two invoices should settle within ten days. Its bank systems hold a more fragmented story. The core loan account records the missed amount. Payments hold incoming and outgoing cash. The relationship platform holds a call note. A collections application has opened a case. The credit bureau file may reflect a different reporting date. The model service has an earlier risk estimate, and the research team that designed a proposed lifecycle challenger has none of the identifiable records.

The conventional response is to ask for an anonymised extract and analyse it elsewhere. That request often compresses a difficult institutional problem into a file-transfer problem. Removing names does not necessarily remove the possibility of re-identification. An MSME's sector, location, facility size, transaction rhythm and uncommon sequence of events can together become identifying. A static extract also loses corrections, decision-time availability, policy versions and the operational actions that changed the later outcome. The file may be safer than a production database, yet less capable of answering the research question and harder to reconcile to the bank's books.

The stronger design brings approved research code to governed data. Customer records remain within a bank-controlled boundary. A purpose-specific analytical view exposes only the fields and history required for the registered study. Stable pseudonymous keys connect facilities, payments, interventions and outcomes, while the identity map remains in a separately controlled service. The researcher can test the pipeline, inspect approved summaries and submit reproducible jobs without acquiring a portable copy of the loan book. Independent validators can replay the same versioned job against the same point-in-time snapshot. Disclosure reviewers decide which aggregate tables, coefficients, diagnostics or model artefacts may leave the environment.

This arrangement is both an evidential and a rights architecture. Kenya's Data Protection Act requires lawful, fair and transparent processing, explicit and legitimate purposes, data minimisation, accuracy, storage limitation and appropriate safeguards. It also requires a data-protection impact assessment where processing is likely to create high risk, addresses significant solely automated decisions, embeds protection by design and permits research processing only with the applicable conditions and safeguards [1]. The Data Protection (General) Regulations make these duties operational through retention schedules, pseudonymisation, automated-decision controls, accuracy measures and DPIA triggers that include large-scale linkage, significant profiling and innovative processing in the specified circumstances [2].

The ODPC's 2025 research guidance is especially relevant to this programme. It treats research processing as a risk-based activity rather than a blanket exemption, calls for purpose and lawful-basis analysis, data minimisation, secure handling, data-sharing agreements where required and non-identifiable publication, and expressly identifies trusted research environments among available privacy-enhancing measures [3]. The practical question is therefore not whether a dataset has been called anonymous. It is whether the participating institutions can demonstrate why each item is processed, who can act on it, how access is controlled, how outputs are checked, when records are removed and how a customer's rights remain exercisable.

The bank-controlled research sandbox is the place where those controls, the Part 2a evidence contract, the Part 2b model artefact and the Part 3 intervention contract meet. It is an internal research and validation environment. It does not itself confer regulatory permission to launch a new product or customer treatment. A live pilot follows only after the licensed institution has completed the relevant legal, risk, conduct, model, technology and supervisory pathway.

2. The control boundary is an operating model

The word *sandbox* can suggest a disposable technical playground. Here it means a governed collection of environments with named owners, permitted data movements and release gates. The bank remains the data controller or otherwise records the applicable controller and processor roles. It owns the customer relationship, product decision, Credit Policy and Compliance Gate, production credentials and systems of record. A research institution, technology supplier or model developer may supply code, methods and analysis under an approved mandate. It does not acquire the bank's authority merely because its model produces a probability.

That retained authority has a concrete Kenyan banking implication. The CBK Guidance Note on Cybersecurity places responsibility for the institution's cyber strategy, policy and standards with its board and senior management and recognises risks arising through privileged access, interconnected systems and service providers [4]. The CBK Guideline on Outsourcing requires institutions to manage the risks of outsourced activities and identifies information-system management, loan-application processing, research and bad-debt collection among activities that may be material. It also reserves core management and lending decisions to the institution, including the decision to grant loans and determine compliance with applicable requirements [5]. A model-to-data arrangement must therefore preserve bank control over decision rights as carefully as it preserves custody of records.

The architecture contains six connected planes:

1. **Systems of record** hold legally and operationally authoritative customer, facility, payment, servicing, collection and accounting events.
2. **The evidence plane** reconstructs point-in-time features, outcomes and data-quality states from versioned source events.
3. **The research plane** provides synthetic, pseudonymised or approved aggregate views according to the study and user role.
4. **The model plane** trains and validates challengers offline, then registers an approved, signed and reproducible scoring artefact.
5. **The decision plane** combines model output with the versioned Credit Policy and Compliance Gate, human authority and customer communication.
6. **The record and learning plane** sends executed terms, realised cash, complaints, cures, re-defaults and recoveries to operational systems and future validation.

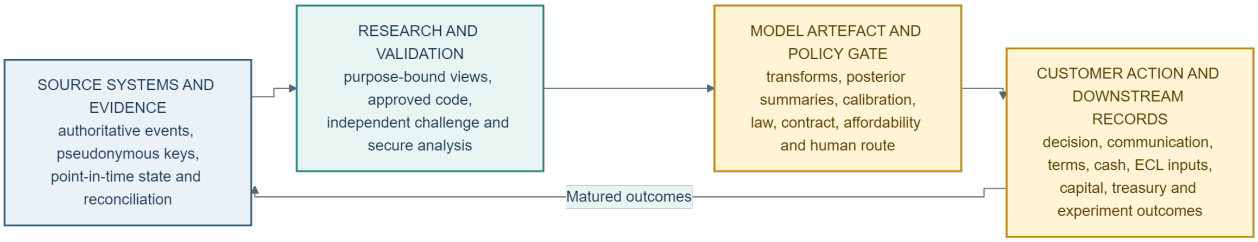


Figure 1: The bank-controlled lifecycle-credit architecture. The research team works through an approved analytical boundary. The licensed institution retains the source records, model release, policy gate, customer action and downstream financial record. Matured outcomes return to the evidence plane without rewriting the information available at the original decision.

The planes are logical boundaries, not a demand for six technology products. A smaller institution may implement several planes on one private-cloud platform with separate identities, storage policies and deployment accounts. A larger bank may use different systems and teams. The control requirement remains the same: every material data movement, transformation, approval and decision must have one accountable owner and a reproducible contract.

3. One event fabric, several accountable records

Part 2a defined why a payment is not one timestamp. Part 4 assigns that temporal truth to operating systems. A transaction has an economic time, an ingestion time, a time when the bank could first use it, and sometimes a correction time. A customer decision has a cutoff, a policy effective time and an execution time. If those clocks are compressed into one database update timestamp, a later correction can leak into a historical training row and an old score can appear more informed than it was.

Each accepted event therefore enters the evidence plane through a canonical envelope:

$$\mathcal{E}_k = \left(e_k,\ s_k,\ t_k^{\text{event}},\ t_k^{\text{known}}, t_k^{\text{effective}},\ v_k^{\text{schema}},\ v_k^{\text{definition}}, q_k,\ r_k,\ h_k\right),$$

where e_k is a unique event identifier, s_k the source system, the three times record occurrence, institutional availability and business effectiveness, v^{schema} and $v^{\text{definition}}$ identify the technical and business meaning, q_k carries quality and correction status, r_k identifies the superseded or related event, and h_k is a content digest used to detect unintended alteration. A reversal or correction creates a related event. It does not silently replace the historical record used by a prior decision.

Canonical identifiers connect customer, enterprise, facility, account, collateral or guarantee, payment, servicing case, intervention and outcome. Direct identifiers remain in the system that needs them. The evidence plane uses stable pseudonymous keys. This gives the research programme longitudinal continuity without making names, telephone numbers, national identifiers or free-text case notes ubiquitous analytical fields. The mapping service is separately permissioned, and its use is itself logged.

System ownership follows the event's meaning. The core lending or product subledger owns contractual principal, interest, charges and status. The payments system owns settlement events. The customer and relationship systems own verified communication and authorised profile changes. Collections owns case actions and promises, while the contract system owns executed modifications. Finance owns journals and reporting classifications. The research evidence plane does not become a competing book of record. It preserves source identifiers, derives registered analytical states and reconciles those states to their owners.

Record family	Authoritative owner	Analytical representation	Required control
Facility and contractual schedule	Product subledger or core lending system	Versioned cash-flow schedule and product state	Executed terms, effective date and amendment lineage
Payment and settlement	Payments and account ledger	Eligible cash event with availability time	Duplicate, reversal, value-date and settlement reconciliation

Record family	Authoritative owner	Analytical representation	Required control
Delinquency and collection	Servicing and collections systems	State transition, case and contact event	Approved DPD definition, channel, outcome and complaint link
Modification and restructure	Contract and credit workflow	New schedule and modification classification	Authority, affordability, customer acceptance and accounting flag
Model and policy decision	Scoring service and decision engine	Immutable decision packet	Evidence cutoff, artefact, policy, reasons, override and execution
Recovery and closure	Recovery subledger and finance	Gross cash, direct cost, net recovery and closure	Cash receipt, cost allocation, write-off and control-total reconciliation

Table 1: Proposed source-of-record and analytical-ownership matrix. A derived lifecycle state remains traceable to the authoritative events that support it. Ownership is assigned by economic and legal meaning, not by whichever platform happens to display the result.

Risk data must also be fit for aggregation. The Basel Committee's risk-data principles emphasise governance, common identifiers, metadata, accuracy, completeness, timeliness, reconciliation, authoritative sources and independent validation. The principles were initially directed at systemically important banks, but their architecture provides a useful proportionate benchmark for any institution attempting to connect model evidence to material risk decisions [6]. A small bank may implement fewer platforms; it does not need weaker definitions.

Before a model is trained, the evidence plane tests temporal validity and balance identities. For facility i over period t, a simplified control is:

$$B_{i,t}^{\text{close}} = B_{i,t}^{\text{open}} + D_{i,t} + I_{i,t} + F_{i,t} - P_{i,t} - C_{i,t} - W_{i,t} + A_{i,t},$$

where D is disbursement or draw, I accrued interest, F charges, P eligible payment, C credits or reversals in the customer's favour, W write-off and A other approved adjustments. Each term has a source, sign convention and availability time. Aggregate balances reconcile to portfolio and general-ledger control totals within approved tolerances. An unreconciled population can remain available for data-quality diagnosis, but it cannot quietly become the champion's training population.

At decision time, the scoring service writes a compact decision packet containing the evidence snapshot, feature and model versions, posterior summaries, support and quality states, policy version, available actions, selected route, human authority, customer communication and downstream keys. The packet's identity can be expressed as:

$$d_{i,t} = H\left(x_{i,t}^{\text{PIT}}, v_{\text{feature}}, v_{\text{model}}, v_{\text{cal}}, v_{\text{policy}}, a_{i,t}, o_{i,t}\right),$$

where H is a governed digest, $a_{i,t}$ the executed action and $o_{i,t}$ the authorised override record where one exists. The digest is an integrity control and lookup key. The underlying evidence, approvals and records remain the substantive proof of what happened.

4. Model-to-data access and the trusted research environment

The research workspace begins with a registered purpose, not a broad entitlement to explore. Its protocol names the question, population, variables, outcome horizon, legal basis, retention period, users, processors, publication plan and prohibited uses. The DPIA links that protocol to foreseeable harms, data-subject rights, automated-decision effects, re-identification risk, security, mitigations and residual-risk ownership. A material change in features, linkage, population, partner, infrastructure or customer effect reopens the assessment.

Access is the intersection of several permissions:

$$\mathcal{D}_{u,t}^{\text{allow}} = \mathcal{D}_{u}^{\text{role}} \cap \mathcal{D}_{p}^{\text{purpose}} \cap \mathcal{D}_{z}^{\text{environment}} \cap \mathcal{D}_{a}^{\text{approval}} \cap \mathcal{D}_{t}^{\text{time}}.$$

The user must have the right role, the dataset must serve the registered purpose, the environment must be approved for its sensitivity, the required authorisation must be active, and the credential must remain within its time window. Row and column policies can further constrain products, institutions, regions or outcome maturity. Service identities receive the same discipline as people. Production-scoring credentials cannot be reused in a notebook, and a research job cannot write into a customer decision table.

The architecture separates five zones. The production zone retains direct identity and transactional authority. The linkage zone creates and controls pseudonymous keys. The research zone contains approved variables and tools, normally without unrestricted internet access or removable storage. The validation zone can reproduce the work but remains organisationally independent of development. The release zone checks proposed outputs for small cells, unusual combinations, sensitive text, model leakage and contractual restrictions. Re-identification is available only to an authorised operational function for a defined customer or data-quality purpose, never as a routine research convenience.

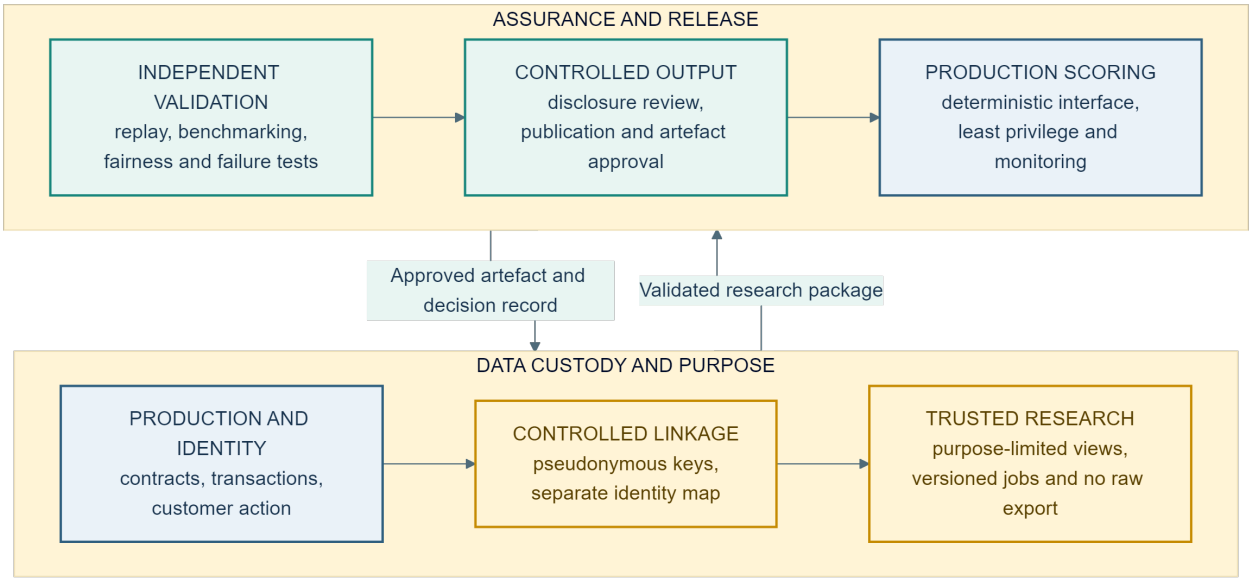


Figure 2: The model-to-data research boundary. Direct identity remains in production, longitudinal research uses controlled pseudonymous linkage, validation is independent, and only disclosure-checked results or approved artefacts leave the research path. Re-identification follows a named operational purpose and authority.

The controls are more than authentication. Workspace images and dependencies are versioned. Data imports are approved and scanned. Queries, jobs and output requests are logged. Secrets come from a managed service and are not written into notebooks. Privileged access is time-limited and reviewed. Backups follow the same sensitivity classification as primary storage. Retention covers source snapshots, derived tables, logs, model artefacts and temporary files rather than deleting only the convenient copy.

The research environment also preserves intellectual honesty. A public or synthetic dataset may be copied into development. A pseudonymised institution dataset remains controlled personal data where attribution remains possible. A table may leave the workspace only after disclosure review. A model artefact can itself reveal sensitive information and is therefore classified, tested and released deliberately. NIST's AI Risk Management Framework provides a useful voluntary lifecycle vocabulary for governing, mapping, measuring and managing these socio-technical risks [7]. The NIST Privacy Framework complements it by linking data processing to privacy risk and organisational outcomes [8]. Kenyan law and the bank's approved controls remain the governing authority.

5. Progress through evidence stages, not data shortcuts

The absence of a portable bank dataset does not stop development. It changes the order of proof. Public KBA, CBK, KNBS and other authoritative series can establish sector context and exercise acquisition controls. A synthetic portfolio can reproduce product schedules, selection, censoring, delayed outcomes, modifications, interventions, macroeconomic regimes and deliberate data faults. These materials are appropriate for testing schemas, code, estimands, power, recovery logic and failure handling. They do not establish the accuracy, fairness or commercial value of a model for Kenyan MSMEs.

Institutional evidence begins with a secure retrospective reconstruction. The bank maps definitions, rebuilds point-in-time snapshots, measures missingness, reconciles balances and tests the transparent benchmark before exposing an advanced challenger. Shadow scoring then runs the approved candidate prospectively without changing a customer's price, limit, treatment or collection path. It reveals latency, coverage, support excursions, calibration as outcomes mature, operational failure and disagreement with the existing process. A live pilot follows as a separate decision, with lawful treatments, eligibility, assignment, customer safeguards, stop rules and accountable owners. This progression prevents a technically successful notebook from being mistaken for a validated credit service.

Privacy-enhancing technology enters only where its protection matches the threat. A single-bank study will usually obtain more control and less complexity from a well-administered private workspace or secure enclave than from federated learning. Federated learning becomes relevant when several institutions seek a shared model while retaining local records. For institution j, a simplified objective is:

$$\min_{\theta} \sum_{j=1}^{J} \pi_j L_j(\theta), \qquad \pi_j \geq 0, \quad \sum_{j} \pi_j = 1,$$

with local losses L_j, agreed weights π_j and a shared parameter vector θ. The founding method shows how local updates can train a common model without centralising the underlying observations [9]. That does not make the updates harmless. Institutions can have materially different sectors, products, definitions and policies; an update can leak information, be poisoned or conceal weak performance in a small participant.

Secure aggregation can prevent the coordinator from seeing each institution's update in the clear while still recovering an authorised aggregate [10]. Differential privacy can bound how much a released statistic or trained artefact changes when one protected unit is added or removed, but only after the privacy unit, adjacency relation, clipping, mechanism and cumulative budget are defined. NIST's guidance also emphasises implementation hazards and the need to evaluate the complete guarantee, not merely the presence of noise [11]. A trusted execution environment can protect code and data during computation from parts of the surrounding infrastructure, while attestation, side-channel analysis, key management, patching and operator responsibilities remain necessary. These mechanisms complement purpose limitation, access control, validation and disclosure review. They do not replace them.

6. Promote a complete artefact and preserve a safe return path

The approved production object is more than a coefficient file. It contains the data contract, feature definitions, point-in-time transforms, spline bases, hierarchy maps, residualiser and encoder versions where used, calibration, posterior scoring quantities, uncertainty rules, support ranges, reason mappings, policy interface, dependencies, test evidence, effective date, expiry, fallback and rollback identifier. Its registry entry links the development dataset, code revision, environment lock, validation report, approvals and known limitations.

Registry states are explicit: development, validation, approved for shadow use, approved for bounded production, restricted, superseded and retired. Promotion requires independent evidence and named authority. A model may pass numerical tests and remain unsuitable because its population is too narrow, its explanations cannot support the action, its latency is unstable or its operational fallback has not been exercised.

Before release, the production implementation is compared with the independently validated reference. For a registered test set $\mathcal{T}$, the equivalence condition may be written:

$$\max_{i \in \mathcal{T}} \left| \hat{p}_i^{\text{prod}} - \hat{p}_i^{\text{ref}} \right| \leq \varepsilon_p, \qquad \max_{i \in \mathcal{T}} \left| \hat{u}_i^{\text{prod}} - \hat{u}_i^{\text{ref}} \right| \leq \varepsilon_u,$$

where $\hat{p}$ is the calibrated risk quantity, $\hat{u}$ an approved uncertainty output and the tolerances are justified before testing. Schema, missing-value, new-group, boundary, concurrency and degraded-dependency fixtures accompany the numerical comparison.

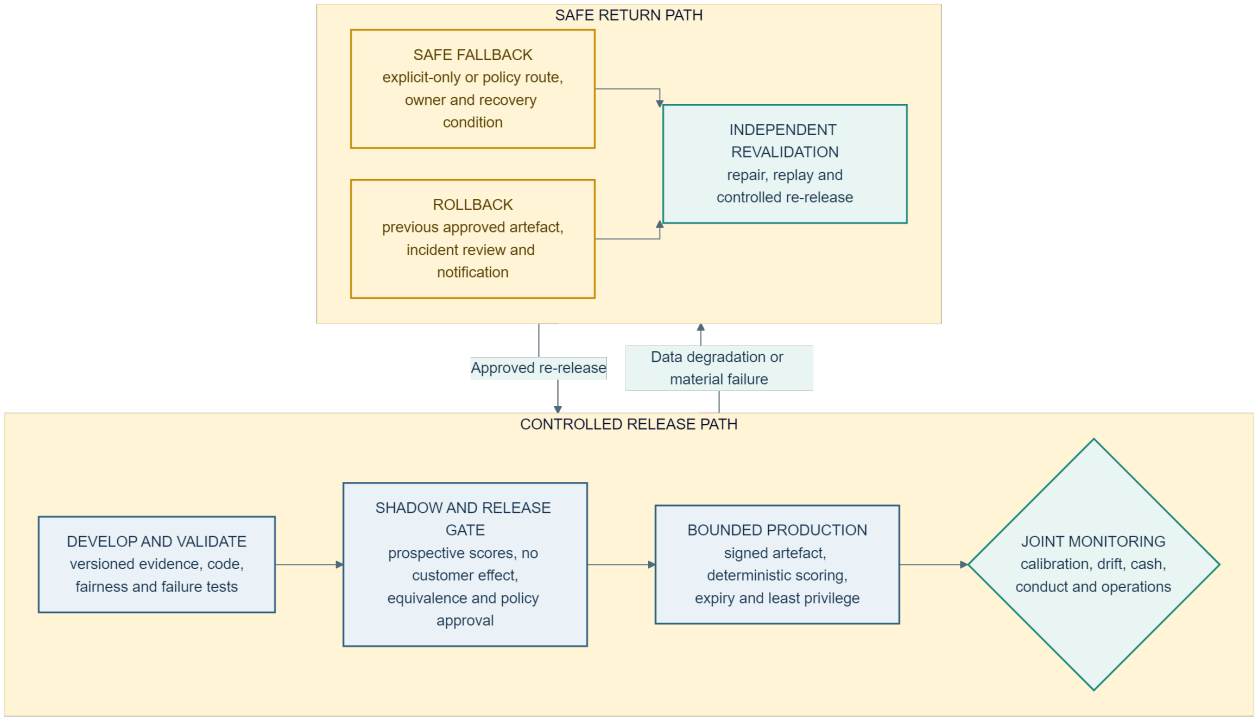


Figure 3: Model promotion, bounded production and safe return. Shadow use generates prospective operating evidence before customer effect. Production retains tested fallback and rollback paths, while material failures return the artefact to independent review rather than triggering an unrecorded repair.

Fallback is designed before failure. If the AI representation is unavailable, the approved explicit-only model may serve where validated. If point-in-time features are stale or incomplete, the policy gate can defer, restrict the available action or route the case to a person. If calibration or conduct monitoring breaches a release threshold, the previous approved artefact can be restored. Every route records affected accounts, start and end time, decision consequences, owner and cure evidence.

7. Experiment and enterprise records must join at the decision

A causal study cannot be reconstructed from a later list of customers who received a message. The enterprise record stores eligibility time, eligibility-rule version, assignment unit, arm, randomisation or propensity, treatment availability, delivery, receipt, adherence, crossover, contamination, human override and withdrawal. Outcomes retain their definitions and maturity windows. Payments, arrears, cure, re-default, modification, complaint, gross recovery, direct workout cost and net recovery then join through the decision-time keys established earlier.

This design supports intention-to-treat analysis from assignment and per-protocol or treatment-received analysis from execution, with their different assumptions stated. It also prevents a servicing action from becoming an accidental predictor of an outcome it helped create. A reminder sent after emerging arrears belongs to treatment history; the later payment is an outcome. Both may become valid time-varying history at a future decision cutoff, but neither may be moved backwards into the earlier evidence set.

Enterprise integration follows a fit-gap rule. For each calculation, the bank determines whether the existing platform can perform it natively and transparently, whether it belongs in a governed product or analytical service, or whether the platform should receive only the approved result and lineage. The HLR and its uncertainty remain in the model service. Policy permissions remain in the decision service. Contractual balances remain in the product subledger. The general ledger receives balanced accounting events. Part 5's ECL, capital and treasury processes consume approved data and results under their own methodologies rather than inheriting a credit score as an accounting conclusion.

An integration contract specifies event identity, producer, consumer, schema, units, currency, timestamp semantics, idempotency, correction method, service objective, authentication, failure behaviour and reconciliation owner. A successful API response is not proof that the economic event posted correctly. The executed action must reconcile from customer communication to product terms, cash settlement, subledger and control total. Manual adjustments remain possible under maker-checker authority, a reason code and subsequent reconciliation.

8. Validation, resilience and ecosystem accountability

Independent validation covers the whole decision service. It challenges the conceptual model, data lineage, point-in-time reconstruction, software implementation, policy interface, customer explanations, group outcomes, operational capacity and failure modes. The output-release process separately checks whether a proposed research table, narrative or artefact is authorised, statistically supportable and suf-

ficiently protected. This permits rigorous publication without turning a public paper into a map of customer records or security controls.

Third-party participation follows the same lifecycle. The bank assesses criticality, data access, subcontractors, concentration, security, service levels, audit rights, incident notification, business continuity, exit and portability. The FSB's third-party risk toolkit provides a useful international structure for identifying critical services and managing them throughout the relationship [12]. The Basel operational-resilience principles add governance, critical-operation mapping, tolerance for disruption, testing, dependency management and incident recovery [13]. These complement the CBK requirements already governing the participating bank.

The RACI is economical but explicit. The bank's business and credit functions own the product and decision. Data owners certify sources and quality. The model owner develops or sponsors the artefact. Independent model risk validates it. Compliance, legal, conduct and the data-protection function approve their respective controls. Technology owns the production service and resilience. Collections and servicing own executed interventions. Finance, impairment, capital and treasury own downstream recognition. Internal audit tests the control framework. Researchers own the registered analysis and reproducibility package, while publication requires bank and disclosure approval under the agreed protocol.

Service levels describe customer and risk consequences, not only uptime. They include event freshness, evidence completeness, scoring latency, fallback activation, human-review response, correction propagation, experiment-log completeness, reconciliation breaks, recovery objectives and incident escalation. A degraded service should fail into a known customer-safe and risk-approved state. Its recovery test must demonstrate that delayed events, duplicate actions and conflicting versions cannot silently enter the lifecycle record.

The phrase *regulatory sandbox* remains reserved for a regulator-operated testing regime with its own admission and safeguards. Kenya's Capital Markets Authority, for example, publishes a Regulatory Sandbox Policy Guidance Note for innovative capital-markets products [14]. That framework does not establish a CBK banking pathway for this programme. The proposed environment is therefore a bank-controlled research sandbox followed, where required, by a regulator-engaged controlled pilot under a pathway agreed in writing with the relevant authority.

9. Conclusion

Part 4 turns lifecycle underwriting from a compelling method into an accountable banking capability. The source records remain authoritative, research moves to governed data, the champion becomes a complete approved artefact, policy and human authority remain visible, and every intervention returns as a reproducible cash and customer outcome. Synthetic development, retrospective reconstruction, shadow scoring and bounded live use form successive standards of evidence. Privacy, cybersecurity, model risk and operational resilience are designed into those stages rather than added after deployment.

This architecture gives Part 5 a dependable financial interface. Contractual and behavioural cash flows, modifications, cures, defaults, recoveries, treatment costs and uncertainty can enter pricing, ECL, capital, liquidity and funding analysis with their definitions, timestamps and owners intact. The bank can then ask not only whether the model predicted well, but whether the complete system produced more sustainable and collectible credit relationships.

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Part 5

KESONIA Pricing, Expected Credit Loss, Capital and Funding

Underwrite for Collection Series

Abstract

A credit facility enters an MSME as working capital and returns to a lender as a sequence of uncertain cash flows. Between those moments, the applicable benchmark may change, invoices may arrive late, a repayment may fail and cure, an intervention may alter behaviour, a guarantee claim may settle, or a recovery may take longer than its ultimate value suggests. Part 5 develops a financial architecture for that uncertainty. It begins with compounded KESONIA and the governed customer premium, then carries contractual and behavioural cash flows into IFRS 9 expected credit loss, internal economic capital, liquidity, asset-liability management, funds-transfer pricing, intervention economics and credit-risk transfer. The same lifecycle evidence supports each use, while horizon, probability measure, discount rule, recognition criteria and accountable owner remain distinct. Guarantees and optional ring-fenced receivables funding enter through contractually defined cash rather than assumed protection. The result is a joined control plane in which transparent pricing and sustainable collection reinforce each other: customer terms can be traced to approved components, accounting loss to probability-weighted cash shortfalls, capital to adverse portfolio outcomes, treasury to the timing of liquidity, and risk transfer to enforceable coverage and settled claims.

1. A benchmark reaches the enterprise through a contract

A wholesale distributor receives a variable-rate working-capital facility shortly before its busiest trading season. The enterprise does not experience the benchmark as an abstract monetary-policy signal. It experiences a quoted rate, a repayment amount, a reset date and the amount of cash that remains after suppliers, wages, tax and debt service. If the facility is repaid from daily sales, a small change in interest may be absorbed. If most cash arrives through two invoices near month-end, the same economic rate can produce a more difficult payment path. The benchmark starts the price; the contract and the enterprise's cash conversion cycle determine how that price is lived.

The Kenya Shilling Overnight Interbank Average, or KESONIA, is administered and published by the Central Bank of Kenya. It reflects the volume-weighted average rate on unsecured overnight Kenya shilling interbank transactions. CBK began publishing KESONIA and a KESONIA Compounded Index on 1 September 2025 and publishes the overnight observation for the previous business day. On weekends and bank holidays, the last working-day observation is held constant for the relevant calendar interval [1].

That daily rate is a reference observation. A loan accrual still requires a complete convention. The facility agreement or incorporated product terms must identify the authoritative source, value date, publication time, accrual start and end dates, day-count basis, observation shift or lookback where used, payment delay, rounding, floor, cap, reset frequency, fallback, correction policy and treatment of non-business days. Two facilities can both refer to compounded KESONIA while producing different payment timing because their observation windows or payment conventions differ. Those differences belong in the product record introduced in Part 4.

For an accrual period from a to b, let r_i be a published KESONIA observation expressed as a decimal, let n_i be the calendar days for which that observation applies under the contract, and let $d = \sum_i n_i$. A transparent compounded factor and annualised rate are:

$$F_{a,b} = \prod_{i=1}^{d_b} \left(1 + \frac{r_i n_i}{365}\right), \qquad R_{a,b}^{\text{KESONIA}} = \left(F_{a,b} - 1\right) \frac{365}{d}.$$

The amount of benchmark interest on principal (B) is (B(F_{a,b}-1)) before the contractual premium and other permitted components. A production engine should calculate from the official published observations or the official compounded index where the chosen convention permits it. An independently reconstructed factor is a control calculation. It is reconciled to the authorised source, calendar and contract before posting.

This calculation carries several operational controls. A received observation has publication date, value date, source digest and validation status. A later source correction creates a versioned event and follows the contractual correction window. Missing publication does not invite an invented rate. CBK's KESONIA materials identify CBR as the alternative reference where KESONIA is impractical and as a fallback when KESONIA data are unavailable, while contracts are expected to address the contingency [1]. The product owner, legal function and treasury function therefore agree the fallback before a customer is charged.

The revised Risk-Based Credit Pricing Model expresses the total lending rate for the relevant variable-rate facility as:

$$r_{i,t}^{\text{lend}} = R_{a,b,t}^{\text{KESONIA}} + K_{\text{RBCP},i,t},$$

and the total cost of credit includes the applicable disclosed fees and charges in addition to that lending rate [2]. CBK's implementation material records the revised model as fully operational from March 2026 and identifies KESONIA compounded in arrears as the common reference for covered variable-rate lending, with CBR available as the alternative reference rate in the stated circumstances [3]. Fixed-rate loans and foreign-currency loans require their own pricing basis.

The formula creates transparency by separating a common market reference from the bank's premium. It does not make every customer's instalment change daily. The contractual accrual and reset rules decide when accumulated benchmark observations affect an amount due. A daily ingestion service may update accrued interest each business day, while customer communication and collection follow the agreed billing cycle. This distinction lets the bank reconcile monetary transmission, contractual entitlement, accounting accrual and customer affordability without forcing them into one timestamp.

2. The premium is governed across the credit lifecycle

CBK describes the premium K as containing lending-related costs, the return to shareholders and the borrower's risk profile, together with other relevant lending costs under the framework [2], [3]. This

paper writes it as K_{RBCP} so that it cannot be confused with regulatory capital K_{reg}. It is a published bank-specific or product-specific commercial premium under an approved methodology. The posterior default probability from Part 2b is one possible input to its borrower-risk analysis. It is not the premium by itself.

A bank can govern the premium through a component bridge such as:

$$K_{\text{RBCP},g,t} = c_{\text{operate},g,t} + c_{\text{service},g,t} + k_{\text{borrower},g,t} + c_{\text{capital},g,t} + \pi_{g,t} + \varepsilon_{g,t},$$

where g is an approved product or customer segment, the cost terms use documented allocation rules, k_{borrower} is the governed borrower-risk contribution, c_{capital} is the institution's pricing charge for allocated capital, π is the approved shareholder-return component and ε reconciles other permitted components. This is an internal management decomposition, not a replacement for CBK's definition or the bank's published methodology. Every component needs one owner, an as-of date, an allocation basis and a reconciliation to the approved premium.

2.1 Reconcile the customer premium to the Treasury transfer price

The customer rate and the internal transfer price answer related but different questions. The customer rate records the price agreed under the product, contract and approved RBCPM methodology. The transfer price attributes the economic value and cost of funding to the business that originates or holds the asset. A joined pricing architecture therefore preserves both quantities and reconciles them through a visible bridge.

Suppose the contractual benchmark for product group g during period t is $R_{a,b,t}^{\text{KESONIA}}$, and Treasury assigns the group an approved matched transfer rate $FTP_{g,t}$. Define the Treasury increment over the customer benchmark as:

$$\Delta FTP_{g,t} = FTP_{g,t} - R_{a,b,t}^{\text{KESONIA}}.$$

The increment can be positive or negative. It reflects the relationship between the contractual benchmark and the institution's approved transfer-pricing curve for the asset's behavioural maturity, repricing frequency, liquidity use and embedded options. Its interpretation depends on the FTP policy. A transfer curve based on marginal funding cost answers a different management question from one based on average funding cost, and a matched-maturity method produces a different attribution from a single pooled rate. The selected method should remain consistent across comparable products and should be accompanied by a reconciliation showing how the curve was constructed and applied.

For internal product analysis, the premium can then be viewed through a more granular reconciliation:

$$\widehat{K}_{\text{bridge},g,t} = \Delta FTP_{g,t} + EL_{g,t}^{\text{price}} + c_{\text{operate},g,t} + c_{\text{service},g,t} + c_{\text{capital},g,t} + \pi_{g,t} + \varepsilon_{g,t}^{\text{other}},$$

subject to:

$$\widehat{K}_{\text{bridge},g,t} = K_{\text{RBCP},g,t} \quad \text{after approved reconciliation adjustments.}$$

This bridge is an analytical control beneath the formal CBK expression, not an alternative customer-pricing rule. The external rate continues to follow KESONIA plus the approved premium [2], [3]. Internally, the bridge shows which part of the increment over KESONIA is associated with matched funding and liquidity, which part reflects expected credit loss, which part meets operating and servicing cost, which part remunerates allocated capital, and which part represents the approved commercial return.

The expected-loss term in the granular bridge is the explicit allocation of the borrower-risk contribution already represented by k_{borrower} in the higher-level decomposition. A bank may retain the broader borrower-risk term where its approved methodology combines expected loss with other calibrated risk adjustments. It may instead display expected loss and each additional risk adjustment separately. Whichever presentation is selected, the bridge should count each economic component once. IFRS 9 allowance, regulatory capital and internal economic-capital charge retain their separate meanings even when related information contributes to product economics.

The corresponding risk-adjusted contribution before tax can be displayed as:

$$M_{g,t}^{\text{risk-adjusted}} = r_{g,t}^{\text{lend}} - FTP_{g,t} - EL_{g,t}^{\text{price}} - c_{\text{operate},g,t} - c_{\text{service},g,t} - c_{\text{capital},g,t},$$

with fee income and fee-related cost presented in separate lines where the product includes them. Substituting $r_{g,t}^{\text{lend}} = R_{a,b,t}^{\text{KESONIA}} + K_{\text{RBCP},g,t}$ makes the control relationship visible:

$$M_{g,t}^{\text{risk-adjusted}} = K_{\text{RBCP},g,t} - \Delta FTP_{g,t} - EL_{g,t}^{\text{price}} - c_{\text{operate},g,t} - c_{\text{service},g,t} - c_{\text{capital},g,t}.$$

The formula gives Product, Finance, Credit Risk and Treasury a shared reconciliation without merging their mandates. Treasury owns the transfer curve, liquidity and optionality attribution. Credit Risk owns the calibrated loss input and its limitations. Finance owns accounting recognition and the relationship between management economics and reported results. Capital Management owns the approved allocation and cost-of-capital method. Product owns the customer proposition, while Legal and Compliance confirm that the terms, disclosures and repricing permissions match the approved methodology and contract.

Pricing or funding element	Primary question	Accountable owner	Required control
Compounded KESONIA	What benchmark applies under the contract?	Product, Legal and Treasury	Authoritative source, observation window, calendar, fallback and correction history
Treasury transfer rate	What funding and balance-sheet value is attributed to this asset?	Treasury	Approved curve, behavioural maturity, repricing convention and effective date

Pricing or funding element	Primary question	Accountable owner	Required control
Treasury increment over KESONIA	How does internal matched funding differ from the customer benchmark?	Treasury and Product Finance	Reconciliation of curve, liquidity, basis and option effects
Expected-loss pricing input	What central credit loss is attributed to the product horizon?	Credit Risk	Compatible PD, EAD, LGD, recovery timing, scenario and exposure basis
Operating and servicing cost	What resources are consumed from origination through collection?	Finance and Operations	Activity-based allocation, volume basis, treatment cost and periodic back-testing
Capital charge	What return is required for the capital allocated to adverse loss?	Capital Management	Capital measure, allocation method, horizon, confidence standard and cost of capital
Commercial return	What approved residual return remains after the recognised costs?	Product and Finance	Hurdle, portfolio strategy, competition, affordability and governance approval
Fees and charges	What separately disclosed amounts apply to the customer?	Product, Legal and Finance	Contractual entitlement, disclosure, tax treatment and total-cost reconciliation

The bridge is established at a governed pricing date. A new offer uses the benchmark convention, transfer curve, risk estimates, cost allocation and capital charge approved for that date. Contractual benchmark resets then follow the facility terms. Treasury may update its internal view of funding cost and behavioural maturity for management reporting without silently changing the customer's contractual premium. Where the contract permits repricing, the customer-facing change follows the documented product process, disclosures, notice period and approval route.

Price also changes behaviour. A wider premium can reduce take-up, change utilisation, encourage early repayment by stronger borrowers, increase payment pressure on marginal borrowers, or alter refinancing demand. Those responses affect the size, composition and timing of the asset book that Treasury must fund. Product planning should therefore connect price bands to acceptance, utilisation, prepayment, first-payment default, cure and early closure. Part 6 can estimate those relationships prospectively, while Part 5 carries the resulting cash-flow distributions into margin and liquidity analysis.

Randomised evidence from a South African consumer lender found downward-sloping credit demand and a stronger loan-size response to maturity than to price, consistent with binding liquidity constraints [4]. Its contribution here is experimental design, while Kenyan elasticity remains a quantity to be estimated from the relevant population. A pricing experiment should vary and measure rate, tenor and repayment burden together, then follow acceptance, utilisation, arrears and realised contribution. Treasury can therefore distinguish a rate that improves unit margin from a rate or maturity combination that changes the funded portfolio, customer selection and cash-flow timing.

The monthly pricing and transfer-pricing control pack should reconcile at least five views: the customer contract, the product subledger, the Treasury transfer record, the profitability ledger and the general ledger. Differences are classified by benchmark timing, curve version, behavioural assumption, modification, fee treatment, accounting basis or operational error. Material unexplained differences move

through an assigned investigation and approval workflow. This turns the pricing bridge into an operating control rather than a formula displayed only at product approval.

The borrower-risk component can use expected loss, transition risk, recovery cost and concentration evidence. Its units must remain consistent. A one-year PD cannot be multiplied by a lifetime LGD and a point-in-time balance and then compared directly with an annual interest margin. For a facility i, product horizon h and physical probability measure $\mathbb{P}$, a simple expected-loss rate is:

$$EL_i(h) = \frac{\mathbb{E}_{\mathbb{P}}\left[L_i(h) \mid \mathcal{I}_{i,t}\right]}{E_i(h)},$$

where $L_i(h)$ is the defined cash loss over the horizon, $\mathcal{I}_{i,t}$ is the evidence available at pricing time and $E_i(h)$ is a compatible exposure or average-balance denominator. A richer calculation follows monthly default, cure, drawdown, recovery and expense cash flows. Either approach produces a pricing input after calibration and governance. It does not determine the whole K_{RBCP}.

Lifecycle underwriting improves that input by distinguishing customers whose origination profile looks similar but whose later cash conversion and repayment evidence diverge. The explicit features, residual AI representation and hierarchical effects described in Part 2b can update the distribution of future cash loss. Part 3 then chooses from authorised actions. A rising posterior risk estimate may prompt a review, an affordable payment-date alignment, a limit change or a collections intervention. The existing facility's customer premium changes only when the contract and approved policy permit repricing or when new terms are lawfully agreed.

Peer-reviewed transaction-level evidence from a Chinese fintech found that machine-learning models using non-traditional information improved loss and default prediction relative to traditional comparators, particularly under the studied aggregate credit shock, while the comparative advantage narrowed as customer credit history lengthened [5]. The institutional setting and data differ from Kenya. The study therefore strengthens the manuscript's regime-aware challenger design: measure the residual contribution of AI by relationship depth, portfolio segment and stress period, and retain it where local out-of-time cash-loss forecasts improve after calibration, stability and governance tests.

This creates four different commercial moments:

1. **New-offer pricing** applies the current benchmark, published premium methodology, product economics and customer assessment to a proposed facility.
2. **Contractual repricing** follows the benchmark and reset mechanics already agreed for an existing variable-rate facility.
3. **Limit or eligibility management** governs additional exposure, renewals and new draws without retrospectively rewriting accrued terms.
4. **Modification** changes contractual cash flows through an authorised process with affordability, conduct and accounting consequences.

The Credit Policy and Compliance Gate owns the permissible route. It receives posterior risk, uncertainty, evidence quality, affordability, product exposure and portfolio constraints, then combines them with law, contract and institutional policy. This keeps the role of AI economically useful and opera-

tionally bounded. The model improves the estimate of future outcomes; the institution decides which terms and actions are permitted.

Pricing governance also asks whether the premium remains sustainable for the enterprise. A higher charge may cover a greater expected loss on paper while increasing payment stress, adverse selection or early exit. The bank should therefore test acceptance, utilisation, first-payment default, cure, net recovery, complaints and repeat borrowing by price band. Part 6 will treat those outcomes as evidence to be measured rather than assuming that a wider margin always improves risk-adjusted return.

3. One lifecycle distribution, several financial measures

Once a facility has been disbursed, its financial meaning is carried by cash and time. A scheduled instalment is a contractual claim. A settled payment is liquidity. A missed payment is a state transition whose eventual loss depends on cure, modification, recovery and cost. A guarantee is a contingent cash flow whose value depends on coverage, documentation, claim acceptance and settlement timing. A model that stops at default probability leaves most of that economic path unrepresented.

For facility i, month m and scenario s, define net lender cash as:

$$CF_{i,m}^{(s)} = C_{i,m}^{P,(s)} + C_{i,m}^{I,(s)} + C_{i,m}^{F,(s)} + R_{i,m}^{(s)} + G_{i,m}^{(s)} - O_{i,m}^{\text{serv},(s)} - O_{i,m}^{\text{work},(s)} - T_{i,m}^{(s)},$$

where the three C terms are collected principal, interest and fee cash; R is recovery net of amounts already represented elsewhere; G is an eligible guarantee or insurance settlement; O^{serv} is servicing cost; O^{work} is direct workout cost; and T is the cost of the intervention assigned to that account. Refunds, collateral proceeds and recoveries are separate event families until their legal owner, gross amount, cost and allocation have been reconciled.

The HLR supplies conditional transition probabilities and uncertainty under the physical measure $\mathbb{P}$. A timing challenger may refine when default, cure or recovery occurs. The product subledger supplies contractual schedules, outstanding balances and draw availability. The causal layer supplies an estimated intervention effect where the assignment design supports one. A cash-flow engine combines these objects without changing their provenance. It can then produce a distribution of principal, interest, expense, shortfall and recovery by month, product, sector, geography and institution.

That distribution has several destinations. Customer pricing asks which approved terms are sustainable at origination or renewal. IFRS 9 asks for probability-weighted discounted cash shortfalls at the reporting date. Economic capital asks how much adverse loss the institution chooses to withstand over a defined horizon and confidence standard. Treasury asks when cash arrives, when funding reprices and which gaps remain under stress. Risk transfer asks which losses are contractually covered and when a claim becomes cash. An investor valuation may ask for market-consistent discounting or a required return. Each use starts from common events but applies a different lens.

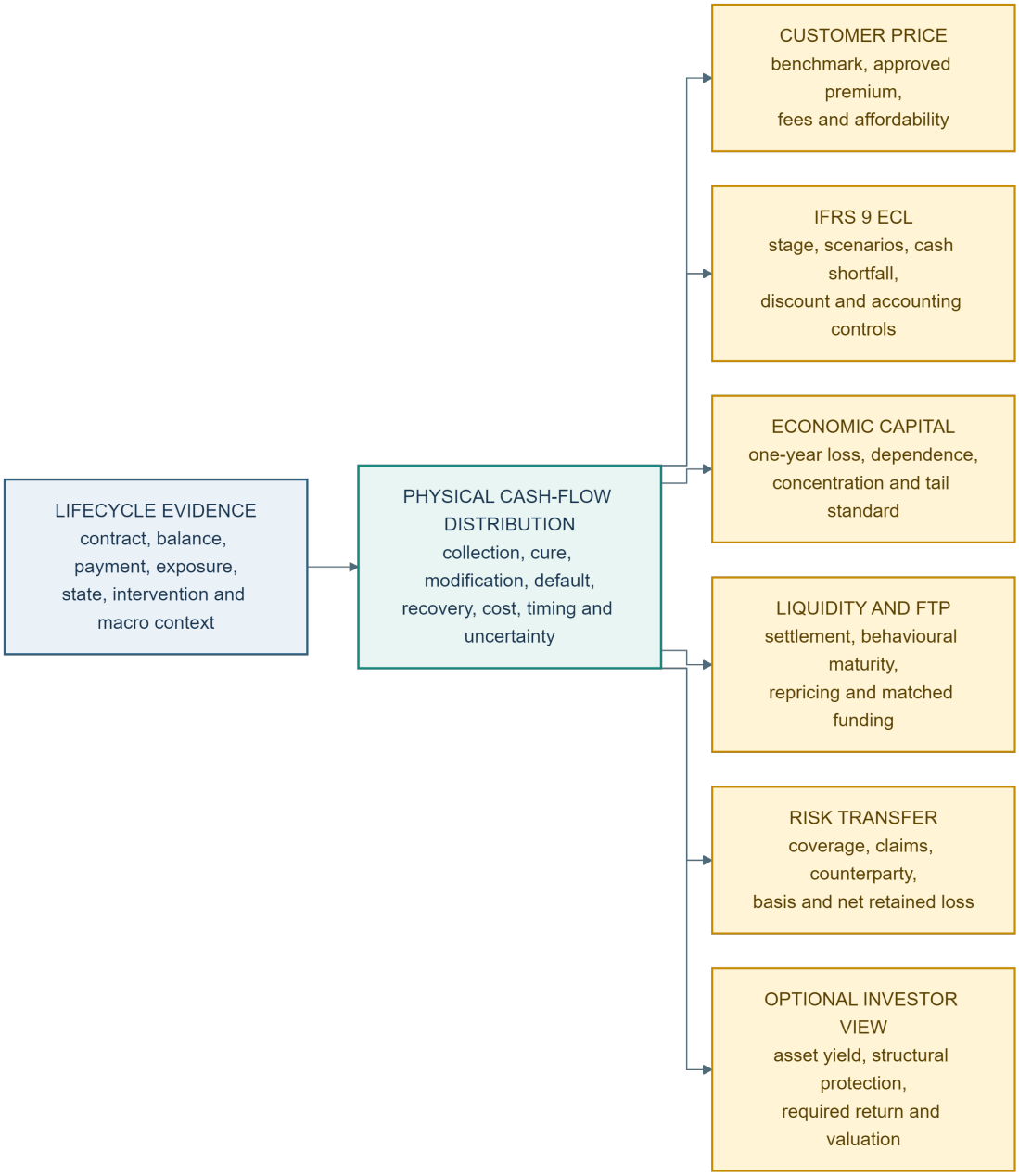


Figure 1: From lifecycle evidence to distinct financial measures. One physical cash-flow distribution preserves the relationships among prediction, collection and loss. Each destination then applies its own horizon, recognition rule, discounting, governance and accountable owner.

The probability measure matters most where valuation departs from forecasting. Under $\mathbb{P}$, default and recovery estimates describe outcomes expected in the operating world represented by observed data and approved scenarios. A market-consistent measure $\mathbb{Q}$ is useful where an institution values a tradable or contingent claim against sufficiently informative market prices and a documented calibration method. Adding a spread to KESONIA does not perform that transformation. In a market with limited comparable MSME credit instruments, the investor hurdle may be a transparent required return with sensitivity ranges rather than an asserted risk-neutral intensity.

Keeping the measures distinct supports innovation. The bank may use the full posterior distribution to stress future cash, a simpler approved quantity for a customer offer, a probability-weighted shortfall for impairment and a separately calibrated tail distribution for internal capital. Consistency means that the same borrower cannot be assumed current in one engine and defaulted at an earlier date in another without a documented reason. It does not require every function to use one number.

4. IFRS 9 follows probability-weighted cash shortfalls

IFRS 9 measures expected credit losses as probability-weighted credit losses that incorporate the time value of money and reasonable and supportable information available without undue cost or effort about past events, current conditions and forecasts of future economic conditions. The framework is principles-based and permits methods that satisfy those requirements [6]. The lifecycle architecture contributes event history, calibrated transitions and cash-flow timing. The reporting entity's accounting policy determines how those inputs become an allowance.

Let $CS_{i,m}^{(s)}$ be the cash shortfall in month m under forward-looking scenario s, measured as the contractual cash due to the reporting entity less the cash it expects to receive, including cash flows from eligible collateral and contractual credit enhancements where the accounting requirements include them. With scenario weight π_s, original effective interest rate EIR_i and time fraction τ_m, a cash-shortfall expression is:

$$ECL_i = \sum_{s=1}^{S} \pi_s \sum_{m \in \mathcal{H}_i} \frac{CS_{i,m}^{(s)}}{(1 + EIR_i)^{\tau_m}}, \qquad \sum_{s=1}^{S} \pi_s = 1.$$

The horizon $\mathcal{H}_i$ depends on the applicable impairment stage and instrument requirements. Twelve-month ECL is the portion of lifetime cash shortfalls associated with defaults possible within the next twelve months. It is not limited to cash that happens to be missed during the next twelve months. Lifetime ECL reflects default risk across the expected life. The distinction is important for short MSME facilities, revolving lines and facilities whose behavioural use extends beyond one contractual billing interval [6].

The model-to-ECL bridge therefore contains more than a PD field. For each reporting date it stores the instrument population, stage, default definition, scenario-conditioned marginal or conditional PD term structure, EAD path, LGD or expected cash-shortfall path, prepayment, cure, recovery timing, directly attributable recovery assumptions, contractual credit enhancements, original EIR, currency,

source versions and overlay. The bank reconciles opening allowance, new originations, repayments, transfers between stages, changes in risk, modifications, write-offs, recoveries and closing allowance.

Significant increase in credit risk, or SICR, is an accounting assessment of the change in lifetime default risk since initial recognition. A lifecycle score may provide relevant evidence through payment status, cash-flow deterioration, utilisation, adverse restructuring, sector stress or a material increase in calibrated lifetime risk. The approved policy combines those signals with qualitative indicators, backstops and reasonable forward-looking information. It also distinguishes a temporary data outage from economic deterioration.

Operational cure and accounting improvement are related but separately governed. A customer can clear an arrear and become operationally current today. The bank may require a sustained performance period, supporting evidence and reconsideration of forward-looking risk before concluding that credit risk has improved enough for an accounting stage transfer. The durable-cure definition from Part 1 provides evidence for that assessment without becoming a universal accounting rule. The same separation applies to re-default: a renewed delinquency after cure updates the observed transition history and tests whether the cure policy was economically durable.

Modification is another cash-flow event rather than a label for failure. An affordable term extension may improve collection and reduce ultimate loss while changing the present value of the asset. Where a modification of a financial asset does not result in derecognition, IFRS 9 requires the gross carrying amount to be recalculated as the present value of modified contractual cash flows discounted at the original EIR, with the applicable modification gain or loss recognised [6]. The contract workflow records the authorised new schedule, customer acceptance, purpose, effective date and accounting classification. The ECL engine then measures the remaining credit risk under the approved policy.

Write-off records the reduction of gross carrying amount when there is no reasonable expectation of recovery for all or part of the asset under the applicable requirements. It does not end the need to record later recovery cash, legal status or customer treatment. Recovery after write-off remains a distinct event with receipt date, gross amount, direct cost and accounting destination. This is why Part 4 required collection and finance records to remain joined after the credit decision.

The financial-stability consequence of forward-looking impairment belongs in the same control pack without changing the accounting measure. The European Systemic Risk Board found that expected-loss recognition can concentrate effects on profit and regulatory capital near the beginning of a deteriorating phase, depending on migration and scenario dynamics [7]. Used as comparative financial-stability evidence, the finding supports coherent but separately reported ECL, capital and liquidity stresses. An earlier allowance signal can then inform funding and capital planning while retaining its accounting identity rather than being relabelled as a cash outflow or regulatory-capital requirement.

The BCBS guidance on credit-risk practices for expected credit losses emphasises robust methodologies, experienced credit judgement, common processes, data and validation, while recognising that accounting frameworks determine the allowance [8]. The bank should therefore validate ECL components at their proper levels: discrimination and calibration for risk estimates, transition and cure accuracy, recovery amount and timing, scenario sensitivity, stage allocation, overlay governance and total allow-

ance back-testing. A good account-level score can coexist with a poor ECL if timing, exposure, recovery or scenario assumptions are weak.

The accounting owner also controls the relationship between model output and journals. A production HLR artefact can send approved PD term structures, uncertainty, evidence quality and lineage into the impairment platform. It does not post its own allowance. Finance and credit-risk functions approve the scenario weights, stage logic, LGD, EAD, recovery assumptions and overlays, and the general ledger receives a balanced journal with reconciliation keys. This preserves the value of continuous evidence while keeping financial reporting reproducible.

5. Economic capital carries the adverse portfolio story

Expected loss describes the centre of a defined loss distribution. The accounting allowance applies IFRS 9 recognition and measurement to expected cash shortfalls. Regulatory capital follows the rules in the applicable CBK prudential framework. Internal economic capital is the institution's assessment of capital needed to absorb adverse loss at a stated horizon, confidence standard and solvency objective. The four quantities can use related data, but they have different mandates and cannot be substituted for one another.

The CBK Prudential Guidelines set the local capital-adequacy and asset-classification framework for regulated institutions [9]. The bank maps each exposure, mitigant and calculation to the currently applicable requirements and supervisory treatment. A better behavioural model may improve risk management before it changes any regulatory parameter. Recognition follows validation, governance and the method permitted for that institution.

For internal analysis, let L_{1y} be the one-year portfolio loss after ordinary contractual collections and recoveries but before capital absorbs the outcome. Let ρ_q be the board-approved risk functional at confidence level q. Economic capital can be represented as:

$$EC_q = \rho_q(L_{1y}) - \mathbb{E}[L_{1y}],$$

with $\rho_q = VaR_q$ under a quantile standard or $\rho_q = ES_q$ where the institution uses expected shortfall. The horizon, confidence level, treatment of expected loss, tax, diversification, pension or operational risks and capital definition must be stated. A number without those choices is not a capital measure.

The simulation or analytical engine draws coherent outcomes. Borrower transitions respond to common macroeconomic and sector factors, facility-level evidence and hierarchical effects. EAD reflects scheduled amortisation, further draw, cancellation and conversion. LGD reflects cure, collateral or guarantee cash, workout cost and recovery delay. Parameter draws propagate posterior uncertainty. Dependence challengers test residual clustering after observable common drivers have been represented. Model-form scenarios compare the champion with credible alternatives.

The relationship between systematic factors and apparent diversification is structural, not merely a reporting preference. Gordy's risk-factor foundation for ratings-based capital shows the special conditions under which an exposure-level capital charge can be independent of the rest of the portfolio,

including a single systematic factor and an asymptotically fine-grained portfolio [10]. An MSME book with platform, sector, supply-chain or geographic clusters can violate the economic spirit of those conditions even when it contains many accounts. Part 5 therefore preserves portfolio simulation, concentration diagnostics and factor-specific reverse stress rather than treating borrower count as diversification.

Where the approved risk measure is positively homogeneous and sufficiently differentiable, marginal or Euler allocation can decompose portfolio capital into contributions that reconcile back to the portfolio total. Tasche's credit-portfolio work develops estimators for such contributions [11]. The implementation should state the risk measure and test allocation additivity, numerical stability and tail-sample error. A convenient segment allocation is not accepted merely because it sums: it must also represent how that segment changes portfolio risk at the margin.

This matters for MSME portfolios because diversification can be overstated by counting borrowers rather than economic drivers. Restaurants in different neighbourhoods may depend on the same delivery platform. Retailers may share an importer, currency exposure or landlord. Farmers and processors may share rainfall and commodity shocks. Transport businesses may share fuel-price sensitivity. The current Basel supervisory principles identify sector, geography, product, collateral, currency, funding source and economic interdependence as relevant forms of concentration and require forward-looking stress testing across material risks [12], [13].

The internal capital pack should therefore report expected one-year loss, selected quantiles, expected shortfall where used, scenario loss, concentration add-ons, marginal and allocated capital, and the contribution of parameter and model uncertainty. Reverse stress identifies the combination of default timing, utilisation, recovery delay and common shock that first breaches a capital or risk-appetite boundary. Management can then change limits, sector mix, collateral, guarantee use, collection capacity or pricing through approved policy.

Allocated economic capital also supports product economics. For product or segment g, a risk-adjusted contribution may be expressed as:

$$RA_g = Y_g - FTP_g - EL_g - OpEx_g - CoC \times EC_g,$$

where each term uses the same period and exposure basis. This is a management view, not the customer-rate formula. It can show that a product with high nominal yield consumes disproportionate collection capacity, liquidity or tail capital. It can also show that an intervention which lowers durable loss and tail dependence creates value even when immediate interest income falls.

6. Collection timing becomes a treasury variable

An instalment can be economically valuable and still arrive too late to fund today's obligation. A promise to pay, a pending mobile-money instruction, an uncleared cheque, an accepted guarantee claim and cash in the settlement account are different liquidity states. The lifecycle ledger therefore records initiation, acceptance, settlement, reversal and final allocation separately. Treasury receives cash only when the account-control and settlement rules say it is available.

Kenyan payment law gives this event chain a precise legal anchor. The National Payment System Act distinguishes payment instructions, clearing, settlement obligations and settlement, and provides that settlement effected through the relevant system is final and irrevocable [14]. The National Payment System Regulations separately define an electronic retail transfer and real-time settlement, including completion of the debit and credit and simultaneous notification [15]. These definitions establish the settlement boundary. Treasury's further tests for legal-entity ownership, account restriction, availability and correct loan allocation remain internal and contractual controls after that boundary.

For time bucket b, a simple cumulative liquidity position is:

$$LG_b = C_0 + \sum_{m \le b} \left(CF_m^{\text{settled}} + F_m^{\text{available}} - D_m^{\text{contractual}} - O_m^{\text{operating}} - U_m^{\text{contingent}} \right),$$

where C_0 is usable opening cash, $F^{\text{available}}$ is committed or reliably available funding, D is debt and depositor cash due, O is operating cash outflow and U is a scenario allowance for undrawn commitments, payment reversals, claim delay or other contingent use. The ladder is run by currency and legal entity. A positive expected ultimate recovery does not cure a negative near-term bucket until its timing and availability are credible.

Behavioural maturity links Part 2b to this ladder. A nominal 30-day facility may remain outstanding after a payment-date alignment, modification or partial cure. A revolving limit can draw when the enterprise's operating cash is already under pressure. Recovery may have a high ultimate percentage and a long legal delay. Treasury stresses those timing channels together with failed-payment recycling, concentration, guarantee settlement and new-funding availability.

CBK's KEPSS disclosure report illustrates the distinction in Kenya's high-value payment infrastructure: payment orders settle in real time when funding and technical conditions are met, and the resulting settlement finality is protected under the National Payment System Act [16]. Retail, mobile-money and platform rails can have different processing and account arrangements. The architecture therefore records the governing rail and its rules instead of applying the KEPSS timing convention to every collection.

The Basel liquidity principles call for prospective cash-flow forecasts across assets, liabilities and off-balance-sheet items, realistic behavioural assumptions, explicit allocation of liquidity costs and stress testing over multiple horizons [17]. CBK's 2025 guidelines add the applicable Kenyan LCR, NSFR and leverage-ratio framework for regulated banks [18]. Lifecycle evidence improves the behavioural inputs to those processes. Treasury and finance retain the approved regulatory calculations.

6.1 Recognise cash through settlement finality, availability and allocation

High-frequency activity becomes a Treasury input through an explicit sequence of states. The merchant records a sale; a payment instruction is initiated; the payment service accepts or rejects it; the relevant rail processes it; settlement becomes final under the applicable arrangement; the collection account receives value; account-control rules determine whether the value is available; and the loan subledger

allocates the receipt to the correct obligation. A reversal, chargeback, disputed payment or reconciliation break can interrupt that sequence. Each timestamp answers a different operational question.

For realised cash event e, define I_e^{final} as the indicator that settlement is final and $I_e^{\text{available}}$ as the indicator that the funds are available to the relevant legal entity under its account and control arrangements. The amount recognised in Treasury's realised cash position is:

$$CF_e^{\text{Treasury}} = CF_e^{\text{gross}} I_e^{\text{final}} I_e^{\text{available}} - C_e^{\text{rail}} - C_e^{\text{reversal}} - C_e^{\text{restricted}},$$

where the cost and restriction terms represent amounts that are separately identified and supported by the ledger. In implementation, multiplication by the indicators applies to the gross receipt, while each deduction follows its own recognition rule. The expression emphasises the governing principle: an observed transaction signal can support forecasting before it qualifies as realised, usable cash.

The lifecycle ledger should therefore expose at least the customer transaction time, payment-instruction time, channel acknowledgement, settlement date, value date, collection-account receipt, availability date, allocation date, reversal status and reconciliation status. Treasury can aggregate the final events into intraday and daily cash positions. Credit Risk can use earlier permitted signals for monitoring, while preserving their status as evidence rather than cash. Finance can reconcile the final allocation to the product subledger and general ledger. This shared event chain prevents a fast behavioural signal from being credited twice, first as expected collection and again as settled liquidity.

Mobile-money and platform-linked repayment require particular attention to cut-off times, weekends, bank holidays, prefunding, trust or settlement-account structure, service outages, failed instructions and concentration in a payment provider. The contractual due date can differ from the date on which the borrower initiates payment and from the date on which the institution can redeploy the value. The cash-flow engine retains all three. Treasury forecasts the actual availability date, Operations manages failed and unmatched items, and customer treatment follows the contractual and conduct framework.

6.2 A Treasury recognition gate converts behaviour into an approved assumption

A behavioural model may predict that daily or weekly collections shorten a portfolio's realised weighted-average life. Treasury recognises that benefit only after the relevant evidence passes a controlled validation gate. The gate is applied to each product, channel, cohort and material customer segment because aggregate stability can conceal a fragile subgroup or a concentrated payment route.

The recognition test addresses:

- settlement success and finality;
- reversal, return and dispute rates;
- time from instruction to available cash;
- weekend, holiday and cut-off effects;
- payment-rail and servicing outages;
- concentration by platform, aggregator, bank account and customer;
- persistence across vintages and economic conditions;
- prepayment, modification, cure and re-default behaviour;

- direct processing and servicing cost;
- customer affordability and working-capital retention;
- forecast error at each liquidity horizon;
- and performance under the approved stress scenarios.

Let $\widehat{CF}_{g,b}^{\text{settled}}$ be the forecast settled cash for group g and bucket b. Treasury can express the available planning amount as:

$$\widehat{CF}_{g,b}^{\text{available}} = \widehat{CF}_{g,b}^{\text{settled}} - H_{g,b}^{\text{operations}} - H_{g,b}^{\text{behaviour}} - H_{g,b}^{\text{concentration}} - H_{g,b}^{\text{model}},$$

where each H is a scenario-dependent deduction supported by evidence and governance. The terms are internal planning adjustments rather than regulatory haircuts. They can be represented through direct cash-flow stresses, conservative percentiles or another approved method. The method should preserve dependence between lower collections, longer settlement, greater drawdown and weaker recovery during the same adverse state.

The approved behavioural maturity follows the distribution of available principal cash rather than the number of payment messages. Thirty daily instructions do not create thirty independent sources of liquidity when they depend on one platform, one settlement account or the same underlying sales cycle. Similarly, a frequent deduction programme can produce an apparently short asset life by removing cash required for inventory and causing later re-default. The recognition test therefore examines durable cure, subsequent utilisation, working-capital sufficiency and total net recovery beside early cash velocity.

Model approval occurs in stages. Shadow operation first measures forecast accuracy and operational readiness without using the new assumption in live Treasury decisions. A controlled live phase then produces realised settlement and customer-outcome evidence. Independent validation assesses calibration, stability, sensitivity and limitations. Treasury proposes a behavioural assumption, Model Risk reviews the supporting method, Finance checks reconciliation, and ALCO approves the use, limit and review date. A material deterioration in settlement, forecast error, concentration or customer outcome can return the assumption to a more conservative state.

6.3 Separate accelerated cash from incremental cash and durable recovery

An intervention can change cash timing without changing the total amount eventually collected. It can also increase ultimate collection, reduce it, or create an early receipt followed by re-default. Treasury and Collections should decompose the observed effect rather than treating every post-intervention payment as new value.

For intervention a and comparator 0, the difference in expected collections can be separated conceptually as:

$$\Delta CF_{a,0} = \Delta CF_{a,0}^{\text{timing}} + \Delta CF_{a,0}^{\text{ultimate}} + \Delta CF_{a,0}^{\text{durable cure}} - \Delta CF_{a,0}^{\text{redefault}} - \Delta C_{a,0}^{\text{operations}}.$$

The timing term measures cash moved between buckets while holding ultimate collection constant. The ultimate term measures a change in the total collected amount over the governed horizon. The durable-cure term represents sustained restoration of payment performance. The re-default term captures early improvement that later reverses, and the operating-cost term captures the additional resources and payment-rail costs required to obtain the result. The components can overlap in raw observations, so the empirical design defines mutually consistent estimands and reconciliation rules.

This decomposition supports three distinct decisions. Treasury can value earlier cash even where ultimate recovery is unchanged. Credit Risk can evaluate whether the intervention changes default, cure and loss. Product and Conduct can evaluate whether the payment path remains affordable and sustainable. Part 6 should estimate the causal quantities where randomisation or a credible quasi-experimental design supports them. Until then, Part 5 can run transparent sensitivities rather than assigning causal meaning to association.

Randomised credit studies show why this decomposition must remain multidimensional. A field experiment with distressed credit-card borrowers in Indonesia found that a specific message treatment reduced delinquency, while experimental microfinance evidence from India found that a repayment grace period increased business investment and profits but also increased default [19], [20]. The studies concern different markets and treatments, yet together they demonstrate the relevant design principle: an intervention can change repayment, enterprise outcomes and credit loss in different directions. The Part 5 value equation therefore follows timing, durable cure, re-default, operating cost and customer outcomes rather than converting an early payment into assumed economic success.

6.4 Connect the operational ladder to LCR, NSFR and ordinary liquidity management

The internal cash ladder, the regulatory liquidity ratios and the stock of high-quality liquid assets are related control views with different definitions. A settled loan repayment increases cash available to the institution under its account arrangements. Its treatment in the Liquidity Coverage Ratio follows the applicable inflow definitions, caps and scenario rules. The repayment itself does not become a high-quality liquid asset merely because it arrived frequently. Once received, the institution may hold the resulting value in an asset that qualifies under the applicable framework, subject to that asset's own criteria and haircut [18].

The LCR view focuses on the prescribed short-term stress horizon. Treasury maps eligible outflows, recognised inflows and the qualifying liquid-asset stock under the current CBK rules. The internal ladder can be more granular, extending from intraday and daily buckets through weekly, monthly and longer horizons. Differences between internal and regulatory treatment are retained as named adjustments. This allows high-frequency collection evidence to improve operational forecasting while the formal ratio remains reproducible under its governing definitions.

The NSFR view addresses the relationship between available stable funding and required stable funding over the applicable structural horizon [18]. A short contractual MSME facility may still require a longer behavioural funding assumption if revolving use, modification or re-default extends its life. Conversely, reliable amortisation can reduce the outstanding balance over time. Treasury maps the ex-

posure according to the regulatory framework and separately maintains the richer behavioural profile used for internal ALM. Neither view is inferred solely from payment frequency.

Ordinary liquidity management also covers risks that sit outside a single ratio. Treasury monitors concentrations of deposits and wholesale funding, unencumbered collateral, committed and uncommitted facilities, currency mismatches, settlement-account exposures, payment-provider dependencies, contingent drawdowns, guarantee-claim delays and the timing of recovery. These factors feed the cash ladder and the institution's risk appetite even where they have limited effect on a reported ratio in the current period.

6.5 Contingency funding begins before a ratio is breached

The contingency funding plan translates adverse cash projections into pre-authorised actions. Its triggers can include persistent forecast error, settlement disruption, unusual deposit outflow, drawdown acceleration, reduced market access, collateral calls, concentration events, guarantee-claim delay, covenant pressure or deterioration in the available liquidity buffer. The escalation process identifies the decision owner, communication route, available actions, execution time and recovery condition.

The scenario set should combine credit and liquidity channels. An MSME revenue shock can reduce repayments, increase revolving draws, lengthen cure, delay recoveries and raise demand for modifications. A payment-platform outage can defer otherwise performing cash. A KESONIA increase can raise wholesale funding cost and customer debt service on different reset dates. A loss of confidence can cause deposit outflow while the asset book extends. Treasury tests these combinations across intraday, short-term and structural horizons rather than moving each variable independently.

Available responses can include using the approved liquidity buffer, drawing a committed facility, adjusting new origination, changing eligible product mix, increasing term funding, pledging eligible collateral, executing an approved hedge, limiting undrawn commitments, accelerating a ring-fenced funding transaction or invoking contractual cash controls. Customer-facing measures remain within the Credit Policy and Compliance Gate and applicable conduct requirements. The contingency plan identifies actions that preserve liquidity without transferring distress to customers through unsuitable collection.

The plan is tested through simulations and operational exercises. Treasury records the time needed to obtain data, convene decision makers, mobilise collateral, access a facility, reconcile settlement accounts and communicate with relevant parties. Lessons update the liquidity model, operational playbook and limits. This closes the gap between a mathematically positive liquidity position and cash that the institution can actually mobilise during stress.

7. ALM and matched-maturity FTP translate cash timing into product economics

Asset-liability management asks how contractual and behavioural asset cash flows interact with funding, repricing, embedded options and liquidity. Funds-transfer pricing attributes the relevant cost or benefit to a product or relationship so that commercial performance is not confused with an unpriced treasury position. Neither function is a synonym for KESONIA or K_{RBCP}.

An international survey of 38 large banks conducted for the BIS Financial Stability Institute found that weak liquidity-transfer-pricing practice often used average rates, failed to charge assets and credit liabilities consistently, and did not transmit current marginal funding costs. Its better-practice conclusion was a centrally governed, marginal and matched-maturity approach, including blended rates for amortising or non-maturing products and charges for contingent liquidity use [21]. That evidence supports the architecture below while leaving the institution to construct a Kenya shilling curve from its own funding capacity, balance sheet and approved policy.

A KESONIA-linked MSME asset may reprice at the end of a monthly observation period while the deposit or wholesale funding that supports it reprices on another date. Fixed-rate assets, floors, caps, prepayments and undrawn revolving commitments add optionality. Stress can extend behavioural maturity at the same time that funding becomes more expensive. ALM therefore models both the scheduled contract and the customer's plausible response under each scenario.

IRRBB, or interest rate risk in the banking book, captures the effect of adverse rate movements on the bank's earnings and economic value. The Basel framework treats earnings and economic-value measures as complementary and requires assumptions about repricing, options and customer behaviour to be governed and stressed [22]. For a defined horizon, the bank can report:

$$\Delta NII_h = NII_h^{\text{shock}} - NII_h^{\text{base}},$$

$$\Delta EVE = \left[PV(A^{\text{shock}}) - PV(L^{\text{shock}})\right] - \left[PV(A^{\text{base}}) - PV(L^{\text{base}})\right].$$

Consider a lender funding short MSME receivables with a mixture of deposits and KESONIA-linked wholesale funding. A rapid increase in KESONIA raises the cost of the floating funding immediately. Customer assets may reset later, may be fixed, or may reach an affordability limit. At the same time, revolving borrowers may draw further and term borrowers may extend through approved modifications. The NII view measures the effect on near-term earnings; the EVE view discounts changed asset and liability cash flows over their lives. The intervention and collection models improve the behavioural assumptions without deciding the bank's IRRBB position.

For facility or cohort g, a transparent matched transfer rate can be written:

$$FTP_{g,t} = r_{\text{curve}}(d_g, q_g, t) + \lambda_{\text{liq},g,t} + \lambda_{\text{option},g,t} + \lambda_{\text{basis},g,t},$$

where d_g is behavioural duration, q_g is repricing frequency, the curve term reflects the approved funding curve, and the remaining terms attribute liquidity, option and basis effects. The bank documents whether credit spread, operating expense and capital are included elsewhere so that the same cost is not charged twice. A 14-day invoice bridge and a revolving line can share a benchmark and receive different FTP because their cash-flow and option profiles differ.

7.1 Construct the funding curve before attributing product performance

The approved funding curve is the central Treasury input to matched-maturity FTP. It represents the institution's internal value of funds by currency, term, repricing structure and liquidity behaviour. It is therefore richer than a copy of the overnight KESONIA observation. KESONIA anchors the relevant customer benchmark and can contribute to an internal floating-rate curve, while Treasury still has to represent the term and funding characteristics of the balance sheet that supports the loan.

A practical curve architecture begins with a governed set of observable market and internal funding points. Treasury can combine the applicable benchmark observations with the institution's marginal secured or unsecured funding evidence, term issuance, committed facilities and stable deposit behaviour. The policy states which points are admissible, how sparse maturities are interpolated, how exceptional transactions are excluded, how stale observations are handled and how the resulting curve is validated. The institution can use a marginal curve for new-business decisions, an average-cost view for complementary planning, or both with clearly separated purposes. A single reported rate should never alternate between those concepts without an identified methodology change.

The FSI evidence gives this choice an economic direction rather than a mechanical template. Matched-maturity marginal pricing makes a long or contingent use of liquidity visible when the product is originated, while a pooled average can transfer that cost to shorter or more stable business [21]. The curve should still be built from institution-specific executable or supportable funding evidence. The external study validates the control principle; local observations, governance and model performance determine the actual Kenya shilling term points and adjustments.

For analytical exposition, write the curve used for behavioural maturity d, repricing frequency q, currency c and valuation date t as:

$$r_{\text{curve}}(d,q,c,t) = R_{\text{ref}}(q,c,t) + s_{\text{source}}(d,c,t) + \ell_{\text{term}}(d,c,t) + c_{\text{balance-sheet}}(d,c,t),$$

where R_{ref} is the approved reference component, s_{source} represents the institution's relevant funding-source spread, ℓ_{term} represents the value of committing liquidity for the behavioural term and $c_{\text{balance-sheet}}$ captures other approved funding effects allocated through the curve. The labels are management categories. Their exact construction follows the bank's FTP policy, accounting design and current regulatory framework.

The more detailed facility FTP equation then adds effects that belong to the asset rather than the general curve. A prepayable term loan contains an option that can shorten cash flows when rates move. An undrawn revolving commitment contains a draw option that can increase cash use during stress. A product whose customer benchmark resets monthly while its supporting funding reprices daily or at a longer interval creates basis and reset risk. Floors and caps can transfer economic value between customer and bank as the curve moves. Those effects belong in explicit λ_{option} and λ_{basis} terms, or in an equivalently documented implementation, so that the business unit can see why its transfer price differs from the simple benchmark.

Behavioural maturity determines the curve point only after the contractual schedule and plausible customer behaviour have been combined. For an amortising loan, Treasury begins with each expected

principal and interest cash flow rather than assigning the legal final maturity to the whole balance. For a revolving line, expected draw, repayment and cancellation behaviour produce a distribution of cash use. For modified or restructured exposures, the approved schedule and re-default assumptions alter the projected profile. For recoveries, the receipt date and availability of cash matter more than the ultimate recovery percentage. Section 7.3 establishes the additional evidence required before high-frequency collection behaviour changes this maturity.

Curve construction also has a currency dimension. A Kenya shilling customer asset funded in Kenya shillings avoids a direct contractual currency mismatch, while its borrower may still carry economic foreign-exchange sensitivity through imported stock or fuel. That borrower sensitivity belongs in credit and stress analysis. If the bank or an optional funding vehicle finances Kenya shilling receivables with foreign-currency liabilities, the contractual currency mismatch enters Treasury directly. The transfer price then reflects the approved hedge, hedge tenor, rollover exposure, collateral or margin requirement, basis and residual open position. Customer credit risk and Treasury currency risk remain separate components even when the same exchange-rate shock affects both.

Restricted balances and other balance-sheet requirements affect how much gross funding is economically available for lending. For internal analysis, Treasury can express usable funding as:

$$F_t^{\text{usable}} = F_t^{\text{gross}} - F_t^{\text{restricted}} - F_t^{\text{collateral}} - F_t^{\text{operational}},$$

where the deductions follow their actual legal, regulatory, contractual or operational treatment. The framework can include applicable reserve balances, pledged or encumbered cash, derivative margin, settlement prefunding and operational buffers. The current requirement and allocation basis are verified for the institution and date. The resulting cost enters the funding curve or a separately disclosed balance-sheet component according to FTP policy, once only. This treatment is more precise than describing restricted balances as idle or dead capital because the amounts can serve distinct monetary, prudential, settlement or risk-management purposes.

7.2 FTP is two-sided: assets receive charges and funding receives credits

A matched-maturity FTP system creates an internal transaction between Treasury and the business units. An asset-originating unit receives a funding charge that reflects the funds and balance-sheet capacity used by the asset. A deposit-gathering or funding unit receives a credit for the stable funding value it contributes. Treasury receives the residual interest-rate, liquidity and basis position that it is mandated to manage. This two-sided structure allows management to distinguish customer franchise value from a favourable or adverse Treasury position.

For an asset i, let y_i be its customer yield on a compatible annualised basis and FTP_i^A its matched asset charge. The asset spread before credit, operating and capital costs is:

$$S_i^A = y_i - FTP_i^A.$$

For a deposit or funding relationship j, let FTP_j^L be the internal credit assigned for its behavioural funding value and c_j^{customer} the customer interest and directly attributed funding cost on the same basis. Its funding contribution is:

$$S_j^L = FTP_j^L - c_j^{\text{customer}}.$$

The asset and liability contributions are management quantities. They support performance attribution while the bank's external income remains governed by customer contracts and financial-reporting policy. Treasury's residual reflects the consequences of the institution's aggregate funding, repricing and hedge decisions. A transfer-pricing process that charges lending units without crediting stable funding units gives an incomplete view of where balance-sheet value is created.

Non-maturity deposits require behavioural modelling because the customer can often withdraw contractually on demand while aggregate balances may display a more persistent economic life. Treasury can segment the observed balance into operationally stable and more volatile components, then estimate decay, retention, repricing sensitivity and stress outflow by product and customer population. The analysis should distinguish granular transactional balances from concentrated deposits, and should identify balances whose stability depends on promotional rates, a small number of customers, payment-platform relationships or correlated commercial activity.

Peer-reviewed evidence on the U.S. deposits channel shows why administered deposit rates and balances cannot be assumed to move one-for-one with the policy rate: banks widened deposit spreads as the policy rate increased, and deposit outflows varied with local market concentration [23]. The mechanism informs the design of beta, migration and outflow tests. A Kenyan institution estimates its own response by product, channel, customer type and rate regime, then stresses the possibility that digital competition or a loss of confidence shortens the historical deposit life.

Let $D_{j,t}$ be the observed balance of deposit segment j. A transparent analytical decomposition is:

$$D_{j,t} = D_{j,t}^{\text{core}} + D_{j,t}^{\text{volatile}},$$

where the core label represents the amount supported by the bank's approved behavioural evidence rather than a permanent guarantee of funding. The behavioural study examines survival or decay across horizons, migration between segments, rate sensitivity, concentration, seasonality and performance during institution-specific and market-wide stress. The model also estimates the deposit beta, meaning the proportion of a reference-rate movement that passes into the administered customer rate, together with the timing of that pass-through. Floors, non-linear customer response and competitive actions can make the beta vary across rate regimes.

These assumptions affect both IRRBB and liquidity. A longer modelled deposit life can support a longer FTP credit, while a higher stress outflow reduces its liquidity value. A slow repricing beta can initially support earnings when market rates rise, yet customer migration or delayed repricing can reverse that benefit. The bank therefore maintains a base behavioural profile, approved alternatives and stress assumptions rather than relying on one estimated average. The Basel IRRBB framework requires mater-

ial behavioural and modelling assumptions to be governed and stressed [22]. The liquidity framework separately determines the regulatory treatment under the applicable CBK rules [18].

Deposit FTP should also preserve the distinction between economic value and operating cost. Branch, channel and relationship expenses remain in the operating-cost allocation. Deposit insurance, statutory balances, collateral requirements or other institution-specific effects enter the transfer framework only through the approved policy and applicable treatment. Treasury should be able to reconcile the gross funding credit, each adjustment and the net amount delivered to the business unit.

7.3 Select the FTP method according to the decision

Different management decisions can require different funding views. A marginal matched-maturity curve is generally informative for pricing a new facility because it asks what the next unit of funding and balance-sheet use is worth at the relevant maturity. An average-cost view helps explain reported portfolio performance and the historic cost of the existing funding stock. A liquidity planning view asks whether cash is available under stress. An IRRBB view asks how repricing and option behaviour change earnings and economic value. These views can share data and still produce different figures because they answer different questions.

The FTP policy should specify at least:

- the purpose of every curve and transfer rate;
- the currencies and legal entities covered;
- the products and portfolios assigned to each method;
- the benchmark and funding instruments used to build the curve;
- interpolation and extrapolation rules;
- the treatment of behavioural maturity and repricing;
- liquidity, basis and embedded-option adjustments;
- origination-date locking and subsequent reset conventions;
- the treatment of modifications, prepayments and early closures;
- deposit credits and asset charges;
- exception and override authority;
- and reconciliation to management accounts and the general ledger.

An origination-date transfer rate allows the bank to preserve the commercial performance of the original decision and to attribute later market movements to Treasury. A variable-rate asset may instead contain reset components whose transfer treatment follows the repricing structure. A modification can require a new management view while the accounting EIR continues to follow IFRS 9. These records are stored separately so that a product report does not reconstruct historical profitability using today's curve.

The Part 4 ledger should therefore pass a defined FTP event set into the Treasury environment. The minimum fields include legal entity, product, currency, origination date, contractual maturity, payment schedule, benchmark, reset dates, floor, cap, undrawn amount, prepayment rights, modification history, settlement status and approved behavioural segment. Treasury adds the curve identifier, curve date, method, behavioural maturity, adjustments, transfer rate, approval and later restatement

reason. Oracle's FTP and cash-flow architecture provides one possible implementation of instrument-level matched-maturity allocation [24], while the control model remains applicable to other systems.

7.4 Attribute performance without moving risk between departments

The objective of FTP is disciplined attribution. The credit function remains accountable for borrower selection, limits, expected loss and concentration within its mandate. Operations remains accountable for servicing and collection cost. Product remains accountable for customer design and commercial performance. Treasury owns the funding, liquidity, interest-rate and hedge position transferred to it under policy. Capital Management owns the capital measure and allocation. Finance reconciles the combined result.

For product group g, the management bridge can be expressed as:

$$\begin{aligned} \text{Customer yield} &- \text{FTP charge} \\ &= \text{matched asset spread}, \\ \text{matched asset spread} &- \text{expected loss} \\ &- \text{operating and servicing cost} \\ &- \text{capital charge} \\ &= \text{risk-adjusted product contribution}. \end{aligned}$$

The bridge is produced by vintage, sector, product, channel and behavioural segment. It can reveal that a high headline yield is compensating for expensive funding, collection intensity or capital consumption. It can also show that a lower-yielding facility generates attractive contribution because its cash-flow timing, loss performance and operating burden are stable. These findings inform future product and funding decisions through governance rather than retrospectively redefining the customer's contractual rate.

Back-testing completes the cycle. Treasury compares the contractual and modelled cash-flow profile with actual settlement, prepayment, drawdown, modification, cure and recovery. It measures the effect of forecast error on liquidity gaps, NII, EVE and product margin. Material differences are allocated to data, model, behaviour, operations, market movement or policy. The relevant owner then corrects the source, assumption or method. This prevents favourable behavioural assumptions from remaining embedded in FTP after the portfolio has ceased to exhibit them.

7.5 IRRBB connects repricing gaps, behavioural options and basis risk

The NII and EVE measures introduced above become useful when their underlying repricing and behavioural channels are visible. Treasury first maps each asset, liability and material off-balance-sheet position to the date on which its economic rate can next change, while retaining contractual maturity as a separate field. A monthly KESONIA-linked asset with a floor can reprice at the next contractual reset, yet the floor can keep its effective rate unchanged. A deposit may be withdrawable on demand while its administered rate changes only after a management decision. A revolving commitment can create cash use before it creates interest income. These differences form the repricing profile.

For time bucket b, a simple repricing gap is:

$$RG_b = A_b^{\text{reprice}} - L_b^{\text{reprice}},$$

where A_b^{reprice} and L_b^{reprice} are the rate-sensitive asset and liability amounts whose rates can change in that bucket under the approved assumptions. The gap is a diagnostic rather than a complete risk measure. It does not capture the size of the rate change, yield-curve shape, discounting, embedded options, basis relationships or future business. NII and EVE scenarios extend the analysis by repricing and valuing the full cash-flow set.

The MSME portfolio creates several material IRRBB channels:

- **Reset mismatch:** customer assets and supporting funding observe or reset against rates on different dates.
- **Basis risk:** KESONIA-linked assets can be supported by administered deposits, fixed-rate funding or another market instrument whose price moves differently.
- **Floor and cap risk:** contractual boundaries change the sensitivity of customer income as rates move.
- **Prepayment risk:** stronger borrowers may repay or refinance when rates and competition change, removing higher-yielding assets earlier than projected.
- **Extension risk:** modification, partial cure or stressed repayment can lengthen the asset profile when funding is expensive.
- **Drawdown risk:** revolving customers may use undrawn limits when liquidity weakens, increasing the balance and funding need.
- **Deposit optionality:** customers can withdraw balances or seek higher rates even where the bank's base model assigns behavioural stability.
- **Pipeline risk:** approved but undrawn offers can enter the balance sheet after the market curve has moved.

The scenario design should include changes in the level, slope and shape of the relevant rate curves, together with basis and behavioural variants that are plausible for the institution [22]. A parallel increase illustrates immediate funding pressure but cannot represent every MSME outcome. A short-end increase may affect overnight and short wholesale funding before longer customer assets reset. A flatter or inverted curve can change the economics of maturity transformation. A lower-rate scenario can activate asset floors, increase prepayment and reduce the value of deposit-rate lag. Each market scenario is paired with coherent assumptions for deposit beta, drawdown, prepayment, modification and new business.

Treasury should also report sensitivities by product, currency and legal entity. A key-rate or bucket sensitivity shows where economic value is concentrated along the curve. A basis sensitivity shows the effect of imperfect movement between the customer benchmark and the funding or hedge index. An option sensitivity shows the consequence of alternative prepayment, withdrawal and drawdown behaviour. Aggregation follows the bank's approved method and recognises diversification only where the risk relationships support it.

Hedging begins with the identified exposure. Treasury may reduce a reset gap, basis position, currency mismatch or option exposure through changes in funding mix, asset terms, balance-sheet limits or an approved financial instrument. The decision evaluates hedge cost, liquidity, collateral or margin requirements, rollover risk, counterparty credit and residual basis. Management-risk reduction and hedge-accounting qualification are separately assessed. Finance owns the accounting treatment, while Treasury owns the economic hedge and its ongoing effectiveness within the mandate.

The intervention architecture contributes to IRRBB by improving the distribution of behavioural cash flows. It can reveal that a payment-date alignment lengthens one bucket while improving ultimate repayment, or that a draw restriction reduces contingent funding use. Those effects enter Treasury after the recognition and validation process in section 6.2. The credit model supplies evidence about customer behaviour; the bank's IRRBB framework determines how that behaviour changes NII, EVE and limits.

7.6 ALCO governs assumptions, limits and management action

ALCO receives a joined view of liquidity, funding, IRRBB and product economics. Its role extends beyond reviewing a finished report. It approves the material methodologies and assumptions through which contractual positions become behavioural cash flows, transfer prices and stress results. The committee's mandate, delegated authorities and escalation routes determine which decisions are made by Treasury management, which require ALCO approval and which move to a board committee.

The governance cycle begins with an assumption register. Each material assumption records:

- definition and economic purpose;
- products, segments and legal entities covered;
- source data and observation window;
- estimation method and model owner;
- base, alternative and stress values;
- sensitivity and limitation;
- independent-review status;
- approval date and expiry or review date;
- performance threshold;
- and fallback value when evidence becomes unreliable.

The register covers deposit decay and beta, prepayment, drawdown, cure, modification, recovery timing, payment-rail settlement, behavioural maturity, new-business volume, wholesale-funding availability, guarantee settlement and optional SPV cash-flow assumptions. A model can supply several of those quantities, while accountable Treasury and risk functions decide whether they are fit for each intended use.

Limit design turns the analysis into operating discipline. The bank can set limits for cumulative liquidity gaps, survival horizon, funding concentration, unencumbered buffer, wholesale maturity concentration, currency mismatch, NII sensitivity, EVE sensitivity, basis exposure, behavioural-assumption reliance and model exceptions. Product and sector limits can also respond to concentration, drawdown

and collection volatility. Every limit identifies its measurement frequency, warning level, breach level, authority, cure period and available actions.

ALCO actions may include changing the funding tenor, diversifying counterparties, increasing available liquidity, adjusting hedge positions, limiting a product's new originations, changing undrawn commitment policy, revising the approved behavioural assumption, altering the FTP curve or referring customer-term changes through Product and Credit governance. The action log records the evidence, decision, owner, implementation date and expected effect. Subsequent reporting compares the realised outcome with that expectation.

7.7 The Treasury scorecard makes forecast quality visible

A lifecycle Treasury scorecard should show the connection between customer cash, funding and balance-sheet risk. Its purpose is to make assumptions testable and to reveal whether an apparent product margin depends on unrecognised liquidity or interest-rate exposure.

The core scorecard includes:

- contractual, forecast and settled principal and interest cash by horizon;
- instruction-to-settlement and settlement-to-availability time;
- reversal, failed-payment and reconciliation-break rates;
- contractual and behavioural weighted-average life;
- prepayment, drawdown, modification, cure and re-default variance;
- cumulative liquidity gaps by currency and legal entity;
- liquidity-buffer availability and encumbrance;
- LCR and NSFR movements under the applicable calculations;
- deposit concentration, decay and repricing beta;
- wholesale-funding maturity and counterparty concentration;
- NII and EVE sensitivity by scenario;
- basis, floor, cap and option exposure;
- asset FTP charge, deposit FTP credit and Treasury residual;
- customer yield, matched spread and risk-adjusted contribution;
- guarantee-claim receivables and settlement delays;
- and forecast error, override, breach and remediation status.

Vintage reporting gives the scorecard an evidential spine. Each origination cohort retains its original product terms, curve, transfer price, risk estimate and expected cash profile. Actual settlement, prepayment, modification, default, cure and recovery are then compared with that baseline. The bank can distinguish a weak original assumption from a later market shock or operational failure. Reconstructing historical performance with the latest behavioural curve would obscure that distinction.

The scorecard also separates management and regulatory views. Internal expected cash, regulatory inflow recognition, accounting cash shortfalls and investor cash can differ because their rules and purposes differ. The report reconciles the populations and adjustments rather than forcing one number across all uses. That approach allows Treasury, Finance, Credit Risk, Capital Management and the

optional SPV team to begin with the same underlying events while retaining their accountable calculations.

The operating sequence begins with reconciled account schedules from Part 4. Treasury maps contractual cash flows, applies approved prepayment, drawdown, cure, modification and recovery behaviours, generates repricing and liquidity ladders, and runs benchmark, basis and behavioural shocks. Finance and product teams receive the transfer rate and matched spread. ALCO receives aggregate NII, EVE, liquidity and concentration views. The customer price remains governed through the RBCPM and product process.

Oracle Financial Services Funds Transfer Pricing is one possible enterprise implementation. Oracle describes matched-maturity transfer pricing and instrument-level allocation of asset, funding and interest-rate spreads, while its cash-flow engine supports contractual schedules, repricing and ALM scenarios [24]. A bank may use that platform, another treasury system or a validated internal engine. The authoritative product terms and settlement events remain in their source systems, and the treasury engine reconciles its output rather than becoming a second loan ledger.

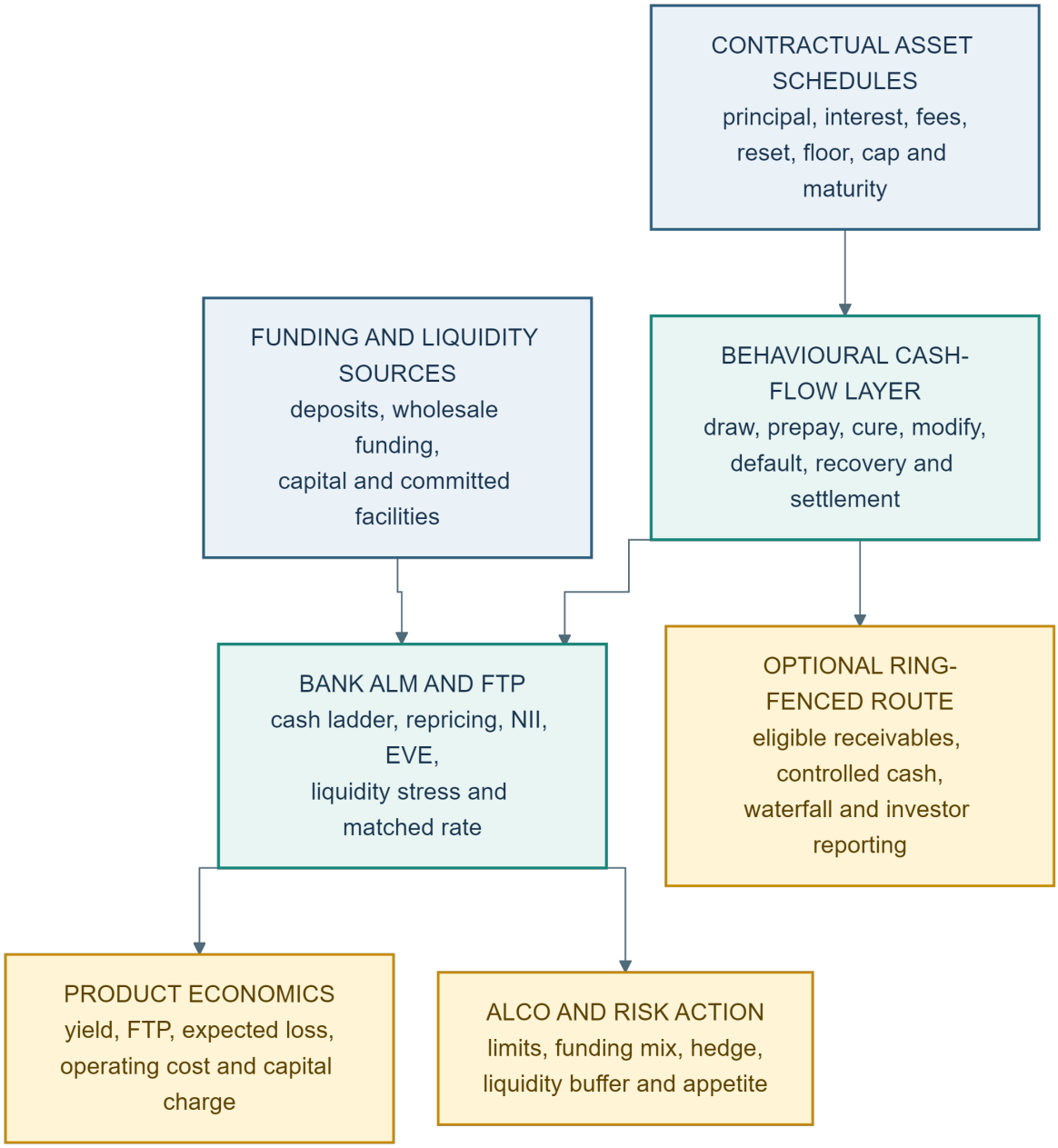


Figure 2: Contractual cash flows, behavioural maturity and matched funding. The bank's ALM and FTP functions own the prudential funding view. The same reconciled asset cash flows can support an optional ring-fenced route without making that route the default architecture for every MSME portfolio.

8. Price the intervention as an economic decision

Part 3 treated an intervention as a governed action with a defined purpose, population, outcome and human route. Part 5 measures its economic consequence. A message that produces an immediate payment can still be poor policy if it merely moves cash by one day, increases complaints or is followed by re-default. A payment-date alignment may reduce near-term contractual receipts while improving durable cure and net recovery.

For intervention a against comparator 0, define causal net economic value over horizon h as:

$$NEV_h(a,0) = \Delta PV(C_h^{\text{net}}) - \Delta Cost_h^{\text{treat}} - \Delta Cost_h^{\text{fund}} - CoC \times \Delta EC_h,$$

where $\Delta PV(C^{\text{net}})$ is the identified change in discounted collections after direct workout cost, and the other terms measure incremental treatment, funding and allocated capital effects. Conduct, fairness, vulnerability and customer-sustainability limits are tested alongside the financial quantity. They are not converted into a price that permits harmful treatment.

The estimate requires the randomised or quasi-experimental design established in Part 3 and recorded in Part 4. Predictive model scores can target the eligible population, but observed payments after treatment do not identify the untreated counterfactual by themselves. Promotion requires durable-cure, re-default, net-recovery, cost, complaint and group-outcome evidence. The result informs servicing strategy and product design. It enters customer pricing only through the approved prospective methodology and never as an undisclosed surcharge for receiving assistance.

9. Risk transfer is a monitored cash-flow contract

MSME credit risk can be shared through a public or private guarantee, credit insurance, funded participation, first-loss facility or synthetic protection. Each form changes the lender's loss only through its executed terms. The model therefore represents protection as a contingent claim with an eligible population, coverage percentage, attachment, limit, exclusions, waiting period, reporting duties, claim event, settlement rule, recovery sharing, expiry and counterparty.

For scenario s, define gross covered-pool loss L_s^{gross}, attachment A, exhaustion E, coverage fraction c, and eligibility indicator I_s^{elig}. A stylised claim is:

$$Claim_s = I_s^{\text{elig}}\, c \min\left\{\max\left(L_s^{\text{gross}} - A, 0\right), E - A\right\},$$

and retained loss after settled protection is:

$$L_s^{\text{net}} = L_s^{\text{gross}} - Claim_s + P^{\text{prot}} + C_s^{\text{claim}} + C_s^{\text{delay}},$$

where P^{prot} is protection premium or fee, C^{claim} is the cost of reporting and pursuing a claim, and C^{delay} measures the funding effect of settlement delay. The executed agreement may instead apply facility-

level cover, pari-passu sharing, first-loss cover, deductibles or claim caps. The formula is configured to the contract rather than forcing every programme into one layer.

Kenya's Credit Guarantee Scheme provides a concrete public example. The Public Finance Management (Credit Guarantee Scheme) Regulations establish a scheme intended to encourage additional MSME lending through partial guarantees and define participating financial intermediaries, eligible credit and reporting responsibilities [25]. The National Treasury describes the current programme as risk sharing between the Scheme and participating financial institutions, with qualifying facilities originated and monitored by those institutions [26]. Scheme participation therefore complements underwriting and collection. It does not replace lender due diligence, servicing or recovery.

International principles add two tests to nominal coverage: additionality and sustainability. The World Bank and FIRST Initiative principles call for a clear mandate, sound governance and risk management, defined eligibility, partial coverage consistent with prudential treatment, transparent risk-based pricing, efficient claims management and systematic evaluation [27]. The scheme should therefore report whether it expanded viable credit or improved terms beyond the counterfactual, while also measuring claims, recoveries, concentration, operating cost and capital adequacy. A low observed claim rate is not sufficient evidence of success if the guarantee reached borrowers who would have received the same facility without it.

Cross-country peer-reviewed evidence on 76 partial credit guarantee schemes found wide variation in governance and risk-reduction tools, limited use of risk-based pricing, and a substantial government role in funding and management but less involvement in risk assessment and recovery [28]. The finding supports a deliberate allocation of responsibilities: the guarantor sets coverage, eligibility, pricing and claims discipline; the lender retains appraisal, monitoring, servicing and recovery duties defined by the programme; and the evaluation separates credit additionality, economic additionality and financial sustainability. Kenyan programme results remain an empirical question measured under the Kenyan scheme.

Economic protection, IFRS 9 recognition and regulatory-capital recognition are tested separately. IFRS 9 includes cash flows from collateral and other credit enhancements in ECL when the relevant requirements are met and the enhancement is part of the contractual terms and not recognised separately [6]. Prudential recognition follows the applicable framework. The Basel credit-risk-mitigation standard requires legal enforceability and operational conditions for eligible guarantees and identifies residual risks such as claim delay, documentation weakness, maturity mismatch and counterparty dependence [29]. The participating bank maps those conditions to the current CBK treatment before recognising capital benefit.

Monitoring continues after cover is purchased. The dashboard follows covered and uncovered EAD, premium or fee, attachment utilisation, claim frequency, acceptance and rejection, settlement lag, recoveries, subrogation, expiry, counterparty credit, sector concentration and basis between the portfolio loss definition and the covered event. Gross and net calibration are both retained. A falling net loss can reflect a strong portfolio, generous protection or delayed recognition of an uncovered deterioration, so management reads the movement with exposure and claim status.

9.1 Treasury distinguishes loss protection from available liquidity

Credit protection can reduce an institution's ultimate loss while leaving a near-term cash requirement. A guarantee claim may require delinquency, notice, documentation, a waiting period, verification and approval before settlement. Credit insurance can contain exclusions or contestable evidence. A funded participation transfers cash and economic exposure differently from an unfunded guarantee. A synthetic arrangement can transfer defined credit loss while leaving the underlying asset and its funding on the bank's balance sheet. Treasury therefore maps the executed form of protection into cash timing rather than inferring liquidity from the coverage percentage.

The liquidity ladder records at least four states: potential covered loss, notified claim, accepted claim receivable and settled claim cash. Only the last state enters realised available liquidity. A highly probable claim can support a scenario forecast while retaining its settlement uncertainty. The forecast records claim eligibility, expected amount, settlement distribution, currency, account destination, recovery-sharing rule and counterparty. A disputed or incomplete claim follows a separate path and receives no automatic cash credit.

For protection programme p, let ΔL_p be the scenario reduction in retained credit loss, P_p the protection premium or fee, $C_p^{\text{operation}}$ the administration and claim cost, C_p^{funding} the value of funding the claim delay, and C_p^{basis} and $C_p^{\text{counterparty}}$ the modelled cost of uncovered basis and counterparty deterioration. A management-value expression is:

$$V_p^{\text{protection}} = PV(\Delta L_p) - PV(P_p) - PV(C_p^{\text{operation}}) - PV(C_p^{\text{funding}}) - PV(C_p^{\text{basis}}) - PV(C_p^{\text{counterparty}}).$$

The calculation is performed across the scenario distribution. It reveals the difference between nominal coverage and economic value. A low-fee guarantee can have limited liquidity value when settlement is slow. A protection contract with fast settlement can still provide weak coverage if exclusions create significant basis. A strong counterparty can become correlated with the covered portfolio through sovereign, sector or macroeconomic stress. The institution therefore considers amount, timing, basis, concentration and counterparty capacity together.

Funded and unfunded structures also affect balance-sheet management differently. Upfront funded protection or a funded participation can provide liquidity at inception or transfer part of the asset funding, subject to the executed structure. An unfunded guarantee generally provides contingent loss reimbursement. Collateralised protection introduces collateral eligibility, valuation, margining, custody and liquidity requirements. Treasury, Credit Risk, Capital Management, Finance and Legal each record the treatment appropriate to their mandate.

9.2 Protection pricing and renewal use gross and net portfolio views

The protection decision begins with the gross loss distribution. Management then applies the exact contract to every scenario to calculate covered claim, retained loss, claim timing, premium and operational cost. The result is a net distribution that supports pricing, capital and liquidity analysis. Retaining the

gross distribution prevents apparent improvement caused by broader protection from being mistaken for stronger underlying underwriting.

Pricing should examine:

- expected and tail claim value;
- attachment and exhaustion utilisation;
- timing of premium and claim cash;
- claim documentation and servicing cost;
- historical acceptance, rejection and dispute patterns;
- recovery and subrogation allocation;
- counterparty credit and concentration;
- currency and basis mismatch;
- collateral or margin requirements;
- maturity and renewal risk;
- and the effect on product and portfolio limits.

Renewal is a forward-looking decision. Protection that expires before the underlying asset matures leaves a residual period. A guarantee programme can change eligibility or capacity as market conditions deteriorate. Premium can rise precisely when the institution wishes to renew. Treasury includes those paths in funding and liquidity stress; Capital Management measures the effect on retained tail risk; Credit Risk evaluates the protected and unprotected portfolio; and Product assesses whether the resulting economics remain sustainable.

The reporting pack therefore reconciles gross loss, contractual claim, accepted claim, settled claim and net retained loss. It shows the portion of any capital or accounting benefit that has met the applicable recognition criteria, while separately showing economic protection and liquidity. This allows a decision maker to see, for example, that a programme provides meaningful ultimate loss protection but limited 30-day cash support.

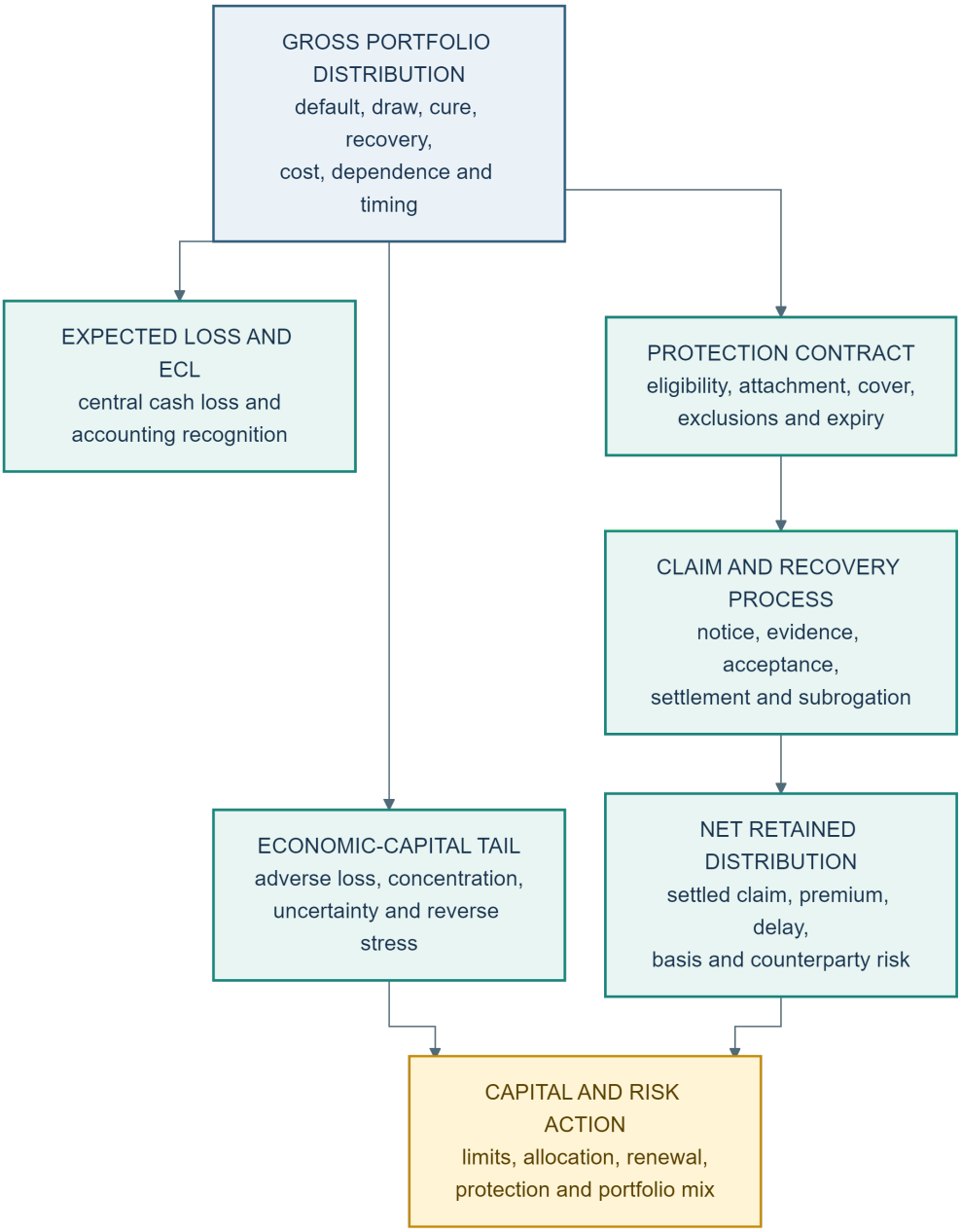


Figure 3: Expected, tail and transferred loss. Pricing, ECL and economic capital begin with the gross portfolio. Risk transfer changes the retained distribution only after coverage, claims, settlement delay, recoveries, basis and counterparty performance have been represented.

10. Ring-fenced receivables funding is an optional extension

A bank can apply the lifecycle architecture while retaining loans on its balance sheet. Where scale, legal feasibility and investor demand justify another funding channel, eligible receivables may support a ring-fenced securitisation or private funding vehicle. The Capital Markets Act permits cash-flow-generating assets that are legally capable of sale, transfer or assignment to enter a securitisation trust and addresses true sale under Kenyan law [30]. The Asset-Backed Securities Regulations define the originator, issuer, trustee, servicer, eligible assets, credit enhancement and investor cash-flow structure for covered offers [31]. Transaction counsel, accountants, tax advisers, the relevant regulators and investors determine the structure in practice.

The optional vehicle purchases only receivables and associated rights that the originator owns and can transfer. An eligibility schedule tests documentation, balance, arrears, currency, sector, geography, prior encumbrance, data quality, concentration and enforceability. The servicer collects through controlled accounts. A waterfall pays permitted expenses, senior interest and principal, subordinate obligations, reserve cures and residual amounts in the contractual order. Overcollateralisation, cash reserves, subordination and early-amortisation triggers are modelled as finite protections.

Ring-fencing changes legal rights and cash priorities, but its economic analysis must still include the sponsor and servicer. Gorton and Souleles show, in credit-card securitisation evidence, that investors can price sponsor risk and implicit support alongside the formal SPV structure [32]. The Kenyan transaction remains governed by its own law and documents. The research strengthens the scenario design: model originator distress, servicer interruption, commingling, asset-representation breach, replacement cost and voluntary support separately from the legal true-sale conclusion, making operational dependence visible while preserving the vehicle's legal distinction.

10.1 The vehicle has its own cash architecture

The ring-fenced vehicle converts eligible receivable cash flows into a contractually ordered funding structure. Its Treasury task begins with the legal entity and transaction documents. The vehicle receives only the assets and associated rights validly transferred to it, funds purchases through the agreed notes, equity or facilities, holds cash in controlled accounts and applies available funds through the priority of payments. The originator's bank-level ALM and FTP continue separately.

A practical account structure can include:

- a receivable purchase or funding account;
- a collection account receiving borrower and recovery cash;
- a principal account and an interest or revenue ledger where the documents distinguish them;
- a cash reserve or liquidity reserve account;
- permitted expense and tax accounts;
- a hedge or collateral account where applicable;
- and a distribution account for amounts released after the required tests.

The legal documents determine ownership, control, permitted withdrawals, account-bank standards and replacement triggers. The model mirrors those terms. It avoids treating an operational subledger as a bank account or using cash that is trapped, restricted or allocated to another priority.

Collections enter the vehicle through a defined servicing route. Where the servicer temporarily receives cash before remitting it to the controlled account, the structure measures commingling exposure, remittance frequency and replacement or sweep mechanics. A backup-servicing plan identifies the data, authority, payment instructions, customer communication and operational capacity needed to continue collection if the primary servicer fails. Servicing continuity is a liquidity protection because an otherwise performing receivable produces no usable cash when the collection mechanism is interrupted.

10.2 The borrowing base governs how much funding the assets support

The eligible receivable balance and the amount lenders are willing to advance are related but distinct. Let $ER_{j,t}$ be the eligible balance of asset group j, a_j its contractual advance rate, R_t required reserves or retained amounts, and DD_t other documented borrowing-base deductions. A stylised borrowing-base capacity is:

$$BBC_t = \sum_j a_j ER_{j,t} - R_t - DD_t.$$

Outstanding funded debt is tested against that capacity under the executed facility. The borrowing base can recognise different advance rates for product, seasoning, delinquency, concentration or other eligible classifications. Ineligible assets remain visible in the portfolio but contribute no borrowing-base value. A breach can require a cure, additional eligible assets, repayment, cash trapping or another contractual action.

Overcollateralisation provides a complementary view. For eligible receivables ER_t and the notes or debt included in the transaction definition N_t:

$$OC_t = \frac{ER_t}{N_t}.$$

The denominator must follow the transaction documents. A test based on senior notes answers a different question from one based on senior plus subordinate debt or total capitalisation. The report displays the contractual threshold, actual ratio, headroom and cure requirement. Advance rate and overcollateralisation remain separate controls even where their numerical effects appear similar.

Eligibility is recalculated from point-in-time asset data. The calculation records receivable ownership, balance, product, currency, seasoning, arrears, modification, platform or channel, geography, obligor concentration, documentation, policy or guarantee status where relevant, and any prior encumbrance. Reversals and corrections produce a versioned borrowing-base result. The trustee or calculation agent can reproduce the result from the agreed data tape and rules.

10.3 SPV ALM matches receivable cash to note and reserve obligations

The vehicle's asset-liability management is narrower than the originating bank's prudential ALM but still economically important. It maps monthly and, where necessary, daily asset collections against note interest, note principal, fees, taxes, reserve requirements, hedge payments and other waterfall obligations. It monitors timing, basis, currency, optionality and concentration within the transaction boundary.

For bucket b, the vehicle's projected cumulative cash position can be written:

$$LG_b^{\mathrm{SPV}} = C_0^{\mathrm{SPV}} + \sum_{m \le b} \left(CF_m^{\mathrm{asset}} + F_m^{\mathrm{committed}} - DS_m^{\mathrm{notes}} - E_m^{\mathrm{senior}} - H_m^{\mathrm{hedge}} - R_m^{\mathrm{required}} \right),$$

where asset cash includes settled principal, interest, recoveries and eligible credit-enhancement proceeds; committed funding is recognised according to its availability conditions; note debt service follows the transaction documents; senior expenses follow their priority; hedge cash follows the executed agreement; and reserve requirements reflect the current test. The ladder is run under base and stressed asset scenarios.

The cash profile can contain several mismatches. Receivables may collect daily while notes pay monthly or quarterly. Assets may amortise while notes remain outstanding until a payment date. Customer assets and notes can observe compounded KESONIA over different windows or include different floors, lags and day-count conventions. Recoveries and guarantee claims can arrive after note obligations. A revolving purchase period can consume cash for new receivables while debt service continues. The model identifies each mismatch and the contractual mechanism that absorbs it.

10.4 Note pricing and the waterfall remain distinct from bank FTP

For floating-rate note class k, a simplified coupon can be expressed as:

$$r_{k,t}^{\mathrm{note}} = R_{a_k, b_k, t}^{\mathrm{KESONIA}} + m_{k,t},$$

where the note's observation period, payment date, day count, fallback, floor and margin follow its own documents. The asset contract can use another observation window even where both sides reference KESONIA. The resulting basis and timing difference belong in the SPV cash model.

The note margin is an investor funding term. It is separate from the originating bank's internal FTP rate, the customer premium, expected loss, excess spread and investor hurdle. The vehicle can maintain a matched-funding attribution showing how asset yield covers note cost, expenses, expected loss, reserve drag and residual return. Calling that analysis SPV matched-funding attribution preserves the distinction from a bank-wide FTP system that prices business units and deposits.

On each payment date, the waterfall applies available funds in the contractual order. A representative sequence can include taxes and statutory amounts, trustee and account-bank costs, servicing fees, hedge payments, senior interest, senior principal, subordinate interest and principal, reserve replenishment and residual distribution. The actual order, caps, deferrals and pro rata or sequential rules

come from the executed transaction. The calculation engine records cash entering each step, amount paid, shortfall, carried amount and ending balance.

10.5 Reserves, prefunding and permitted investments create real economics

A cash reserve provides time for asset collections to meet senior obligations during disruption. Its target can be a fixed amount, a percentage of notes, a defined number of months of senior expenses and debt service, or a formula that changes with the portfolio. The model reproduces the contractual definition and tests opening balance, required amount, deposits, releases, permitted investment income, draws, shortfall and cure.

Prefunding can reduce execution risk by providing cash before receivables are purchased, while unused cash creates negative carry when its permitted investment return is below the note cost and transaction expenses. A revolving structure adds reinvestment risk because available principal may purchase new assets whose yield, term or credit quality differs from the original pool. Eligibility, concentration and minimum-yield tests can govern purchases. The model measures the value of liquidity protection beside its cost.

Permitted investments are treated as assets with their own maturity, credit quality, liquidity and counterparty exposure. Reserve or collection cash is not assumed to earn the customer asset yield. Account-bank concentration, set-off, replacement conditions and access during operational disruption form part of the cash-risk analysis.

10.6 Triggers convert deterioration into contractual cash action

Performance triggers can include delinquency or default ratios, cumulative net loss, collection performance, excess spread, reserve shortfall, borrowing-base deficiency, overcollateralisation, concentration, servicer failure, account-bank downgrade or other documented events. A trigger is calculated from defined data, numerator, denominator, observation date and cure rule. It then activates the exact contractual response.

Early amortisation commonly changes the vehicle from purchasing or revolving into paying down debt. Cash that might otherwise acquire new receivables or reach the residual position is redirected through the protective waterfall. A senior sequential-pay rule can accelerate Class A principal. Reserve releases can become restricted. Equity or residual distributions can stop. The model shows both the trigger date and the resulting change in every later cash flow.

Trigger design requires balance. A late trigger can permit asset deterioration to consume structural protection. An excessively volatile trigger can force unnecessary amortisation after a temporary data or settlement event. The documents can include averaging, persistence, cure periods and data-quality fallbacks where appropriate. The calculation remains transparent so that investors can distinguish economic deterioration from an operational reporting break.

10.7 Currency, hedging and liquidity facilities follow executed capacity

A vehicle that acquires Kenya shilling receivables and issues Kenya shilling notes has limited direct contractual currency mismatch, although borrower cash can remain sensitive to exchange-rate shocks. If liabilities, expenses or investor commitments are denominated in another currency, the vehicle models principal, interest, collateral, margin and rollover cash under exchange-rate scenarios. A hedge can reduce the mismatch while introducing counterparty, basis, collateral and termination exposure.

A liquidity facility can bridge timing disruption under specified drawing conditions. It is not a substitute for absorbing credit loss unless the agreement expressly provides that protection. The model records commitment fee, available amount, draw conditions, interest, maturity, repayment priority, provider credit and termination events. An uncommitted facility enters stress as a contingent possibility rather than assured funding.

10.8 The SPV reporting pack reconciles assets, cash and investor claims

The monthly or payment-date pack should include:

- opening, purchased, collected, defaulted, recovered and closing receivables;
- eligible and ineligible asset balances with reason codes;
- borrowing-base capacity, debt and headroom;
- advance rates and overcollateralisation;
- principal, interest, recovery and expense collections;
- collection-account, reserve-account and distribution-account movements;
- asset yield, note coupon, basis and excess spread;
- note opening balance, interest, principal and closing balance by class;
- waterfall allocation and unpaid amounts;
- reserve target, balance, draw and cure;
- covenant, trigger and early-amortisation status;
- servicing remittance, reconciliation and backup-servicer readiness;
- guarantee or protection claim status;
- hedge, currency and counterparty exposure where applicable;
- and one-year and ultimate loss, principal shortfall and payment timing by scenario.

Each total reconciles to the asset tape, controlled bank accounts, note register and transaction ledger. The pack distinguishes expected, due, received, available and distributed cash. Investors, trustee, servicer, originator and calculation agent can therefore examine the same economic events through their contractual roles.

The vehicle's analysis starts with the same monthly cash-flow distribution but retains separate funding quantities. Asset yield is the return collected on purchased receivables. A note margin m_{fund} is an investor funding term. Bank FTP is the originating institution's internal funding attribution. The investor hurdle is a required return or valuation input. Excess spread is the cash remaining after realised asset collections, losses, expenses and note costs under the waterfall. These quantities are reconciled without renaming one another.

For tranche k, attachment A_k and exhaustion E_k, scenario loss can be represented as:

$$TL_{k,s} = \min\{\max(L_s - A_k, 0)\,, E_k - A_k\}\,.$$

The lender pack reports one-year and ultimate loss, payment timing, reserve use, covenant breaches and principal shortfall by scenario. Legal transfer, investor protection and capital relief remain consequences of the executed transaction and applicable approvals. The model supplies reproducible evidence for those decisions.

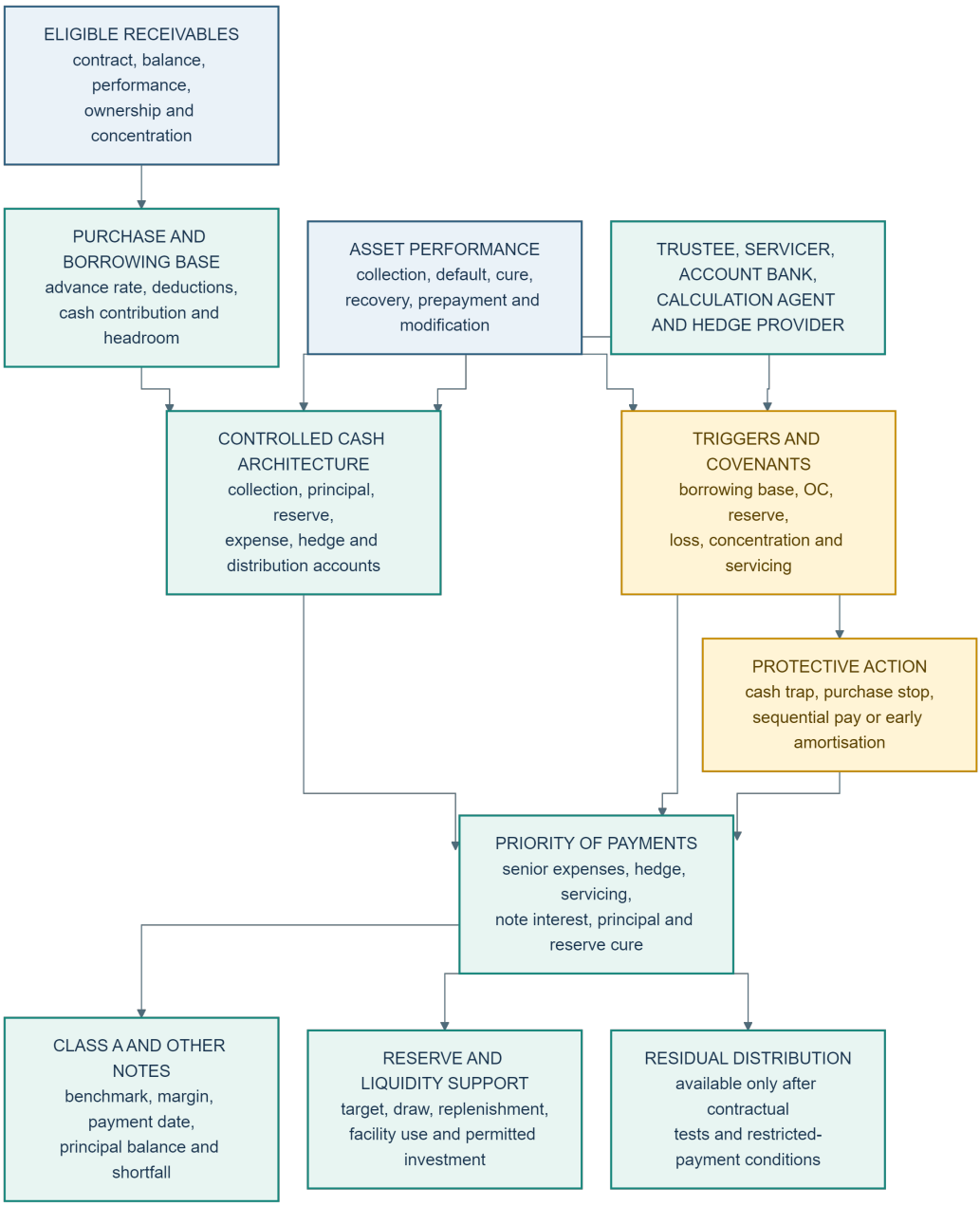


Figure 4: Ring-fenced receivables cash, note and control architecture. Eligible assets support a governed borrowing base, while settled collections move through controlled accounts and the priority of payments. Reserves, covenants, service continuity and protective triggers determine when cash reaches noteholders or the residual position.

11. One control plane, several accountable owners

Part 5 completes the journey from a quoted benchmark to realised and transferred loss. Credit owns risk appetite and customer selection. Product and legal own terms. Finance owns ECL and journals. Capital owns internal and regulatory assessment. Treasury owns ALM, liquidity and FTP. Collections owns executed recovery strategy. The protection owner manages guarantees and claims. An optional vehicle's trustee, servicer and calculation agent own its contractual cash waterfall.

The joined evidence plane lets those owners reconcile without collapsing their mandates. A customer payment has one economic amount and timestamp, while pricing, ECL, capital and liquidity interpret it for different decisions. Part 6 will test whether the proposed lifecycle system improves durable collection, net recovery, customer outcomes and risk-adjusted economics under a bank-controlled experimental programme.

11.1 Each function owns a defined financial interpretation

The control plane begins with common event identity rather than one universal financial number. An account, contract, payment, modification, default, recovery or guarantee claim has an authoritative event history. Each accountable function applies the horizon, probability measure, recognition rule and policy required for its decision.

Function	Principal Part 5 decision	Inputs received from the lifecycle	Controlled output
Product and Pricing	Which terms create a sustainable customer and portfolio proposition?	Benchmark, customer cash cycle, expected loss, FTP, operating cost and capital charge	Approved price, limit, tenor, repayment design and disclosure
Credit Risk	What loss and transition distribution is supported by the evidence?	PD, EAD, LGD, cure, recovery, concentration and uncertainty	Risk estimates, limits, stress results and portfolio action
Collections and Workout	Which authorised action improves durable cash and customer outcomes?	State, affordability, treatment eligibility, causal evidence and account history	Treatment, restructuring, recovery plan and outcome record
Finance	What amount is recognised and reported?	Contractual cash, original EIR, stage, scenarios, shortfall, modification and write-off	ECL, income, modification result, journals and disclosures
Capital Management	How much adverse loss should the institution withstand or allocate?	Gross and net loss distributions, dependence, concentration and mitigants	Regulatory assessment, economic capital, allocation and risk-adjusted return
Treasury and ALCO	When is cash available and how is funding, rate and liquidity risk managed?	Contractual and behavioural cash, deposits, funding, options, settlement and stress	Curves, FTP, liquidity position, NII, EVE, limits, hedge and funding action
Protection Management	What contractual loss transfer is available and performing?	Covered exposure, eligibility, attachment, claim, settlement and counterparty	Protection value, claim action, renewal decision and net retained loss

Function	Principal Part 5 decision	Inputs received from the lifecycle	Controlled output
Structured Finance Parties	What cash belongs to the vehicle and how is it distributed?	Eligible assets, controlled accounts, borrowing base, notes, reserves and triggers	Waterfall, covenant result, investor report and protective action
Model Risk and Validation	Is a model or behavioural assumption fit for its intended financial use?	Development evidence, validation, stability, limitations and change request	Approval, restriction, monitoring threshold or remediation
Data and Technology	Can every result be reproduced from governed events and versions?	Source events, mappings, corrections, model artefacts and access controls	Reconciled data products, lineage, availability and operational evidence
Legal and Compliance	Are the terms, transfer, processing and actions authorised and enforceable?	Contract, regulation, policy, consent or lawful basis, and proposed action	Legal interpretation, control requirement, approval condition and exception

The matrix prevents a model improvement from silently changing every downstream process. A new cure model can improve credit monitoring while Finance retains the approved ECL method, Treasury retains the approved behavioural maturity and Product retains the contractual pricing process. Each function can adopt the model after completing its intended-use assessment.

11.2 The monthly financial cycle reconciles before it explains

The reporting cycle starts by freezing the governed event population for the period, including late events, corrections and reversals according to the close policy. Operations and Finance reconcile due, paid, allocated, modified, defaulted, written-off and recovered amounts to controlled accounts, the product subledger and the general ledger. Differences remain visible with owners and resolution status.

Credit Risk then produces the approved risk and cash-flow distributions, including scenarios, concentration and model uncertainty. Finance calculates ECL and other accounting consequences under its policy. Capital Management updates internal and regulatory views. Treasury produces liquidity, funding, FTP, NII, EVE and hedge reporting. Protection Management reconciles covered exposure, claims and settled proceeds. Where a ring-fenced vehicle exists, its calculation agent applies the asset eligibility, borrowing base, accounts, waterfall and covenant rules.

The joined management bridge then explains movement from the prior period. It separates new originations, contractual amortisation, benchmark movement, premium and fee changes, model changes, stage migration, drawdown, modification, prepayment, cure, default, recovery, settlement delay, funding-curve movement, protection and capital allocation. Management can therefore see whether a change in risk-adjusted contribution came from customer performance, market rates, funding, accounting recognition, operational cost, model revision or structural protection.

11.3 Model and assumption changes have intended-use approval

Every material model or behavioural assumption has an inventory entry that identifies its purpose. The HLR and timing challengers may support underwriting, ECL inputs, collection targeting, liquidity fore-

casting, economic capital or SPV scenarios, but approval for one use does not automatically establish approval for another. Each use has its own materiality, validation evidence, fallback and accountable owner.

The intended-use discipline is also reflected in current comparative supervisory practice. The 2026 U.S. interagency Revised Guidance on Model Risk Management begins model development with a clear purpose, aligns validation rigor with model use and materiality, and requires users to understand limitations when a model is extended to another application [33]. Used as an external benchmark alongside the applicable Kenyan framework, it reinforces the Part 5 control: a model validated for underwriting enters ECL, capital, Treasury or vehicle reporting after the receiving function has assessed that specific use, data, performance range and fallback.

A material change package should contain:

- the economic question and intended use;
- the affected products, portfolios, legal entities and customers;
- source-data and point-in-time controls;
- development sample and observation horizon;
- comparative performance against the current method;
- calibration, discrimination and cash-flow forecast accuracy;
- sensitivity, stress and stability analysis;
- fairness, conduct and customer-outcome assessment where relevant;
- impact on ECL, capital, liquidity, FTP, pricing and SPV cash where applicable;
- operational readiness, parallel-run evidence and reconciliation;
- limitations, use restrictions and human-review requirements;
- rollback criteria and fallback method;
- independent-validation findings;
- and approvals from each affected function.

For Treasury use, the package includes settlement accuracy, forecast cash error, weighted-average-life error, prepayment and drawdown stability, liquidity-gap impact, NII and EVE sensitivity, funding-curve interaction and performance under operational and market stress. For Finance use, it includes stage, cash-shortfall and allowance consequences. For Capital use, it includes tail, dependence and concentration effects. For a vehicle, it includes borrowing-base, reserve, waterfall, trigger and note-payment effects.

11.4 The integrated scorecard preserves reconciliation across uses

The final Part 5 scorecard brings the departmental views together without ranking one as the master of the others. It reports customer price and total cost, expected and realised cash, ECL and stage movement, economic and regulatory capital, liquidity and funding, IRRBB, FTP and margin, treatment economics, protection, and optional vehicle performance. Every metric declares population, date, horizon, currency, numerator, denominator, scenario and version.

The scorecard includes reconciliations between:

- customer benchmark and charged rate;

- customer rate and internal FTP;
- expected contractual cash and settled available cash;
- physical expected loss and IFRS 9 ECL;
- expected loss and economic-capital tail;
- gross loss, covered claim and net retained loss;
- bank-held assets and assets sold or assigned to a vehicle;
- vehicle asset cash, waterfall cash and note balances;
- and expected intervention benefit and realised durable outcome.

Exceptions are part of the report. A missing settlement, failed reconciliation, model override, expired assumption, limit breach, disputed claim or unavailable funding source carries an owner, age, financial effect and remediation date. This gives executives a view of the architecture's reliability alongside its reported economics.

Part 6 then tests the mechanisms that can be learned prospectively: price response, repayment design, treatment uplift, durable cure, re-default, collection timing, customer outcomes and forecast accuracy. Part 5 retains ownership of how validated evidence enters pricing, accounting, capital, Treasury and funding after the relevant function has approved it.

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